Structure-Preserving Modeling, Complex-Representation Hamiltonian Neural Networks, and KAM Stability Analysis for Magnetically Actuated Flexible Catheters

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Abstract: This study develops a unified framework for the three-dimensional large-deformation dynamics of a magnetically actuated flexible catheter subject to magnetic, fluid, and vessel-wall contact effects in microgravity. A geometrically exact Cosserat-rod model, a joint translational–rotational Galerkin discretization, local constraint reduction, and a consistent Legendre transformation provide a finite-dimensional canonical Hamiltonian system and COMSOL-compatible state and port interfaces. The fixed $N_{\mathrm{diag}} = 2$ diagnostic basis uses the paired-coordinate contract $q = \Pi_{\mathrm{conf}}(U)$, $p = B_N^T P_{\mathrm{full}}$, $\dot{q} = M_N^{-1} p$, and $\dot{p} = \dot{B}_N^T P_{\mathrm{full}} + B_N^T \dot{P}_{\mathrm{full}}$. It is used to test consistency of the data mapping and does not establish $N_{\mathrm{diag}} = 2$ as the final sufficient modal order. All three candidate trajectories in the COMSOL–Galerkin bridge satisfy the independent 2% force-v2 joint-L2 criterion, whereas the original 0.5% strict perturbation-energy criterion is not satisfied overall; the two criteria are not interchangeable. The low-dimensional P0 study and the three-trajectory P1/P1.5/P1.6 studies show a conditional advantage in the derivative-supervised pilot, substantial gray-box residual gains over physics-only without stable one-step-field superiority, and a marked loss of predictive accuracy when exact equilibrium constraints are imposed, respectively. After correcting a cross-trajectory comparison, isolating the error source, identifying the cotangent projection contract, comparing solver-output semantics, and correcting fixed-basis initialization, a three-level time-step comparison yields observed orders of 1.64186–2.00088. For the 1 s amplitude-sentinel case, G2, pointwise G3, impulse G3, and peak-to-peak perturbation-energy drift are 0.00146226, 0.000221162, 0.0000563606, and $0.00307057 < 0.005$, respectively. The finite evidence supports at most this single diagnostic sentinel as a candidate input to a subsequent A1 qualification experiment; it does not certify data-generation qualification across modes, amplitudes, or operating conditions, and it neither establishes BDF symplecticity nor constitutes numerical certification of P2 or KAM conclusions. The planned 96-trajectory data set, full-scale P2 training, open-port and high-fidelity studies, numerical invariant tori, and engineering safe domain remain subjects of future work.

Keywords: magnetically actuated flexible catheter; geometrically exact rod; Hamiltonian neural network; complex-valued neural network; symplectic algorithm; KAM theory

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Introduction

Magnetically actuated flexible catheters in a microgravity environment are simultaneously subjected to magnetic actuation, background-flow effects, vessel-wall contact, and boundary manipulation. Their dynamics therefore combine large rotations, strong geometric nonlinearity, bending–torsion coupling, and open-system ports. If the continuous model, finite-dimensional reduction, data representation, and long-time stability analysis use inconsistent variables or energy conventions, model error, learning error, and numerical discretization error cannot be compared within a common theoretical framework. Consistent mathematical interfaces are therefore required across continuum mechanics, structure-preserving discretization, and verifiable machine learning.

Geometrically exact beam and rod theories provide a unified kinematic and variational foundation for flexible structures undergoing three-dimensional large deformation, while magnetic, fluid, and contact models describe the surrounding multiphysics environment[1–18]. However, retaining translation and cross-sectional orientation jointly, enforcing mutually exclusive energy accounting for conservative and nonconservative loads, and obtaining a finite-dimensional canonical system suitable for Hamiltonian learning and symplectic analysis require a consistent construction of the weak form, Galerkin discretization, constraint reduction, and Legendre transformation.

Hamiltonian neural networks introduce a conservative inductive bias through the symplectic gradient of a real-valued energy function. Dissipative and port-based networks extend structured identification to forced and dissipative systems, whereas complex-valued networks provide structured representations for amplitude–phase coupling[19–24]. A complex-valued internal representation, however, does not imply that the physical Hamiltonian may be an arbitrary complex-valued function, nor can dissipation and external forcing be generated unconditionally by a single complex scalar. Moreover, a multi-branch architecture does not guarantee identifiability by construction; it must be validated using a known physical backbone, port-power supervision, paired operating conditions, and ablation studies.

KAM theory and symplectic geometric algorithms provide rigorous tools for analyzing quasiperiodic structures in nearly integrable Hamiltonian systems and their numerical maps[25–33]. Classical continuous KAM theory, KAM theory for symplectic maps, two-dimensional twist-map theory, and infinite-dimensional KAM theory concern different mathematical objects. Neither a training loss nor a trajectory residual can directly replace the required smallness condition in a high-order or analytic function norm. Learning errors must therefore be measured together with physical perturbations, coordinate-transformation errors, Galerkin truncation, and time-discretization errors in compatible norms.

The main contributions of this work are summarized as follows.

(1) A multiphysics geometrically exact catheter model is established that retains centerline translation, cross-sectional orientation, independent torsion, and rotary inertia, while magnetic, fluid, contact, and boundary ports are organized through a unified energy–power identity.

(2) A joint translational–rotational Galerkin discretization, local reduction of the Kirchhoff constraints, and a consistent Legendre transformation are used to construct a finite-dimensional canonical Hamiltonian system compatible with COMSOL data interfaces.

(3) A structured neural network with complex-valued internal representations and a real-valued Hamiltonian readout is developed, with separate branches for conservative corrections, passive dissipation, external inputs, and general residuals.

(4) Interfaces are established among statistical learning error, high-order regularity certificates, symplectic-discretization error, and KAM conditions, yielding separate analysis routes for continuous flows, symplectic maps, and a posteriori invariant-torus validation.

The numerical study comprises the low-dimensional P0 architecture comparison, the three-trajectory P1/P1.5/P1.6 structural ablations, the COMSOL–Galerkin bridge, and diagnosis of

the canonical data contract. The finite evidence supports at most the single 1 s diagnostic sentinel as a candidate input to a subsequent A1 qualification experiment; it does not certify data-generation qualification across modes, amplitudes, or operating conditions. The planned 96-trajectory data set, full-scale P2 training, open-port identification, high-fidelity model comparison, and numerical KAM analysis have not yet been completed; consequently, the present pilot results are not used to claim full-scale model superiority, invariant tori, or an engineering safe domain.

The remainder of the paper is organized as follows. Sections 1 and 2 develop the continuous multiphysics model and the joint Galerkin discretization. Section 3 constructs the finite-dimensional Hamiltonian structure and canonical interfaces. Section 4 presents the complex-representation structured HNN and its statistical error analysis. Section 5 discusses symplectic discretization, Birkhoff structures, and invariant-torus validation. Numerical results are reported in Section 6, applicability and limitations are discussed in Section 7, and Section 8 concludes the paper.

1. Geometrically Exact Bending–Torsion Dynamics of a Flexible Catheter under Multiphysics Loading

1.1. **Modeling framework.** Consider the three-dimensional large-deformation motion of a magnetically actuated flexible catheter in a space-station microgravity environment under the combined action of magnetic fields, fluid flow, and vessel-wall contact. To retain the complete mechanical information required for the subsequent finite-dimensional Hamiltonian structure, complex-representation Hamiltonian neural network, and KAM stability analysis, the centerline position and cross-sectional orientation are treated as independent primary fields. The model retains axial extension, transverse shear, bending about two directions, independent torsion, cross-sectional rotary inertia, and externally distributed couples.

The continuous model is first formulated within a geometrically exact Cosserat-rod framework. For a slender catheter, a Kirchhoff shear-free constraint may subsequently be imposed; this constraint removes transverse shear strain but does not eliminate cross-sectional orientation, torsional degrees of freedom, angular velocity, or rotary inertia. The complete translational–rotational model is therefore not replaced by a centerline-only field equation.[1, 2, 17]

Let the reference material coordinate be

$$S \in [0, L], \qquad t \in [0, T], \tag{1.1}$$

where L is the catheter length in the reference configuration. Unless otherwise stated, all cross-sectional parameters may vary with S to represent nonuniform geometry, spatially varying material properties, or localized magnetic components.

Magnetic, fluid, and contact effects enter the rod equations as distributed forces and couples per unit reference length. A strictly coupled multiphysics PDE system would require simultaneous solution of the electromagnetic, fluid, and structural field equations. The formulation developed in this section is instead a geometrically exact rod-dynamics model driven by multiphysics load inputs.

1.1.1. *Model hierarchy and computational branches.* The overall route is as follows: a continuous model is constructed from Kirchhoff/geometrically exact rod mechanics, Maxwell magnetic actuation, Morison-type fluid loading, and catheter–vessel-wall contact; Galerkin discretization yields finite-dimensional dynamics; and the resulting system supports a Hamiltonian backbone, a complex-representation-enhanced structured Hamiltonian neural network, and KAM/structure-preserving stability analysis. To avoid conflating the continuous parent model, the COMSOL training model, the learned structured model, and the KAM analysis object, the following four-level hierarchy is used throughout.

1. **Geometrically exact physical parent model.** The coupled translational–cross-sectional-orientation Cosserat rod, independent torsion, rotary inertia, distributed

forces, and distributed couples are retained. This level preserves the complete mechanical information and provides the common energy and virtual-work basis for all reduced branches[1, 2].

2. **Strict Kirchhoff theoretical branch.** For slender, low-shear catheters, the shear-free constraint is imposed without removing cross-sectional orientation, torsion, angular velocity, or rotary inertia. Before CHNN or KAM analysis, the constrained finite-dimensional system must be expressed in independent local coordinates on the constraint manifold or approximated by a regularized model whose convergence to the constrained system has been verified. A differential–algebraic system with Lagrange multipliers must not be treated directly as a canonical Hamiltonian system on the full coordinate space.
3. **COMSOL Kirchhoff-consistent training branch.** The primary training data preferentially use the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom, which is the closest built-in COMSOL realization of the Kirchhoff shear-free assumptions. Because this interface permits large displacements and rotations while retaining a small-strain, element-local linearization, it is referred to as a "Kirchhoff-consistent engineering model rather than being identified unconditionally with the geometrically exact parent model[4–6]. When term-by-term reproduction of the coupled weak form is required, custom Weak Form PDE or Weak Contribution equations are used[7, 8].
4. **High-fidelity validation and open-port branch.** Timoshenko beams, three-dimensional Solid Mechanics, or a custom Cosserat weak form are used to assess shear effects, local cross-sectional effects, and model discrepancy. Magnetic control, background-flow input, passive drag, contact dissipation, friction, and boundary actuation enter the finite-dimensional model through mutually exclusive power ports. High-fidelity data are used for external validation and model-discrepancy learning and are not mixed with conservative Kirchhoff samples without explicit labels. The complete open system is used for structured CHNN identification and safe-domain computation; only the autonomous conservative backbone obtained by closing the ports, or an independently proven exact symplectic extension, is a direct object of classical KAM theory.

Accordingly, in addition to the continuous equations and Galerkin system, the first two sections provide Sections 3 to 5 with mutually consistent conservative energies, constraint-reduced states, port decompositions, model-level labels, and numerical-representation labels. All subsequent engineering reductions are performed within this common physical–mathematical framework.

1.2. **Geometry and kinematics.**

1.2.1. *Centerline and cross-sectional frame.* Let the catheter centerline be described by the position vector

$$\mathbf{r} = \mathbf{r}(S,t) \in \mathbb{R}^3. \tag{1.2}$$

Introduce a right-handed orthonormal frame on each cross section,

$$\{\mathbf{d}_1(S,t), \mathbf{d}_2(S,t), \mathbf{d}_3(S,t)\}, \tag{1.3}$$

and define the rotation matrix

$$\mathbf{R}(S,t) = [\mathbf{d}_1\ \mathbf{d}_2\ \mathbf{d}_3] \in SO(3), \tag{1.4}$$

which satisfies

$$\mathbf{R}^T\mathbf{R} = \mathbf{I}, \qquad \det \mathbf{R} = 1. \tag{1.5}$$

The full model does not impose $\mathbf{d}_3$ to be parallel to the centerline tangent; transverse shear is therefore admissible. The centerline stretch and unit tangent are defined, respectively, by

$$\lambda(S,t) = \| \mathbf{r}_{,S} \|, \qquad \mathbf{t}(S,t) = \frac{\mathbf{r}_{,S}}{\lambda}. \tag{1.6}$$

1.2.2. *Skew-symmetric operator.* For any $\mathbf{a} \in \mathbb{R}^3$, define the skew-symmetric matrix $\widehat{\mathbf{a}} \in \mathfrak{so}(3)$ such that[9, 17]

$$\widehat{\mathbf{a}}\,\mathbf{b} = \mathbf{a} \times \mathbf{b}, \qquad \forall \mathbf{b} \in \mathbb{R}^3. \tag{1.7}$$

Its inverse mapping is denoted by $(\cdot)^\vee$. Hence, both spatial and temporal derivatives of the rotation matrix can be represented by three-dimensional vectors.

1.2.3. *Shear–extension strain.* Define the shear–extension strain vector in the material frame as

$$\boldsymbol{\nu}(S,t) = \mathbf{R}^T \mathbf{r}_{,S} = \begin{bmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{bmatrix}. \tag{1.8}$$

where ν_1 and ν_2 are the two transverse shear-strain components and ν_3 is the axial extension component along the third director. The reference strain is denoted by

$$\boldsymbol{\nu}_0(S) = \begin{bmatrix} \nu_{01} \\ \nu_{02} \\ \nu_{03} \end{bmatrix}. \tag{1.9}$$

For a straight rod with no initial shear and no reference-length extension, one may take

$$\boldsymbol{\nu}_0 = (0,0,1)^T. \tag{1.10}$$

In the fully shearable Cosserat parent model, the axial material-strain component along the third director must be distinguished from the centerline stretch strain. The former is defined by

$$\varepsilon_3 = \nu_3 - \nu_{03}. \tag{1.11}$$

On the other hand, $\mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}$ together with $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ gives $\lambda = \| \mathbf{r}_{,S} \| = \| \boldsymbol{\nu} \|$. The centerline extension strain is therefore defined as

$$\varepsilon_\lambda = \lambda - \lambda_0 = \| \boldsymbol{\nu} \| - \| \boldsymbol{\nu}_0 \|. \tag{1.12}$$

Under the Kirchhoff shear-free condition $\nu_1 = \nu_2 = 0$, if the reference configuration also satisfies $\nu_{03} = \lambda_0$, then $\varepsilon_3 = \varepsilon_\lambda$. The constitutive matrices used below remain formulated in terms of the material strain $\boldsymbol{\nu} - \boldsymbol{\nu}_0$.

1.2.4. *Curvature–torsion kinematics.* Define the curvature–torsion vector in the material frame by

$$\widehat{\boldsymbol{\kappa}} = \mathbf{R}^T \mathbf{R}_{,S}, \qquad \boldsymbol{\kappa} = \begin{bmatrix} \kappa_1 \\ \kappa_2 \\ \kappa_3 \end{bmatrix}. \tag{1.13}$$

where κ_1 and κ_2 are the bending curvatures about the two principal directions and κ_3 is the independent twist rate. The variation of the local frame along the reference coordinate satisfies

$$\mathbf{d}_{i,S} = (\mathbf{R}\boldsymbol{\kappa}) \times \mathbf{d}_i, \qquad i = 1,2,3. \tag{1.14}$$

The reference curvature–torsion vector is denoted by

$$\boldsymbol{\kappa}_0(S) = (\kappa_{01}, \kappa_{02}, \kappa_{03})^T. \tag{1.15}$$

1.2.5. *Cross-sectional angular velocity and compatibility.* Define the cross-sectional angular velocity in the material frame by

$$\widehat{\boldsymbol{\omega}} = \mathbf{R}^T \mathbf{R}_{,t}. \tag{1.16}$$

The corresponding angular velocity in the spatial frame is

$$\boldsymbol{\omega}_s = \mathbf{R}\boldsymbol{\omega}. \tag{1.17}$$

The identities $\mathbf{r}_{,St} = \mathbf{r}_{,tS}$ and $\mathbf{R}_{,St} = \mathbf{R}_{,tS}$ yield the kinematic compatibility relations[1, 2, 9]

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T \dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \tag{1.18}$$

$$\dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.19}$$

The second relation enforces compatibility between the spatial and temporal evolution of the cross-sectional orientation and is indispensable to the complete bending–torsion dynamics.

1.2.6. *Motion of material points within a cross section.* Let the in-plane material coordinate be

$$\boldsymbol{\xi} = \xi_1 \mathbf{e}_1 + \xi_2 \mathbf{e}_2, \tag{1.20}$$

The spatial position of a material point in the rod is then

$$\mathbf{x}\left(S, \boldsymbol{\xi}, t\right) = \mathbf{r}(S,t) + \mathbf{R}(S,t)\boldsymbol{\xi}. \tag{1.21}$$

and its velocity is

$$\dot{\mathbf{x}} = \dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega} \times \boldsymbol{\xi}\right) = \dot{\mathbf{r}} - \mathbf{R}\widehat{\boldsymbol{\xi}}\boldsymbol{\omega}. \tag{1.22}$$

This expression contains both centerline translation and cross-sectional rotational velocity and is the starting point for the complete kinetic energy and the configuration-dependent mass matrix.

1.3. **Complete kinetic energy, elastic potential, and constitutive relations.**

1.3.1. *Cross-sectional mass parameters and the mass-centerline convention.* Let the material density be $\rho\left(S, \boldsymbol{\xi}\right)$. Define the mass per unit reference length as

$$\mu(S) = \int_{A(S)} \rho \, dA, \tag{1.23}$$

Define the first mass moment of the cross section by

$$\mathbf{c}_\rho(S) = \int_{A(S)} \rho \boldsymbol{\xi} \, dA, \tag{1.24}$$

and the mass-inertia tensor per unit reference length in the material frame by

$$\mathbf{J}_c(S) = \int_{A(S)} \rho \left(\| \boldsymbol{\xi} \|^2 \mathbf{I} - \boldsymbol{\xi}\boldsymbol{\xi}^T \right) dA. \tag{1.25}$$

The preceding definitions retain a general cross-sectional mass distribution. To ensure that the subsequent momentum balance, weak form, and discrete mass matrix use a common inertial reference, the centerline $\mathbf{r}(S,t)$ is taken, from Section 1.4 onward, to pass through the mass center of each cross section. The distributed rod therefore satisfies

$$\boxed{\mathbf{c}_\rho(S) = \mathbf{0}.} \tag{1.26}$$

This convention removes neither cross-sectional rotational degrees of freedom nor rotary inertia. A magnetic tip, sensor, or other localized component whose center of mass is offset from the centerline is not absorbed into $\mathbf{c}_\rho$; it is introduced as an attached rigid body with its own coupled translational–rotational kinetic energy. This treatment preserves both closure of the base-rod equations and the physical inertial coupling of eccentric attachments.

If the principal cross-sectional axes coincide with $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, then

$$\mathbf{J}_c = diag\left(\rho I_1, \rho I_2, \rho I_3\right), \tag{1.27}$$

where I_1, I_2 are the area moments of inertia about the two principal directions and $I_3 = I_1 + I_2$ is the polar moment. For a nonuniform density distribution, the area-integral definitions must be used directly; the density cannot in general be factored out of the integrals.

1.3.2. *Complete structural kinetic energy.* The three-dimensional kinetic energy of the base rod follows from the material-point velocity as

$$T_{str} = \frac{1}{2}\int_0^L \int_{A(S)} \rho \left|\dot{\mathbf{r}} + \mathbf{R}\left(\boldsymbol{\omega} \times \boldsymbol{\xi}\right)\right|^2 dA\, dS. \tag{1.28}$$

After integration over the cross section, the general form becomes

$$T_{str} = \frac{1}{2}\int_0^L \left[\mu \left|\dot{\mathbf{r}}\right|^2 + 2\dot{\mathbf{r}}^T \mathbf{R}\left(\boldsymbol{\omega} \times \mathbf{c}_\rho\right) + \boldsymbol{\omega}^T \mathbf{J}_c \boldsymbol{\omega}\right] dS. \tag{1.29}$$

Under the mass-centerline convention of Section 1.3.1, $\mathbf{c}_\rho = \mathbf{0}$, and the kinetic energy of the base rod reduces to

$$\boxed{T_{str} = \frac{1}{2}\int_0^L \left[\mu \left|\dot{\mathbf{r}}\right|^2 + \boldsymbol{\omega}^T \mathbf{J}_c \boldsymbol{\omega}\right] dS.} \tag{1.30}$$

If eccentric attached components are present, their kinetic energy T_e is added separately and generally reintroduces translational–rotational cross terms. Regardless of whether the base rod uses a mass-centerline convention, the kinetic energy retains rotary inertia about both transverse axes and torsional inertia about the tangent direction. The model must not subsequently be reduced to a centerline-only field by deleting the rotational terms.

1.3.3. *Elastic strain-energy density.* Let the shear–extension and bending–torsion stiffness matrices be, respectively,

$$\mathbf{K}_\nu(S) = diag\left(k_1 GA, k_2 GA, EA\right), \tag{1.31}$$

$$\mathbf{K}_\kappa(S) = diag\left(EI_1, EI_2, GJ\right), \tag{1.32}$$

where E is Young's modulus, G is the shear modulus, k_1, k_2 are shear correction factors, and J is the cross-sectional torsional constant. For asymmetric materials or cross sections, $\mathbf{K}_\nu$ and $\mathbf{K}_\kappa$ may be general symmetric positive-definite matrices so that directional coupling is retained.

The elastic strain-energy density per unit reference length is defined by

$$W\left(\boldsymbol{\nu}, \boldsymbol{\kappa}; S\right) = \frac{1}{2}\left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right)^T \mathbf{K}_\nu \left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right) + \frac{1}{2}\left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right)^T \mathbf{K}_\kappa \left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right). \tag{1.33}$$

The internal elastic potential energy is

$$V_{int} = \int_0^L W\, dS. \tag{1.34}$$

1.3.4. *Constitutive laws for internal force and couple.* The resultant internal force and couple in the material frame are obtained from the gradient of the strain-energy density:

$$\mathbf{n}_b = \frac{\partial W}{\partial \boldsymbol{\nu}} = \mathbf{K}_\nu \left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right), \tag{1.35}$$

$$\mathbf{m}_b = \frac{\partial W}{\partial \boldsymbol{\kappa}} = \mathbf{K}_\kappa \left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right). \tag{1.36}$$

Their counterparts in the spatial frame are

$$\mathbf{n} = \mathbf{R}\mathbf{n}_b, \qquad \mathbf{m} = \mathbf{R}\mathbf{m}_b. \tag{1.37}$$

These constitutive equations retain extension, shear in both transverse directions, bending about two axes, and independent torsion. For a general nonlinear hyperelastic material, the quadratic form W may be replaced by the corresponding objective strain-energy density, with the constitutive resultants still obtained from its derivatives.

1.4. **Momentum balance and multiphysics loading.**

1.4.1. *Translational balance.* The continuous balance laws below use the mass centerline specified in Section 1.3.1 as the centerline of the base catheter. The linear-momentum balance per unit reference length is

$$\mu\ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \tag{1.38}$$

where

$$\mathbf{f}_{ext} = \mathbf{f}_{mag} + \mathbf{f}_{hydro} + \mathbf{f}_{cont} + \mathbf{f}_g. \tag{1.39}$$

All terms represent spatially distributed forces per unit reference length. The inertia of eccentric attachments is not contained implicitly in $\mu\ddot{\mathbf{r}}$; it is introduced consistently at the discrete level through their independent kinetic energy. Under the energy-consistent fluid formulation of Section 1.4.4, the added-inertia operator is placed on the left-hand side, while the fluid-acceleration input on the right-hand side contains only the pure Froude–Krylov term and drag. The complete engineering Morison branch and this theoretical branch must not be activated simultaneously.

1.4.2. *Rotational balance.* The angular-momentum balance about the centerline is written as

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \frac{\partial}{\partial t}\left(\mathbf{R}\mathbf{J}_c\boldsymbol{\omega}\right), \tag{1.40}$$

where

$$\mathbf{l}_{ext} = \mathbf{l}_{mag} + \mathbf{l}_{hydro} + \mathbf{l}_{cont}. \tag{1.41}$$

Using $\dot{\mathbf{R}} = \mathbf{R}\widehat{\boldsymbol{\omega}}$, the expanded form is

$$\mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})\right]. \tag{1.42}$$

where $\boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})$ is the gyroscopic inertia term. It follows from the exact time derivative of angular momentum and is not an empirical correction that may be added or removed arbitrarily.

1.4.3. *Magnetic force and couple from the Maxwell stress tensor.* In the air or vacuum region surrounding an infinitesimal magnetic catheter segment, the Maxwell stress tensor is[10, 11]

$$\mathbf{T}_M = \frac{1}{\mu_0}\left(\mathbf{B} \otimes \mathbf{B} - \frac{1}{2}|\mathbf{B}|^2\mathbf{I}\right), \tag{1.43}$$

where $\mathbf{B}$ is the magnetic flux density and μ_0 is the permeability of free space. For a segment of reference length ΔS, the magnetic force per unit reference length and the magnetic couple about the centerline are defined by

$$\mathbf{f}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} \mathbf{T}_M \mathbf{n}_A \, dA, \tag{1.44}$$

$$\mathbf{l}_{mag} = \lim_{\Delta S \to 0} \frac{1}{\Delta S} \int_{\partial\Omega_{\Delta S}} (\mathbf{x} - \mathbf{r}) \times (\mathbf{T}_M \mathbf{n}_A) \, dA, \tag{1.45}$$

where $\mathbf{n}_A$ is the outward unit normal to the closed integration surface.

When a magnetic segment can be represented by a dipole approximation, let $\boldsymbol{\mu}_0^b(S)$ be the magnetic moment per unit reference length in the material frame. The corresponding spatial magnetic moment is[10, 11]

$$\boldsymbol{\mu}(S,t) = \mathbf{R}(S,t)\boldsymbol{\mu}_0^b(S). \tag{1.46}$$

For a prescribed external magnetic field $\mathbf{B}(\mathbf{r},t)$, the magnetic potential energy is

$$V_{mag} = -\int_0^L \boldsymbol{\mu}\cdot\mathbf{B}(\mathbf{r},t)\, dS. \tag{1.47}$$

The associated magnetic force and couple are

$$\mathbf{f}_{mag} = \nabla_{\mathbf{r}}(\boldsymbol{\mu}\cdot\mathbf{B}), \qquad \mathbf{l}_{mag} = \boldsymbol{\mu}\times\mathbf{B}. \tag{1.48}$$

The Maxwell-stress integral and the magnetic-dipole potential are two scale-dependent representations of the same magnetic action. One should be selected as the primary load model according to the required fidelity; they must not be superposed and counted twice.

1.4.4. *Modified Morison loading and energy-consistent added mass.* Let the fluid velocity be $\mathbf{v}_f(\mathbf{r},t)$, with material acceleration

$$\mathbf{a}_f = \frac{\partial \mathbf{v}_f}{\partial t} + (\mathbf{v}_f\cdot\nabla)\mathbf{v}_f. \tag{1.49}$$

Define the fluid velocity relative to the catheter by

$$\mathbf{u}_r = \mathbf{v}_f - \dot{\mathbf{r}}. \tag{1.50}$$

If only the component normal to the current centerline is retained, define

$$\mathbf{P}_\perp = \mathbf{I} - \mathbf{t}\otimes\mathbf{t}, \qquad \mathbf{u}_{r\perp} = \mathbf{P}_\perp\mathbf{u}_r. \tag{1.51}$$

An engineering Morison validation branch may be written as[3, 12, 13]

$$\mathbf{f}_{hydro}^M = \mathbf{f}_{in}^M + \mathbf{f}_{add}^M + \mathbf{f}_{drag}, \tag{1.52}$$

where

$$\mathbf{f}_{in}^M = \rho_f C_M A_d \mathbf{P}_\perp \mathbf{a}_f, \tag{1.53}$$

$$\mathbf{f}_{add}^M = -\rho_f C_A A_d \mathbf{P}_\perp \ddot{\mathbf{r}}, \tag{1.54}$$

$$\mathbf{f}_{drag} = \frac{1}{2}\rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \mathbf{u}_{r\perp}. \tag{1.55}$$

Here, ρ_f is the fluid density, A_d is the characteristic displaced-fluid area per unit length, D is the catheter outer diameter, and C_A, C_M, C_D are the added-mass, inertia, and drag coefficients, respectively; under a common convention, $C_M = 1 + C_A$. This empirical branch must be used as a complete formulation and must not be combined with the functional T_A introduced below.

To incorporate configuration-dependent added mass into the full inertial structure, the theoretical branch is instead defined through a velocity-dependent added-fluid inertia functional. Let[12, 13]

$$m_A(S) = \rho_f C_A A_d, \qquad \mathbf{w}_\perp = \mathbf{P}_\perp(\dot{\mathbf{r}} - \mathbf{v}_f), \tag{1.56}$$

and take

$$T_A[\mathbf{r},\dot{\mathbf{r}};t] = \frac{1}{2}\int_0^L m_A\, \mathbf{w}_\perp^T\mathbf{w}_\perp\, dS. \tag{1.57}$$

The corresponding inertial action is generated consistently by the functional Euler–Lagrange derivative

$$\mathbf{I}_A = \frac{d}{dt}\frac{\delta T_A}{\delta\dot{\mathbf{r}}} - \frac{\delta T_A}{\delta\mathbf{r}} \tag{1.58}$$

When $\mathbf{P}_\perp$ is frozen, the leading term is

$$\mathbf{I}_A = \rho_f C_A A_d \mathbf{P}_\perp \left(\ddot{\mathbf{r}} - \mathbf{a}_f\right), \tag{1.59}$$

Variations of $\mathbf{P}_\perp$ with configuration and spatial variations of the prescribed flow generate additional geometric terms. When the theoretical branch based on T_A is used, the explicit fluid-acceleration input on the right-hand side retains only the pure Froude–Krylov term

$$\boxed{\mathbf{f}_{FK} = \rho_f A_d \mathbf{P}_\perp \mathbf{a}_f}, \tag{1.60}$$

with

$$\mathbf{f}_{hydro,rem} = \mathbf{f}_{FK} + \mathbf{f}_{drag}. \tag{1.61}$$

In this formulation, T_A already supplies the complete added-mass action associated with $C_A\left(\mathbf{a}_f - \ddot{\mathbf{r}}\right)$. Together with the Froude–Krylov term having coefficient 1, it recovers $C_M = 1 + C_A$; consequently, $\mathbf{f}_{in}^M$ must no longer retain its $C_A\mathbf{a}_f$ term, and no additional $\mathbf{f}_{add}^M$ may be included. When $\mathbf{v}_f = \mathbf{0}$ and m_A is time independent, T_A reduces to an added kinetic energy that can be included in the conservative inertial backbone. For a general prescribed flow, its cross terms and explicit time dependence are treated collectively as a flow port.

For quadratic drag, define the nonnegative instantaneous state-dependent coefficient

$$d_f = \frac{1}{2}\rho_f C_D D \parallel \mathbf{u}_{r\perp} \parallel \geq 0. \tag{1.62}$$

The drag force must then be decomposed exclusively into a passive dissipative kernel and a background-flow input:

$$\boxed{\mathbf{f}_{drag} = d_f \mathbf{P}_\perp \left(\mathbf{v}_f - \dot{\mathbf{r}}\right) = \underbrace{-d_f \mathbf{P}_\perp \dot{\mathbf{r}}}_{\mathbf{f}_{drag,diss}} + \underbrace{d_f \mathbf{P}_\perp \mathbf{v}_f}_{\mathbf{f}_{drag,flow}}.} \tag{1.63}$$

Only the passive kernel has a definite dissipative sign with respect to the catheter mechanical energy:

$$\dot{\mathbf{r}}^T \mathbf{f}_{drag,diss} = -d_f \parallel \mathbf{P}_\perp \dot{\mathbf{r}} \parallel^2 \leq 0, \tag{1.64}$$

The term $\dot{\mathbf{r}}^T \mathbf{f}_{drag,flow}$ is generally sign-indefinite and is therefore assigned to the flow-input port. The complete drag force also satisfies the relative-motion passivity identity

$$\left(\dot{\mathbf{r}} - \mathbf{v}_f\right)^T \mathbf{f}_{drag} = -d_f \parallel \mathbf{P}_\perp \left(\dot{\mathbf{r}} - \mathbf{v}_f\right) \parallel^2 \leq 0, \tag{1.65}$$

This identity, however, does not replace the port accounting in the structural mechanical-energy balance. Thus, when $\mathbf{v}_f \neq \mathbf{0}$, the complete quadratic drag must not be assigned entirely to $\mathbf{Q}_{diss}^{bulk}$. Because d_f depends on the relative velocity, the passive kernel is a statewise positive-semidefinite nonlinear operator rather than a constant damping matrix.

If damping of cross-sectional spin by the fluid is relevant, an additional fluid couple per unit reference length may be introduced, for example,

$$\mathbf{l}_{hydro} = -c_\omega(S) \left|\omega_3 - \omega_{f3}\right| \left(\omega_3 - \omega_{f3}\right) \mathbf{d}_3, \tag{1.66}$$

where ω_{f3} is the effective fluid angular velocity about the catheter tangent. If this effect is negligible compared with the magnetic and structural torsional moments, one may set $\mathbf{l}_{hydro} = \mathbf{0}$ for the operating condition under consideration, while retaining the cross-sectional rotational degree of freedom.

The baseline computational route assumes a stationary vessel wall and Morison loading associated with normal cross-flow, neglecting axial tangential skin friction; the effective fluid spin about the catheter tangent is set to $\omega_{f3} = 0$. These are medium-fidelity reductions of the environmental loads and do not remove the structural variables describing cross-sectional orientation, independent torsion, angular velocity, or rotary inertia. Moving walls, tangential friction, or fluid spin can be incorporated later through additional ports following the same input–dissipation accounting.

1.4.5. *Catheter–vessel-wall contact force and couple.* Let $g(S,t)$ be the gap function between the catheter surface and the vessel wall, and let g_0 be the contact-detection threshold. Define the penetration measure

$$\delta_c = \max\left(g_0 - g, 0\right). \tag{1.67}$$

To represent Hertzian elasticity, normal-impact dissipation, and regularized friction simultaneously, let $\dot{\delta}_c$ denote the material time derivative of the penetration at the contact point. When contact is active, $\delta_c > 0$, the normal force is taken as[14–16]

$$f_n = k_c \delta_c^{\alpha} + c_n \dot{\delta}_c, \qquad \delta_c > 0. \tag{1.68}$$

If the preceding expression is negative or if $\delta_c = 0$, set $f_n = 0$. This nonnegative truncation prevents nonphysical tensile contact forces during separation or noncontact. The tangential friction force and total contact force are, respectively,

$$\mathbf{f}_t = -\mu_c f_n \frac{\mathbf{v}_t}{\sqrt{\mathbf{v}_t \cdot \mathbf{v}_t + v_0^2}}, \tag{1.69}$$

$$\mathbf{f}_{cont} = f_n \mathbf{n}_w + \mathbf{f}_t. \tag{1.70}$$

Here, k_c is the contact stiffness, c_n is the normal contact-damping coefficient, α is the contact exponent, and Hertzian contact commonly uses $\alpha = 3/2$. The vector $\mathbf{n}_w$ is the unit normal directed away from the wall, $\mathbf{v}_t$ is the tangential velocity of the contact point relative to the wall, μ_c is the friction coefficient, and v_0 is a regularization parameter. For automatic differentiation or a consistent Newton iteration, the contact activation and nonnegative truncation may be replaced by smooth regularizations. For conservative contact, one may set $c_n = \mu_c = 0$, in which case the elastic contact term is derivable from a potential.

If the contact force acts at an offset vector $\boldsymbol{\rho}_c$ from the centerline, the contact couple is

$$\mathbf{l}_{cont} = \boldsymbol{\rho}_c \times \mathbf{f}_{cont} + \mathbf{l}_{spin}, \tag{1.71}$$

where $\mathbf{l}_{spin}$ may represent additional damping or frictional torque associated with cross-sectional spin in the contact region. Normal damping, tangential friction, and $\mathbf{l}_{spin}$ are all included in the nonconservative power.

1.4.6. *Microgravity term.* If a residual acceleration $\mathbf{g}_{res}(t)$ is present in the space station, the residual gravitational force per unit reference length is

$$\mathbf{f}_g = \mu \mathbf{g}_{res}. \tag{1.72}$$

Under the ideal microgravity approximation, set

$$\mathbf{g}_{res} = \mathbf{0}. \tag{1.73}$$

1.4.7. *Complete strong-form system.* Collecting the kinematics, balance laws, and constitutive equations gives the geometrically exact system for the base rod:

$$\begin{cases} \mathbf{r}_{,S} = \mathbf{R}\boldsymbol{\nu}, \\ \mathbf{R}_{,S} = \mathbf{R}\widehat{\boldsymbol{\kappa}}, \\ \mathbf{R}_{,t} = \mathbf{R}\widehat{\boldsymbol{\omega}}, \\ \mu \ddot{\mathbf{r}} = \mathbf{n}_{,S} + \mathbf{f}_{ext}, \\ \mathbf{m}_{,S} + \mathbf{r}_{,S} \times \mathbf{n} + \mathbf{l}_{ext} = \mathbf{R}\left[\mathbf{J}_c \dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c \boldsymbol{\omega})\right], \\ \mathbf{n} = \mathbf{R}\mathbf{K}_{\nu}\left(\boldsymbol{\nu} - \boldsymbol{\nu}_0\right), \\ \mathbf{m} = \mathbf{R}\mathbf{K}_{\kappa}\left(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0\right). \end{cases} \tag{1.74}$$

with the compatibility relations

$$\dot{\boldsymbol{\nu}} = \mathbf{R}^T \dot{\mathbf{r}}_{,S} - \boldsymbol{\omega} \times \boldsymbol{\nu}, \qquad \dot{\boldsymbol{\kappa}} = \boldsymbol{\omega}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\omega}. \tag{1.75}$$

Under the energy-consistent added-mass formulation, $\mathbf{I}_A$ is placed on the left-hand side together with the structural inertia, and only $\mathbf{f}_{FK}$, drag, and the remaining fluid ports are retained on the right-hand side. Under the engineering Morison validation branch, the complete combination $\mathbf{f}_{in}^M + \mathbf{f}_{add}^M$ is retained and T_A is omitted. Exactly one of these two representations must be used.

The primary unknowns are $\mathbf{r}, \mathbf{R}, \boldsymbol{\nu}, \boldsymbol{\kappa}, \boldsymbol{\omega}, \mathbf{n}, \mathbf{m}$. The system is a geometrically nonlinear mixed partial differential–algebraic system with Lie-group-valued orientation variables.

1.5. **Variational formulation and complete weak form.**

1.5.1. *Virtual displacement and virtual rotation.* Let $\delta\mathbf{r}$ be the virtual centerline displacement and represent the orientation variation by the material virtual rotation $\boldsymbol{\eta}$:

$$\delta\mathbf{R} = \mathbf{R}\widehat{\boldsymbol{\eta}}. \tag{1.76}$$

The corresponding spatial virtual rotation is

$$\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}. \tag{1.77}$$

The strain definitions give

$$\delta\boldsymbol{\nu} = \mathbf{R}^T\delta\mathbf{r}_{,S} - \boldsymbol{\eta} \times \boldsymbol{\nu}, \tag{1.78}$$

$$\delta\boldsymbol{\kappa} = \boldsymbol{\eta}_{,S} + \boldsymbol{\kappa} \times \boldsymbol{\eta}. \tag{1.79}$$

1.5.2. *Internal virtual work.* The internal virtual work is

$$\delta W_{int} = \int_0^L (\mathbf{n}_b \cdot \delta\boldsymbol{\nu} + \mathbf{m}_b \cdot \delta\boldsymbol{\kappa})\, dS. \tag{1.80}$$

Substituting the strain variations and integrating the spatial derivatives by parts gives

$$\delta W_{int} = [\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}]_0^L - I_r - I_R, \tag{1.81}$$

where

$$I_r = \int_0^L \mathbf{n}_{,S} \cdot \delta\mathbf{r}\, dS, \tag{1.82}$$

$$I_R = \int_0^L (\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b) \cdot \boldsymbol{\eta}\, dS. \tag{1.83}$$

The material-form rotational balance term satisfies

$$\mathbf{m}_{b,S} + \boldsymbol{\kappa} \times \mathbf{m}_b + \boldsymbol{\nu} \times \mathbf{n}_b + \mathbf{l}_b = \mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega}), \tag{1.84}$$

and $\mathbf{l}_b = \mathbf{R}^T\mathbf{l}_{ext}$.

1.5.3. *Inertial and external virtual work.* The inertial virtual work of the base rod is

$$\delta W_{in,str} = \int_0^L [\mu\ddot{\mathbf{r}} \cdot \delta\mathbf{r} + (\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})) \cdot \boldsymbol{\eta}]\, dS. \tag{1.85}$$

If the energy-consistent added-mass formulation of Section 1.4.4 is used, the added-inertia virtual work is

$$\delta W_{in,A} = \left\langle \frac{d}{dt}\frac{\delta T_A}{\delta\dot{\mathbf{r}}} - \frac{\delta T_A}{\delta\mathbf{r}},\, \delta\mathbf{r} \right\rangle, \tag{1.86}$$

where $\langle\cdot,\cdot\rangle$ denotes the duality pairing along the rod. If eccentric attachments are present, their inertial virtual work is generated by the Euler–Lagrange derivative of the corresponding rigid-body kinetic energy T_e. The total inertial virtual work is therefore

$$\delta W_{in} = \delta W_{in,str} + \delta W_{in,A} + \delta W_{in,e}. \tag{1.87}$$

The virtual work of distributed external loads is

$$\delta W_{ext} = \int_0^L \left(\mathbf{f}_{ext,rem} \cdot \delta\mathbf{r} + \mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s\right) dS + \delta W_{\partial}, \tag{1.88}$$

where $\mathbf{f}_{ext,rem}$ no longer contains the added inertia already represented by T_A. In the energy-consistent fluid branch, the fluid contribution contains only $\mathbf{f}_{FK}$, drag, and other inputs not represented by T_A. Because $\boldsymbol{\eta}_s = \mathbf{R}\boldsymbol{\eta}$, it may equivalently be written as

$$\mathbf{l}_{ext} \cdot \boldsymbol{\eta}_s = \mathbf{l}_b \cdot \boldsymbol{\eta}. \tag{1.89}$$

1.5.4. *Coupled weak form.* By D'Alembert's principle, every admissible pair $\delta\mathbf{r}$ and $\boldsymbol{\eta}$ satisfying the essential boundary conditions obeys

$$\delta W_{in} + \delta W_{int} - \delta W_{ext} = 0. \tag{1.90}$$

Expansion yields the complete coupled translational–rotational weak form:

$$\begin{gathered} \int_0^L \mu\ddot{\mathbf{r}} \cdot \delta\mathbf{r}\, dS + \int_0^L \left[\mathbf{J}_c\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}_c\boldsymbol{\omega})\right] \cdot \boldsymbol{\eta}\, dS \\ + \delta W_{in,A} + \delta W_{in,e} + \int_0^L \left(\mathbf{n}_b \cdot \delta\boldsymbol{\nu} + \mathbf{m}_b \cdot \delta\boldsymbol{\kappa}\right) dS \\ - \int_0^L \left(\mathbf{f}_{ext,rem} \cdot \delta\mathbf{r} + \mathbf{l}_b \cdot \boldsymbol{\eta}\right) dS - \delta W_{\partial} = 0. \end{gathered} \tag{1.91}$$

If neither the added-fluid inertia functional nor eccentric attachments are included, setting $\delta W_{in,A} = \delta W_{in,e} = 0$ recovers the base-rod weak form. Because the weak form contains only first spatial derivatives of the centerline and orientation fields, C^0-continuous approximation spaces for both fields are sufficient; no fourth-order centerline equation need be constructed by eliminating the rotational equation.

In functional terms, the centerline and orientation may be viewed as mappings in $\mathbf{r} \in H^1\left(0, L; \mathbb{R}^3\right)$ and $\mathbf{R} \in H^1\left(0, L; SO(3)\right)$, respectively. The virtual displacement $\delta\mathbf{r}$ and material virtual rotation $\boldsymbol{\eta}$ belong to the corresponding H^1 test spaces satisfying homogeneous essential boundary conditions. Thus, the coupled weak form requires only first-order spatial regularity.

1.5.5. *Boundary terms and boundary conditions.* Integration by parts produces the natural boundary term

$$\left[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m} \cdot \boldsymbol{\eta}_s\right]_0^L. \tag{1.92}$$

At an endpoint $S = S_e$, one of the following two classes of boundary conditions may be prescribed:

$$\mathbf{r}\left(S_e, t\right) = \overline{\mathbf{r}}(t) \quad \text{or} \quad \mathbf{n}\left(S_e, t\right) = \overline{\mathbf{n}}(t), \tag{1.93}$$

$$\mathbf{R}\left(S_e, t\right) = \overline{\mathbf{R}}(t) \quad \text{or} \quad \mathbf{m}\left(S_e, t\right) = \overline{\mathbf{m}}(t). \tag{1.94}$$

At a clamped end, $\delta\mathbf{r} = \mathbf{0}$ and $\boldsymbol{\eta} = \mathbf{0}$; at a free end without applied terminal loads, the natural boundary conditions are $\mathbf{n} = \mathbf{0}$ and $\mathbf{m} = \mathbf{0}$. Boundary forces and moments must remain in the weak form unless the boundary conditions have been specified explicitly.

The complete initial–boundary-value problem also requires the initial conditions

$$\mathbf{r}(S, 0) = \mathbf{r}^0(S), \qquad \dot{\mathbf{r}}(S, 0) = \mathbf{v}^0(S), \tag{1.95}$$

$$\mathbf{R}(S, 0) = \mathbf{R}^0(S) \in SO(3), \qquad \boldsymbol{\omega}(S, 0) = \boldsymbol{\omega}^0(S). \tag{1.96}$$

The initial fields must satisfy the geometric compatibility relations $\boldsymbol{\nu}^0 = \left(\mathbf{R}^0\right)^T \mathbf{r}^0_{,S}$ and $\widehat{\boldsymbol{\kappa}}^0 = \left(\mathbf{R}^0\right)^T \mathbf{R}^0_{,S}$, and they must be consistent with the essential boundary conditions. The

strict Kirchhoff branch additionally requires the initial position constraint and its velocity-level compatibility condition.

1.6. **Kirchhoff shear-free constraint.** For a slender catheter, impose the extensible Kirchhoff shear-free constraint

$$\nu_1 = 0, \qquad \nu_2 = 0, \qquad \nu_3 = \lambda. \tag{1.97}$$

which gives

$$\mathbf{r}_{,S} = \lambda \mathbf{d}_3, \qquad \mathbf{d}_3 = \mathbf{t}. \tag{1.98}$$

This condition aligns the third director with the centerline tangent but does not determine the rotation of $\mathbf{d}_1, \mathbf{d}_2$ about $\mathbf{d}_3$. Consequently, κ_3, ω_3, GJ, the polar rotary inertia, and the magnetic couple must all be retained.

In the strong form, the transverse shear force is the reaction enforcing the shear-free constraint and does not vanish merely because $\nu_1 = \nu_2 = 0$. The constraint may be imposed exactly with Lagrange multipliers or regularized by retaining a large but finite shear stiffness. In either case, the cross-sectional orientation field must remain in the model.[17, 18]

1.7. **Continuous energy balance.**

1.7.1. *Structural energy and extended inertial energy.* The mechanical energy of the base rod is defined as

$$E_{str} = T_{str} + V_{int}. \tag{1.99}$$

If eccentric attachments are present, add their kinetic energy T_e. In the energy-consistent added-mass branch with a stationary flow $\mathbf{v}_f = \mathbf{0}$ and time-independent m_A, define the total kinetic energy as

$$T_{tot} = T_{str} + T_A + T_e. \tag{1.100}$$

If magnetic loading, frictionless contact, or residual gravity is derivable from potentials, define

$$V = V_{int} + V_{mag} + V_{cont} + V_g, \tag{1.101}$$

and the extended mechanical energy

$$\boxed{E = T_{tot} + V.} \tag{1.102}$$

For a general prescribed flow, T_A may depend explicitly on time and contain cross terms involving relative velocity. It is then a velocity-dependent added-fluid inertia functional in the Lagrangian rather than the ordinary mechanical kinetic energy of a closed system. Its contribution must remain explicit as an independent flow port in the power balance; it does not imply conservation of the mechanical energy of the combined structure–prescribed-flow system.

1.7.2. *Power balance.* Taking the inner product of the translational equation with $\dot{\mathbf{r}}$, the rotational equation with the spatial angular velocity $\boldsymbol{\omega}_s$, and integrating along the rod gives

$$\frac{d}{dt}\left(T_{tot} + V_{int}\right) = P_\partial + P_{mag} + P_{flow} + P_{cont} + P_g, \tag{1.103}$$

where the boundary power is

$$P_\partial = \left[\mathbf{n} \cdot \dot{\mathbf{r}} + \mathbf{m} \cdot \boldsymbol{\omega}_s\right]_0^L. \tag{1.104}$$

After all conservative external loads have been incorporated into V, one obtains

$$\boxed{\frac{dE}{dt} = P_\partial + P_{diss}^{bulk} + P_{contact} + P_{in} + P_{flow} + \frac{\partial V_{ext}}{\partial t}.} \tag{1.105}$$

Here, P_{diss}^{bulk} contains only the passive bulk/drag components obtained after the input–dissipation split, excludes every contact term, and satisfies $P_{diss}^{bulk} \leq 0$. $P_{contact}$ is the unique nonpotential contact-port power for elastic contact not represented in the potential, normal contact damping, tangential friction, impact, and contact-spin couples; if elastic contact has already been included in the potential, it must not be counted again in $P_{contact}$, and contact damping, friction, and impact must not also appear in P_{diss}^{bulk}. The term P_{flow} collects the background-flow drag power $P_{drag,flow}$, the Froude–Krylov input, and reactive-flow power assigned to the port under the selected representation; it is generally sign-indefinite. The term P_{in} contains only nonfluid nonconservative body inputs and control inputs. Terminal actuation contributes only through P_{∂}. If a nonautonomous magnetic action has already been represented by $V_{mag}(\mathbf{r}, \mathbf{R}, t)$, its explicit-time power appears only through $\partial V_{ext}/\partial t$ and must not be counted again in P_{in}. The extended mechanical energy is conserved only when $P_{diss}^{bulk} = P_{contact} = P_{in} = P_{flow} = P_{\partial} = 0$ and the potential is time independent.

1.8. **Section summary.** With the centerline $\mathbf{r}$ and cross-sectional orientation $\mathbf{R}$ as independent primary fields, this section established a geometrically exact rod-dynamics model retaining extension, shear, bending about two axes, independent torsion, cross-sectional rotary inertia, magnetic couples, fluid loading, and contact couples. The base catheter uses a mass-centerline convention, whereas eccentric magnetic tips and sensors retain translational–rotational coupling through independent rigid-body kinetic energies. In the energy-consistent fluid representation, T_A generates the added inertia and the pure Froude–Krylov term supplies the remaining fluid-acceleration input; this branch is mutually exclusive with the engineering Morison representation, thereby avoiding double counting of $C_A\mathbf{a}_f$.

The complete weak form retains translational and rotational test functions as well as terminal force and moment terms. The passive kernel of quadratic drag, normal contact damping, and friction are assigned to dissipative power, while background-flow drag input and prescribed-flow inertia are assigned to an independent flow port. The Kirchhoff shear-free condition is enforced as a constraint rather than by deleting cross-sectional orientation, independent torsion, or rotary inertia. This coupled weak form provides the theoretical basis for the joint translational–rotational Galerkin discretization in Section 2.

2. Joint Translational–Rotational Galerkin Discretization

2.1. **Discretization principles.** Section 1 established the complete coupled weak form with $\mathbf{r}(S,t)$ and $\mathbf{R}(S,t) \in SO(3)$ as the primary fields. Rather than reducing it to a fourth-order centerline-only equation, this section performs a joint Galerkin discretization directly in the translational space and in a local coordinate chart of the rotation group.

The discretization follows five principles. First, the centerline position and cross-sectional orientation retain independent generalized coordinates. Second, rotation matrices cannot be superposed linearly as ordinary vectors and must be updated through the exponential map or an equivalent multiplicative rule. Third, the mass–inertia matrix must be derived from the complete three-dimensional material-point kinetic energy rather than assumed a priori to be constant or configuration dependent. Fourth, geometric inertial terms must follow rigorously from the Euler–Lagrange derivative of the discrete kinetic energy. Fifth, internal forces are obtained from the gradient of the discrete potential, while boundary forces and moments enter the generalized forces naturally through the weak form.[9, 17, 18]

2.2. **Joint finite-dimensional approximation.**

2.2.1. *Translational approximation.* Let $\mathbf{N}(S)$ be matrix-valued basis functions satisfying the essential translational boundary conditions. Approximate the centerline by

$$\mathbf{r}_h(S,t) = \mathbf{r}_0(S) + \mathbf{N}(S)\mathbf{a}(t), \tag{2.1}$$

where $\mathbf{a}(t) \in \mathbb{R}^{n_a}$ contains the translational generalized coordinates. Hence,

$$\dot{\mathbf{r}}_h = \mathbf{N}\dot{\mathbf{a}}, \qquad \mathbf{r}_{h,S} = \mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}. \tag{2.2}$$

If the essential boundary conditions are time dependent, a known lifting function may be added to the approximation; its first and second time derivatives are treated as prescribed generalized loads. To emphasize the structural formulation, the boundary conditions are assumed homogeneous below.

2.2.2. *Approximation of cross-sectional orientation.* Let $\mathbf{R}_0(S)$ be the reference orientation. In a neighborhood of the reference state, use exponential coordinates for the rotation vector:

$$\mathbf{R}_h(S,t) = \mathbf{R}_0(S) \exp\left(\widehat{\boldsymbol{\theta}_h(S,t)}\right), \tag{2.3}$$

$$\boldsymbol{\theta}_h(S,t) = \boldsymbol{\Psi}(S)\mathbf{b}(t), \tag{2.4}$$

where $\mathbf{b}(t) \in \mathbb{R}^{n_b}$ contains the rotational generalized coordinates and $\boldsymbol{\Psi}(S)$ is a matrix-valued basis satisfying the essential rotational boundary conditions. Define

$$\mathbf{E}_h = \exp\left(\widehat{\boldsymbol{\theta}_h}\right). \tag{2.5}$$

so that

$$\mathbf{R}_h = \mathbf{R}_0\mathbf{E}_h. \tag{2.6}$$

Exponential coordinates form a local chart on $SO(3)$. Provided that the rotational neighborhood does not cross the singular set of the exponential map, this parameterization is smooth and one-to-one and is suitable for finite-dimensional modal truncation and local nonlinear analysis. Global time integration through large rotations may instead employ chart switching or unit quaternions without changing the underlying mechanical derivation.

2.2.3. *Right Jacobian and angular velocity.* Let $\theta = \| \boldsymbol{\theta}_h \|$. The right Jacobian of the rotation vector is defined by

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1-\cos\theta}{\theta^2}\widehat{\boldsymbol{\theta}_h} + \frac{\theta - \sin\theta}{\theta^3}\widehat{\boldsymbol{\theta}_h}^2. \tag{2.7}$$

As $\theta \to 0$, continuous extension gives

$$\mathbf{J}_r(\boldsymbol{\theta}_h) = \mathbf{I} - \frac{1}{2}\widehat{\boldsymbol{\theta}_h} + \frac{1}{6}\widehat{\boldsymbol{\theta}_h}^2 + O\left(\theta^3\right). \tag{2.8}$$

The exact differential relation of the exponential map gives the material angular velocity

$$\boldsymbol{\omega}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\dot{\boldsymbol{\theta}}_h = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}\dot{\mathbf{b}}. \tag{2.9}$$

Define the angular-velocity matrix

$$\mathbf{B}_\omega(S,\mathbf{b}) = \mathbf{J}_r(\boldsymbol{\theta}_h)\,\boldsymbol{\Psi}(S), \tag{2.10}$$

Then

$$\boldsymbol{\omega}_h = \mathbf{B}_\omega(S,\mathbf{b})\,\dot{\mathbf{b}}. \tag{2.11}$$

Thus, $\boldsymbol{\omega}_h = \boldsymbol{\Psi}\dot{\mathbf{b}}$ cannot in general be assumed; that relation is only a zeroth-order small-rotation approximation.

2.2.4. *Joint generalized coordinates and virtual variables.* Define the complete generalized coordinate vector

$$\mathbf{q} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^n, \qquad n = n_a + n_b. \tag{2.12}$$

The translational virtual displacement is

$$\delta \mathbf{r}_h = \mathbf{N}\, \delta \mathbf{a}. \tag{2.13}$$

From $\delta \mathbf{R}_h = \mathbf{R}_h \widehat{\boldsymbol{\eta}_h}$, the material virtual rotation satisfies

$$\boldsymbol{\eta}_h = \mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \delta \boldsymbol{\theta}_h = \mathbf{B}_\omega\left(S, \mathbf{b}\right) \delta \mathbf{b}. \tag{2.14}$$

Accordingly, the translational and rotational test functions are $\mathbf{N}$ and $\mathbf{B}_\omega$, respectively, and both enter the coupled Galerkin condition.

For $\mathbf{q} = \mathbf{0}$ to represent a stress-free reference configuration, the reference centerline, reference orientation, and reference strains must satisfy the compatibility relations

$$\boldsymbol{\nu}_0(S) = \mathbf{R}_0^T(S)\mathbf{r}_{0,S}(S), \tag{2.15}$$

$$\widehat{\boldsymbol{\kappa}}_0(S) = \mathbf{R}_0^T(S)\mathbf{R}_{0,S}(S). \tag{2.16}$$

If these relations are not satisfied, $\mathbf{q} = \mathbf{0}$ describes a pre-extended, pre-sheared, pre-bent, or pre-twisted configuration, with the corresponding initial force and moment generated automatically by $V_{int,h}$. The equilibrium state, tangent stiffness, and center of the KAM expansion must use the same reference convention.

2.3. **Discrete strains and potential energy.**

2.3.1. *Shear–extension strain.* The discrete shear–extension strain is

$$\boldsymbol{\nu}_h\left(\mathbf{q}\right) = \mathbf{R}_h^T \mathbf{r}_{h,S} = \mathbf{E}_h^T \mathbf{R}_0^T \left(\mathbf{r}_{0,S} + \mathbf{N}_{,S}\mathbf{a}\right). \tag{2.17}$$

Its virtual variation can be written as

$$\delta \boldsymbol{\nu}_h = \mathbf{R}_h^T \mathbf{N}_{,S} \delta \mathbf{a} - \boldsymbol{\eta}_h \times \boldsymbol{\nu}_h. \tag{2.18}$$

Thus, there exists a strain–generalized-coordinate matrix $\mathbf{B}_\nu\left(S, \mathbf{q}\right)$ such that

$$\delta \boldsymbol{\nu}_h = \mathbf{B}_\nu\left(S, \mathbf{q}\right) \delta \mathbf{q}. \tag{2.19}$$

Explicitly,

$$\mathbf{B}_\nu = \begin{bmatrix} \mathbf{R}_h^T \mathbf{N}_{,S} & \widehat{\boldsymbol{\nu}_h} \mathbf{B}_\omega \end{bmatrix}, \tag{2.20}$$

because $-\boldsymbol{\eta}_h \times \boldsymbol{\nu}_h = \widehat{\boldsymbol{\nu}_h}\boldsymbol{\eta}_h$.

2.3.2. *Curvature–torsion strain.* Define the curvature–torsion vector of the reference orientation by

$$\widehat{\boldsymbol{\kappa}_R} = \mathbf{R}_0^T \mathbf{R}_{0,S}. \tag{2.21}$$

From $\mathbf{R}_h = \mathbf{R}_0 \mathbf{E}_h$, one obtains

$$\mathbf{R}_h^T \mathbf{R}_{h,S} = \mathbf{E}_h^T \widehat{\boldsymbol{\kappa}_R} \mathbf{E}_h + \mathbf{E}_h^T \mathbf{E}_{h,S}. \tag{2.22}$$

Hence, the discrete curvature–torsion vector is

$$\boxed{\boldsymbol{\kappa}_h\left(\mathbf{q}\right) = \mathbf{E}_h^T \boldsymbol{\kappa}_R + \mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \boldsymbol{\theta}_{h,S}} \tag{2.23}$$

where

$$\boldsymbol{\theta}_{h,S} = \boldsymbol{\Psi}_{,S} \mathbf{b}. \tag{2.24}$$

The corresponding virtual variation satisfies

$$\delta\boldsymbol{\kappa}_h = \boldsymbol{\eta}_{h,S} + \boldsymbol{\kappa}_h \times \boldsymbol{\eta}_h. \tag{2.25}$$

Therefore, there exists $\mathbf{B}_\kappa(S, \mathbf{q})$ such that

$$\delta\boldsymbol{\kappa}_h = \mathbf{B}_\kappa(S, \mathbf{q})\,\delta\mathbf{q}. \tag{2.26}$$

In assembly, $\mathbf{B}_\kappa$ may be obtained by analytic differentiation of the preceding expression or by evaluating the Jacobian of $\boldsymbol{\kappa}_h(\mathbf{q})$ through automatic differentiation. In either implementation, the independent torsional contribution κ_{h3} must be retained.

2.3.3. *Discrete internal potential energy.* Substituting the discrete strains into the potential energy of Section 1, the discrete internal potential energy can be decomposed as

$$V_{int,h}(\mathbf{q}) = V_{\nu,h}(\mathbf{q}) + V_{\kappa,h}(\mathbf{q}), \tag{2.27}$$

where

$$V_{\nu,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)^T \mathbf{K}_\nu (\boldsymbol{\nu}_h - \boldsymbol{\nu}_0)\, dS, \tag{2.28}$$

$$V_{\kappa,h}(\mathbf{q}) = \frac{1}{2}\int_0^L (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)^T \mathbf{K}_\kappa (\boldsymbol{\kappa}_h - \boldsymbol{\kappa}_0)\, dS. \tag{2.29}$$

The discrete internal generalized force is

$$\mathbf{f}_{int}(\mathbf{q}) = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.30}$$

Using the virtual-work expression, it can be written as

$$\mathbf{f}_{int} = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b\right] dS. \tag{2.31}$$

This expression contains the restoring forces associated with extension, shear, bending about two axes, and torsion. In general, $\mathbf{f}_{int}$ is a nonlinear vector-valued function of $\mathbf{q}$ and cannot be written a priori as a constant stiffness matrix multiplied by $\mathbf{q}$.

2.4. **Complete discrete kinetic energy and mass–inertia matrix.**

2.4.1. *Material-point velocity matrix.* The discrete material-point position is

$$\mathbf{x}_h(S, \boldsymbol{\xi}, t) = \mathbf{r}_h(S, t) + \mathbf{R}_h(S, t)\boldsymbol{\xi}. \tag{2.32}$$

and its velocity is

$$\dot{\mathbf{x}}_h = \mathbf{N}\dot{\mathbf{a}} - \mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\dot{\mathbf{b}}. \tag{2.33}$$

Define the material-point velocity matrix

$$\mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q}) = \begin{bmatrix}\mathbf{N} & -\mathbf{R}_h\widehat{\boldsymbol{\xi}}\mathbf{B}_\omega\end{bmatrix}, \tag{2.34}$$

Then

$$\dot{\mathbf{x}}_h = \mathbf{B}_x(S, \boldsymbol{\xi}, \mathbf{q})\,\dot{\mathbf{q}}. \tag{2.35}$$

2.4.2. *Rigorous definition of the mass–inertia matrix* $\mathbf{M}(\mathbf{q})$. The discrete kinetic energy is

$$T_h(\mathbf{q},\dot{\mathbf{q}}) = \frac{1}{2}\int_0^L \int_{A(S)} \rho \dot{\mathbf{x}}_h^T \dot{\mathbf{x}}_h \, dA\, dS. \tag{2.36}$$

Substitution of the velocity matrix gives

$$\boxed{T_h(\mathbf{q},\dot{\mathbf{q}}) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}}} \tag{2.37}$$

where

$$\boxed{\mathbf{M}(\mathbf{q}) = \int_0^L \int_{A(S)} \rho \mathbf{B}_x^T \mathbf{B}_x \, dA\, dS} \tag{2.38}$$

is the mass–inertia matrix derived rigorously from the complete three-dimensional translational–rotational kinetic energy.

2.4.3. *Block form of the mass–inertia matrix.* With $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$, write

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{M}_{ab}(\mathbf{q}) \\ \mathbf{M}_{ba}(\mathbf{q}) & \mathbf{M}_{bb}(\mathbf{q}) \end{bmatrix}. \tag{2.39}$$

Using $\mu, \mathbf{c}_\rho, \mathbf{J}_c$ defined in Section 1, the blocks are

$$\mathbf{M}_{aa} = \int_0^L \mu \mathbf{N}^T \mathbf{N}\, dS, \tag{2.40}$$

$$\mathbf{M}_{ab}(\mathbf{q}) = -\int_0^L \mathbf{N}^T \mathbf{R}_h \widehat{\mathbf{c}}_\rho \mathbf{B}_\omega \, dS, \tag{2.41}$$

$$\mathbf{M}_{ba}(\mathbf{q}) = \mathbf{M}_{ab}^T(\mathbf{q}), \tag{2.42}$$

$$\boxed{\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{B}_\omega^T \mathbf{J}_c \mathbf{B}_\omega \, dS} \tag{2.43}$$

Thus,

$$\mathbf{M}_{bb}(\mathbf{q}) = \int_0^L \mathbf{\Psi}^T \mathbf{J}_r^T(\boldsymbol{\theta}_h)\, \mathbf{J}_c \mathbf{J}_r(\boldsymbol{\theta}_h)\, \mathbf{\Psi}\, dS. \tag{2.44}$$

Whether the mass matrix depends on the configuration is therefore not an a priori assumption; it is determined jointly by the centerline definition, orientation parameterization, cross-sectional mass distribution, and attached components.

2.4.4. *Mass centerline and eccentric attachments.* Consistently with Section 1, the centerline of the base catheter is taken to pass through the mass center of each cross section, so the distributed rod satisfies

$$\mathbf{c}_\rho = \mathbf{0}, \tag{2.45}$$

and the translational–rotational cross block of the base rod vanishes:

$$\mathbf{M}_{ab} = \mathbf{M}_{ba} = \mathbf{0}. \tag{2.46}$$

In this case,

$$\mathbf{M}_{str}(\mathbf{q}) = \begin{bmatrix} \mathbf{M}_{aa} & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_{bb}(\mathbf{b}) \end{bmatrix}. \tag{2.47}$$

However, $\mathbf{M}_{bb}$ still depends on the rotational generalized coordinates through $\mathbf{J}_r(\boldsymbol{\theta}_h)$, and the complete structural mass–inertia matrix therefore generally remains $\mathbf{M}_{str}(\mathbf{q})$.

If the center of mass of a magnetic tip, sensor, or other attached component is offset from the centerline, its kinetic energy must be added separately. Let the center-of-mass position of the attachment be

$$\mathbf{x}_e = \mathbf{r}_h\left(S_e\right) + \mathbf{R}_h\left(S_e\right)\boldsymbol{\rho}_e, \tag{2.48}$$

If its mass is m_e and its material-frame rotary inertia is $\mathbf{J}_e$, then

$$T_e = \frac{1}{2} m_e \left|\dot{\mathbf{x}}_e\right|^2 + \frac{1}{2}\boldsymbol{\omega}_e^T \mathbf{J}_e \boldsymbol{\omega}_e. \tag{2.49}$$

Let $\mathbf{B}_e\left(\mathbf{q}\right)$ map $\dot{\mathbf{q}}$ to the attachment velocity. Then

$$T_e = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_e\left(\mathbf{q}\right)\dot{\mathbf{q}}, \qquad \mathbf{M}_e = \mathbf{B}_e^T \mathbf{M}_e^{rig} \mathbf{B}_e. \tag{2.50}$$

This contribution generally produces a nonzero translational–rotational coupling block. Thus, use of the mass centerline for the base rod does not imply that the cross-mass block of the complete system must vanish.

2.4.5. *Scope of the small-rotation constant-mass approximation.* When $\| \boldsymbol{\theta}_h \| \ll 1$, if only

$$\mathbf{J}_r\left(\boldsymbol{\theta}_h\right) \approx \mathbf{I}, \tag{2.51}$$

is retained, then

$$\mathbf{M}_{bb} \approx \int_0^L \boldsymbol{\Psi}^T \mathbf{J}_c \boldsymbol{\Psi}\, dS, \tag{2.52}$$

becomes constant. This is a zeroth-order small-rotation inertial approximation rather than a general property of the complete model. The principal model retains the exact $\mathbf{J}_r$ and $\mathbf{M}\left(\mathbf{q}\right)$.

2.4.6. *Energy-consistent discretization of fluid added mass.* Define the joint translational velocity matrix

$$\mathbf{N}_q(S) = \begin{bmatrix}\mathbf{N}(S) & \mathbf{0}\end{bmatrix}, \tag{2.53}$$

such that $\dot{\mathbf{r}}_h = \mathbf{N}_q\dot{\mathbf{q}}$. Let

$$\mathbf{w}_{\perp,h} = \mathbf{P}_\perp\left(\mathbf{q}\right)\left[\mathbf{N}_q\dot{\mathbf{q}} - \mathbf{v}_f\left(\mathbf{r}_h, t\right)\right], \qquad m_A = \rho_f C_A A_d. \tag{2.54}$$

The added-fluid inertia functional of Section 1 is discretized as follows (and becomes an added kinetic energy for a stationary flow):

$$\boxed{T_{A,h} = \frac{1}{2}\int_0^L m_A\, \mathbf{w}_{\perp,h}^T \mathbf{w}_{\perp,h}\, dS.} \tag{2.55}$$

Expansion gives

$$T_{A,h} = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_A\left(\mathbf{q}\right)\dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A\left(\mathbf{q}, t\right) + c_A\left(\mathbf{q}, t\right), \tag{2.56}$$

where

$$\boxed{\mathbf{M}_A\left(\mathbf{q}\right) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{N}_q\, dS,} \tag{2.57}$$

$$\mathbf{b}_A\left(\mathbf{q}, t\right) = \int_0^L m_A\, \mathbf{N}_q^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{v}_f\, dS, \tag{2.58}$$

$$c_A\left(\mathbf{q}, t\right) = \frac{1}{2}\int_0^L m_A\, \mathbf{v}_f^T \mathbf{P}_\perp\left(\mathbf{q}\right)\mathbf{v}_f\, dS. \tag{2.59}$$

The structural kinetic energy, eccentric-attachment kinetic energy, and added-fluid contribution are therefore collected into the velocity-dependent inertial functional

$$T_h = T_{str,h} + \sum_e T_e + T_{A,h}. \tag{2.60}$$

Its Hessian with respect to the generalized velocity gives the total effective mass–inertia matrix

$$\boxed{\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}).} \tag{2.61}$$

Only when $\mathbf{v}_f = \mathbf{0}$ is it physically valid to set $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$. For a general prescribed flow, moving the cross terms to an external port to construct an autonomous conservative backbone merely removes $\mathbf{b}_A$ and c_A from the selected Hamiltonian backbone and assigns their complete Euler–Lagrange action to $\mathbf{Q}_{flow}$; it does not assume that these quantities vanish in the original system. For a general flow, one cannot retain only $\mathbf{M}_A\dot{\mathbf{q}}$. The constant added mass obtained by freezing $\mathbf{P}_\perp$ is used only as an engineering validation branch.

2.4.7. *Symmetry and positive definiteness.* By definition of the kinetic-energy Hessian,

$$\mathbf{M}^T(\mathbf{q}) = \mathbf{M}(\mathbf{q}). \tag{2.62}$$

If the material and added-mass parameters are positive, the orientation coordinates remain in a local chart where $\mathbf{J}_r$ is nonsingular, and the joint velocity map does not degenerate almost everywhere for any nonzero generalized velocity, namely,

$$\dot{\mathbf{q}} \neq \mathbf{0} \quad \Rightarrow \quad T_h^{(2)}(\mathbf{q}, \dot{\mathbf{q}}) > 0, \tag{2.63}$$

then

$$\mathbf{y}^T \mathbf{M}(\mathbf{q})\, \mathbf{y} > 0, \qquad \forall \mathbf{y} \neq \mathbf{0}. \tag{2.64}$$

Therefore, $\mathbf{M}(\mathbf{q})$ is symmetric positive definite. What must be excluded is a redundant generalized coordinate or a degenerate velocity map, not a rigid-body mode under free boundary conditions. A rigid-body mode may give a zero eigenvalue of the stiffness matrix, while a valid rigid-body motion still has positive kinetic energy.

2.5. **Joint Galerkin projection.**

2.5.1. *Inertial term.* Substituting the finite-dimensional approximation into the total inertial virtual work of Section 1 gives

$$\delta W_{in,h} = \delta \mathbf{q}^T \left[\frac{d}{dt}\left(\frac{\partial T_h}{\partial \dot{\mathbf{q}}} \right) - \frac{\partial T_h}{\partial \mathbf{q}} \right]. \tag{2.65}$$

For a stationary flow, or for an autonomous conservative inertial backbone selected after assigning the complete Euler–Lagrange action of the prescribed-flow cross terms to $\mathbf{Q}_{flow}$,

$$T_h = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\, \dot{\mathbf{q}}, \tag{2.66}$$

the inertial generalized force can be arranged as

$$\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}). \tag{2.67}$$

For a general prescribed flow, T_h also contains $-\dot{\mathbf{q}}^T \mathbf{b}_A + c_A$, whose additional contribution is denoted by $\mathbf{g}_f(\mathbf{q}, \dot{\mathbf{q}}, t)$. The unified form is therefore

$$\boxed{\mathbf{M}(\mathbf{q})\, \ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q}, \dot{\mathbf{q}}, t), \qquad \mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f.} \tag{2.68}$$

The terms are derived rigorously in Section 2.6.

2.5.2. *Internal-force term.* The discrete internal virtual work is

$$\delta W_{int,h} = \delta \mathbf{q}^T \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS. \tag{2.69}$$

Therefore,

$$\mathbf{f}_{int}(\mathbf{q}) = \int_0^L \left[\mathbf{B}_\nu^T \mathbf{n}_b + \mathbf{B}_\kappa^T \mathbf{m}_b \right] dS = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}). \tag{2.70}$$

2.5.3. *Joint projection of distributed forces and couples.* First, decompose the distributed loads that remain in the external-force ports through mutually exclusive accounting:

$$\mathbf{f}_{ext,rem} = \mathbf{f}_{cons} + \mathbf{f}_{nc}, \qquad \mathbf{l}_{ext} = \mathbf{l}_{cons} + \mathbf{l}_{nc}. \tag{2.71}$$

The conservative part is represented through an external potential in Section 2.5.5; only the nonconservative part is projected directly onto the right-hand side. A spatially distributed couple is first transformed to the material frame, $\mathbf{l}_{nc}^b = \mathbf{R}_h^T \mathbf{l}_{nc}$, and the nonconservative distributed generalized force is

$$\boxed{\mathbf{Q}_{nc,dist}(\mathbf{q}, \dot{\mathbf{q}}, t) = \int_0^L \begin{bmatrix} \mathbf{N}^T \mathbf{f}_{nc} \\ \mathbf{B}_\omega^T \mathbf{R}_h^T \mathbf{l}_{nc} \end{bmatrix} dS.} \tag{2.72}$$

This expression allows magnetic, contact, and fluid couples to enter the rotational generalized coordinates directly when they belong to nonconservative ports. If a load has already been represented by a potential, it must not be included again in $\mathbf{Q}_{nc,dist}$.

For normal quadratic drag acting only on the centerline translational velocity, define the joint velocity matrix

$$\boxed{\mathbf{N}_q = [\, \mathbf{N} \;\; \mathbf{0} \,] \in \mathbb{R}^{3 \times n}, \qquad \dot{\mathbf{r}}_h = \mathbf{N}_q \dot{\mathbf{q}}.} \tag{2.73}$$

Its consistent generalized-force projection is

$$\mathbf{Q}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp (\mathbf{v}_f - \mathbf{N}_q \dot{\mathbf{q}}) \, dS = -\mathbf{D}_{drag}(\mathbf{q}, \dot{\mathbf{q}}, t) \, \dot{\mathbf{q}} + \mathbf{Q}_{drag,flow}(\mathbf{q}, \dot{\mathbf{q}}, t), \tag{2.74}$$

where

$$\boxed{\mathbf{D}_{drag} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{N}_q \, dS, \qquad \mathbf{Q}_{drag,flow} = \int_0^L d_f \mathbf{N}_q^T \mathbf{P}_\perp \mathbf{v}_f \, dS.} \tag{2.75}$$

For every $\mathbf{y} \in \mathbb{R}^n$,

$$\mathbf{y}^T \mathbf{D}_{drag} \mathbf{y} = \int_0^L d_f \parallel \mathbf{P}_\perp \mathbf{N}_q \mathbf{y} \parallel^2 \, dS \geq 0, \tag{2.76}$$

Hence, $\mathbf{D}_{drag} = \mathbf{D}_{drag}^T \succcurlyeq 0$. The matrix $\mathbf{N}_q^T$, rather than $\mathbf{N}^T$, must be used so that $\mathbf{Q}_{drag} \in \mathbb{R}^n$ and $\mathbf{D}_{drag} \in \mathbb{R}^{n \times n}$. Because $\mathbf{N}_q = [\mathbf{N} \; \mathbf{0}]$, the ab, ba, and bb blocks associated with centerline drag vanish. Eccentric fluid couples and cross-sectional spin damping still enter the orientation channels through independent distributed-couple projections. Since d_f depends on the relative velocity, $\mathbf{D}_{drag}$ is a statewise positive-semidefinite nonlinear matrix.

2.5.4. *Boundary generalized force.* The boundary virtual work in Section 1 is

$$[\mathbf{n} \cdot \delta\mathbf{r} + \mathbf{m}_b \cdot \boldsymbol{\eta}]_0^L . \tag{2.77}$$

Substitution of the finite-dimensional virtual variables gives

$$\boxed{\mathbf{Q}_\partial (\mathbf{q}, t) = \begin{bmatrix} \mathbf{N}^T \overline{\mathbf{n}} \\ \mathbf{B}_\omega^T (\mathbf{q}) \, \overline{\mathbf{m}}_b \end{bmatrix}_0^L .} \tag{2.78}$$

The boundary generalized force generally depends on the configuration because the rotational test matrix $\mathbf{B}_\omega$ and the transformation from a spatial moment to a material moment may both vary with orientation. At an essential boundary, the corresponding test function vanishes and the boundary term disappears automatically. At a natural boundary, the prescribed terminal force and moment must be retained.

2.5.5. *Conservative and nonconservative external loads.* If the conservative distributed and boundary loads admit a discrete external potential $V_{ext,h} (\mathbf{q}, t)$, their total conservative generalized force is

$$\mathbf{Q}_{cons} = -\nabla_{\mathbf{q}} V_{ext,h}, \tag{2.79}$$

and define

$$V_h (\mathbf{q}, t) = V_{int,h} (\mathbf{q}) + V_{ext,h} (\mathbf{q}, t) . \tag{2.80}$$

The nonconservative generalized force is denoted collectively by

$$\mathbf{Q}_{nc} = \mathbf{Q}_{nc,dist} + \mathbf{Q}_{control} + \mathbf{Q}_{disturbance} + \mathbf{Q}_{other,nc}. \tag{2.81}$$

The final equations use only $\nabla_{\mathbf{q}} V_h$ on the left-hand side and $\mathbf{Q}_{nc}$ on the right-hand side. The original load projection containing conservative components must not be added again in its entirety to the right-hand side. If a time-dependent magnetic action has already been included in $V_{ext,h} (\mathbf{q}, t)$, its explicit-time power is represented only by $\partial V_{ext,h} / \partial t$.

2.6. **Rigorous derivation of generalized momentum and geometric inertial terms.**

2.6.1. *Generalized momentum.* For the velocity-dependent inertial functional in Section 2.4.6,

$$T_h = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}} - \dot{\mathbf{q}}^T \mathbf{b}_A (\mathbf{q}, t) + c_A (\mathbf{q}, t) , \tag{2.82}$$

define the generalized momentum

$$\boxed{\mathbf{p} = \frac{\partial T_h}{\partial \dot{\mathbf{q}}} = \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}} - \mathbf{b}_A (\mathbf{q}, t) .} \tag{2.83}$$

For a general prescribed flow, one therefore cannot unconditionally write $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}}$. The backbone momentum is defined only for a stationary flow or after the complete action of $\mathbf{b}_A$ has been moved to $\mathbf{Q}_{flow}$ in a deliberately selected autonomous conservative backbone:

$$\mathbf{p}_0 = \mathbf{M} (\mathbf{q}) \, \dot{\mathbf{q}}. \tag{2.84}$$

2.6.2. *Geometric inertia generated by the mass metric.* Taking the Euler–Lagrange derivative of the purely quadratic velocity term gives

$$\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}), \tag{2.85}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T\mathbf{M}\dot{\mathbf{q}}\right).} \tag{2.86}$$

Define

$$\Gamma_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right), \tag{2.87}$$

Then

$$g_{M,i} = \sum_{j,k=1}^{n} \Gamma_{ijk}\dot{q}_j\dot{q}_k. \tag{2.88}$$

Let $C_{ij} = \sum_k \Gamma_{ijk}\dot{q}_k$. Then $\mathbf{g}_M = \mathbf{C}\dot{\mathbf{q}}$, and $\dot{\mathbf{M}} - 2\mathbf{C}$ is skew symmetric, which guarantees the kinetic-energy power identity.

2.6.3. *Additional action generated by prescribed-flow cross terms.* The term $-\dot{\mathbf{q}}^T\mathbf{b}_A + c_A$ gives

$$\boxed{\mathbf{g}_f = -\dot{\mathbf{b}}_A + (\nabla_{\mathbf{q}}\mathbf{b}_A)^T\,\dot{\mathbf{q}} - \nabla_{\mathbf{q}}c_A.} \tag{2.89}$$

Here, $\dot{\mathbf{b}}_A$ denotes the total derivative along the trajectory. The total inertial action is therefore

$$\boxed{\frac{d}{dt}\frac{\partial T_h}{\partial \dot{\mathbf{q}}} - \frac{\partial T_h}{\partial \mathbf{q}} = \mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \mathbf{g}_f.} \tag{2.90}$$

Equivalently, $\mathbf{g}_f$ may be moved to the right-hand side and denoted as a flow input $\mathbf{Q}_{flow}$, but the entire term must be moved; retaining only selected components is inconsistent.

2.6.4. *Constant-mass special case.* If, under a specific small-rotation, zero-eccentricity, frozen-normal-projection, and fixed-added-mass approximation,

$$\frac{\partial \mathbf{M}}{\partial q_i} = \mathbf{0}, \qquad i = 1, \ldots, n, \tag{2.91}$$

then $\mathbf{g}_M = \mathbf{0}$. If $\mathbf{b}_A = \mathbf{0}$ as well, all geometric inertial terms vanish. This conclusion applies only to the corresponding local approximation and must not be used in reverse to deny configuration-dependent inertia in the complete model.

2.7. **Discrete treatment of the Kirchhoff shear-free constraint.**

2.7.1. *Discrete constraints.* The strict Kirchhoff model requires[17, 18]

$$\nu_{h1}(\mathbf{q}) = 0, \qquad \nu_{h2}(\mathbf{q}) = 0. \tag{2.92}$$

Collect all discrete constraints in

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}. \tag{2.93}$$

Define the constraint Jacobian

$$\mathbf{G}(\mathbf{q}) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}}. \tag{2.94}$$

2.7.2. *Lagrange-multiplier form.* Introduce the Lagrange multiplier $\boldsymbol{\lambda}_c$. The generalized constraint reaction is

$$\mathbf{Q}_c := -\mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c. \tag{2.95}$$

The complete constrained discrete system is

$$\begin{cases} \mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}(\mathbf{q},\dot{\mathbf{q}}) + \nabla_{\mathbf{q}} V_h(\mathbf{q},t) + \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c = \mathbf{Q}_{nc}(\mathbf{q},\dot{\mathbf{q}},t), \\ \mathbf{c}(\mathbf{q}) = \mathbf{0}. \end{cases} \tag{2.96}$$

This is a nonlinear differential–algebraic system. The constraint eliminates shear deformation only; it does not remove rotational generalized coordinates or torsional modes.

2.7.3. *Finite-shear-stiffness form.* If the fully shearable Cosserat rod is used, or if the Kirchhoff constraint is approximated by a large but finite shear stiffness, no Lagrange multiplier is required because the shear energy is already included in $V_{int,h}$. The resulting system is an ordinary second-order nonlinear ODE. The two treatments have different numerical properties but share the same complete translational–rotational discretization.

2.8. **Global finite-dimensional equations of motion.**

2.8.1. *Form without explicit constraints.* Assembling the inertial term, internal conservative force, conservative external potential, nonconservative distributed loads, and boundary loads gives

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_T(\mathbf{q},\dot{\mathbf{q}},t) + \nabla_{\mathbf{q}} V_h(\mathbf{q},t) = \mathbf{Q}_{nc}(\mathbf{q},\dot{\mathbf{q}},t) + \mathbf{Q}_{\partial}(\mathbf{q},t).} \tag{2.97}$$

Here, $\mathbf{q}$ contains both translational and rotational generalized coordinates; $\mathbf{M}$ is generated by the velocity-quadratic terms of the base rod, eccentric attachments, and the energy-consistent added mass; $\mathbf{g}_T = \mathbf{g}_M + \mathbf{g}_f$; and V_h contains extension, shear, bending, torsion, and external potentials satisfying the potential condition.

If the prescribed-flow cross terms are moved collectively to the input port and dissipation is separated from control inputs, the system becomes

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla_{\mathbf{q}} V_h = \mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}. \tag{2.98}$$

The passively dissipative bulk/drag kernel obtained by exclusive splitting constitutes $\mathbf{Q}_{diss}^{bulk}$, with $\dot{\mathbf{q}}^T \mathbf{Q}_{diss}^{bulk} \le 0$. $\mathbf{Q}_{contact}$ is the unique nonpotential contact port for elastic contact not included in V_h, contact damping, friction, impact, and contact-spin couples; if elastic contact is already included in V_h, it must not be included again, and contact damping or friction must not also enter $\mathbf{Q}_{diss}^{bulk}$. The background-flow drag term $\mathbf{Q}_{drag,flow}$ must be assigned to $\mathbf{Q}_{flow}$, and its power is generally sign-indefinite; therefore, no global dissipativity claim may be made for the complete relative drag. Nor should $\mathbf{Q}_{diss}^{bulk}$ be forced into the form of a constant damping matrix times velocity, because the quadratic-drag matrix $\mathbf{D}_{drag}(\mathbf{q},\dot{\mathbf{q}},t)$ itself depends on state and relative velocity. The same fluid action may be counted only once in any selected representation.

2.8.2. *Constrained form.* The strict Kirchhoff shear-free model is finally written as

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_T + \nabla_{\mathbf{q}} V_h + \mathbf{G}^T(\mathbf{q})\,\boldsymbol{\lambda}_c = \mathbf{Q}_{nc} + \mathbf{Q}_{\partial}, \tag{2.99}$$

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}. \tag{2.100}$$

The system retains complete bending–torsion dynamics, cross-sectional orientation, and rotary inertia while enforcing the shear-free condition exactly.

2.9. **Discrete energy and power identities.**

2.9.1. *Discrete mechanical energy of the conservative inertial backbone.* For a stationary flow with $\mathbf{b}_A = \mathbf{0}$ and $c_A = 0$, or for an autonomous conservative backbone selected after the prescribed-flow cross terms have been moved completely to $\mathbf{Q}_{flow}$, define

$$E_h(\mathbf{q}, \dot{\mathbf{q}}, t) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} + V_h(\mathbf{q}, t). \tag{2.101}$$

Because $\mathbf{g}_M$ is generated by the Euler–Lagrange derivative of the complete quadratic kinetic energy,

$$\dot{\mathbf{q}}^T [\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M] = \frac{d}{dt}\left(\frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right). \tag{2.102}$$

2.9.2. *General port-power balance.* Left-multiplication of the global equation by $\dot{\mathbf{q}}^T$ gives

$$\frac{dE_h}{dt} = \dot{\mathbf{q}}^T (\mathbf{Q}_{nc} + \mathbf{Q}_\partial - \mathbf{g}_f) + \frac{\partial V_h}{\partial t}. \tag{2.103}$$

If the complete Euler–Lagrange contribution of the prescribed-flow cross action is treated as part of the flow input, then

$$\boxed{\frac{dE_h}{dt} = \dot{\mathbf{q}}^T \left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_\partial\right) + \frac{\partial V_h}{\partial t}.} \tag{2.104}$$

Here, $\mathbf{Q}_{diss}^{bulk}$ contains only the passive bulk/drag kernel after splitting and satisfies $\dot{\mathbf{q}}^T \mathbf{Q}_{diss}^{bulk} \leq 0$; $\mathbf{Q}_{contact}$ records contact power separately under the preceding mutually exclusive contract. The term $\mathbf{Q}_{flow}$ collects $\mathbf{Q}_{drag,flow}$, the Froude–Krylov input, and the reactive-flow contribution determined by the selected representation; its power is not assigned a fixed sign. The relative-motion identity verifies physical passivity of the drag law, whereas the dissipative structural constraint of the CHNN must be imposed only on the preceding $\mathbf{Q}_{diss}^{bulk}$; the two must not be conflated. For an ideal constraint,

$$\mathbf{G}(\mathbf{q})\,\dot{\mathbf{q}} = \mathbf{0}, \tag{2.105}$$

Therefore,

$$\dot{\mathbf{q}}^T \mathbf{G}^T \boldsymbol{\lambda}_c = 0. \tag{2.106}$$

Only in the absence of nonconservative loads, flow inputs, boundary-power inputs, and explicit time dependence of the potential does $dE_h/dt = 0$.

2.9.3. *Consistent tangent matrix.* For nonlinear solution or local linearization, the consistent internal tangent matrix is defined by

$$\mathbf{K}_T(\mathbf{q}) = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{q}} = \nabla_{\mathbf{q}}^2 V_{int,h}(\mathbf{q}). \tag{2.107}$$

This matrix contains both material stiffness and the geometric stiffness generated by variations in orientation, extension, and curvature. It is not the secant matrix obtained by multiplying a state-dependent scalar stiffness by an integral of fixed basis functions.

2.10. **Local linearization and engineering reductions.**

2.10.1. *Local linearization about an equilibrium configuration.* Use the dissipation–input decomposition

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M + \nabla V_h + \mathbf{G}^T \boldsymbol{\lambda}_c = \mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_\partial. \tag{2.108}$$

Let $\mathbf{q}_e$ be a static equilibrium configuration with $\dot{\mathbf{q}}_e = \mathbf{0}$. Then

Here, a "static equilibrium configuration" requires that the explicit time-dependent input, control parameters, and prescribed flow have been frozen at $t = t_e$, or that the system itself is autonomous. If the magnetic field or flow is periodically driven, a Floquet linearization about a periodic orbit must be used instead of a static equilibrium matrix.

$$\nabla V_h\left(\mathbf{q}_e\right)+\mathbf{G}^T\left(\mathbf{q}_e\right)\boldsymbol{\lambda}_{ce}=\mathbf{Q}_{diss}^{bulk}\left(\mathbf{q}_e,\mathbf{0}\right)+\mathbf{Q}_{contact}\left(\mathbf{q}_e,\mathbf{0}\right)+\mathbf{Q}_{in,e}+\mathbf{Q}_{flow,e}+\mathbf{Q}_{\partial,e}. \tag{2.109}$$

Usually, $\mathbf{Q}_{diss}^{bulk}\left(\mathbf{q},\mathbf{0}\right)=\mathbf{0}$; whether $\mathbf{Q}_{contact}\left(\mathbf{q},\mathbf{0}\right)$ vanishes depends on the contact terms not represented in the potential. Let $\mathbf{q}=\mathbf{q}_e+\boldsymbol{\eta}$ and neglect second- and higher-order perturbation terms. For the unconstrained system,

$$\mathbf{M}_e\ddot{\boldsymbol{\eta}}+\mathbf{C}_e\dot{\boldsymbol{\eta}}+\mathbf{K}_e\boldsymbol{\eta}=\delta\mathbf{Q}(t), \tag{2.110}$$

where

$$\mathbf{M}_e=\mathbf{M}\left(\mathbf{q}_e\right), \tag{2.111}$$

$$\mathbf{C}_e=-\left.\frac{\partial\left(\mathbf{Q}_{diss}^{bulk}+\mathbf{Q}_{contact}\right)}{\partial\dot{\mathbf{q}}}\right|_e, \tag{2.112}$$

$$\boxed{\mathbf{K}_e=\left.\left[\nabla_{\mathbf{q}}^2V_h-\frac{\partial\left(\mathbf{Q}_{diss}^{bulk}+\mathbf{Q}_{contact}\right)}{\partial\mathbf{q}}-\frac{\partial\left(\mathbf{Q}_{in}+\mathbf{Q}_{flow}+\mathbf{Q}_{\partial}\right)}{\partial\mathbf{q}}\right]\right|_e.} \tag{2.113}$$

Here, $\delta\mathbf{Q}(t)$ collects explicit time-dependent input perturbations not absorbed by the state Jacobian. Under the actual dissipative-force convention above, passive dissipation gives $\mathbf{C}_e\succcurlyeq\mathbf{0}$ near the equilibrium. If the load Jacobian contains nonsymmetric follower terms, $\mathbf{K}_e$ need not be symmetric.

If contact is active near the equilibrium, the preceding Jacobian and consistent tangent assume that the contact active set remains fixed. If the equilibrium lies on a contact opening/closing switching surface, a smooth contact regularization, semismooth Newton method, or piecewise linearization must be used; the max function cannot be treated as differentiable everywhere.

For a purely conservative system without boundary follower loads, the expression reduces directly to

$$\boxed{\mathbf{K}_e=\nabla_{\mathbf{q}}^2V_h\left(\mathbf{q}_e\right),} \tag{2.114}$$

No derivative of a conservative generalized force already included in V_h is subtracted again. If strict constraints are present, the system must be projected onto the constraint tangent space or first expressed in independent local coordinates satisfying $\mathbf{c}\left(\mathbf{q}\right)=0$. The tangent stiffness at a constrained equilibrium must also include the second-order contribution of the constraint Lagrangian. A constant-matrix linear system is only a local approximation of the complete coupled nonlinear model near the specified equilibrium configuration.[17, 18]

2.10.2. *Scope of engineering reductions.* Small rotations, constant mass, a frozen normal projection, and low-order modal truncation are used only for numerical checks, parameter identification, or ablation experiments. The Euler–Bernoulli branch has two roles: theoretically, it is the Kirchhoff shear-free reduction of the geometrically exact parent model; within COMSOL, it is the Kirchhoff-consistent engineering realization used for the primary training data. It may be used as the primary training branch only after the slenderness ratio, the fraction of shear energy, and the comparison error relative to Timoshenko/solid models satisfy prescribed tolerances. The corotational Beam geometric-nonlinearity formulation in COMSOL permits large rotations but assumes locally small element strains; the mesh must therefore keep the rotation increment between adjacent nodes sufficiently small[4, 5].

2.10.3. *Modal truncation and selection of the principal computational order.* The CHNN and KAM analyses must use a finite-dimensional system of fixed dimension and cannot remain indefinitely at an abstract modal order n. For candidate truncation orders $n = 1, 2, \ldots, n_{max}$, compute at least the leading natural frequencies, terminal displacement, structural energy, torsional response, and principal port powers. Define the adjacent-order convergence indicators

$$e_n^{freq} = \max_{1\le j\le m} \frac{\left|\omega_j^{(n+1)} - \omega_j^{(n)}\right|}{\left|\omega_j^{(n+1)}\right|}, \qquad e_n^{traj} = \frac{\| \mathbf{q}^{(n)} - \Pi_{n+1\to n}\mathbf{q}^{(n+1)} \|}{\| \mathbf{q}^{(n)} \|}, \tag{2.115}$$

Here $\mathbf{q}^{(n)} \in \mathbb{R}^n$, $\mathbf{q}^{(n+1)} \in \mathbb{R}^{n+1}$, and $\Pi_{n+1\to n} \in \mathbb{R}^{n\times(n+1)}$; the two trajectories must use the same physical operating condition and time grid. If the denominator vanishes, report the unnormalised absolute error. Define analogous indicators for energy and torsion. The formal production order is denoted n_{main} and is the first order at which all prescribed indicators simultaneously fall below their tolerances. Until this computation has been completed, the manuscript may retain n_{main} only as an undetermined parameter and must not describe an empirically selected order as proven sufficient.

The canonical data mapping is examined using a fixed $N_{\text{diag}} = 2$ diagnostic basis. It is extracted from the mesh-80 near-frequency cluster, orthonormalized in the consistent finite-element mass metric, and fixed with respect to sign, normalization, and degree-of-freedom row order. It defines the paired-coordinate contract used in the P1 series and the subsequent consistency analysis. The complete bridge retains the first 12 candidate production modes and the mesh-40/80 comparison, so $N_{\text{diag}} = 2$ neither replaces the convergence criterion for n_{main} nor establishes high-fidelity sufficiency.

2.11. **Numerical implementation and finite-dimensional model outputs.**

2.11.1. *COMSOL model hierarchy, projection, and validation.* The COMSOL data use a three-level hierarchy: "primary training–strict reproduction–high-fidelity comparison.

1. **Kirchhoff-consistent primary training model.** Use the Euler–Bernoulli formulation of the three-dimensional Beam interface with Time Dependent dynamics and geometric nonlinearity, retaining axial deformation, bending about two axes, three-dimensional cross-sectional rotations, and torsion. The three-dimensional Euler–Bernoulli section stiffness in COMSOL includes axial force, torque, and two bending moments; shear-free therefore does not mean torsion-free[4, 6]. This model generates data under assumptions consistent with the strict Kirchhoff branch.
2. **Strict weak-form reproduction model.** When a term-by-term check of the coupled translational–orientation weak form, nonstandard constraints, or custom distributed couples is required, Weak Form PDE or Weak Contribution is used to specify test functions, boundary terms, and auxiliary variables[7, 8]. This branch checks discrepancies between the built-in Beam interface and the present equations; it is not the default model for every operating condition.
3. **High-fidelity comparison model.** A Timoshenko Beam model quantifies corrections from transverse shear and rotary inertia, while three-dimensional Solid Mechanics checks local cross-sectional stresses, contact regions, and attachment effects. These models output X_{HF}. Their discrepancy from the Kirchhoff-consistent model X_K is used for validation and model-discrepancy learning rather than directly replacing the primary KAM branch.

Every model should output the centerline position $\mathbf{r}_C(S,t)$, cross-sectional orientation $\mathbf{R}_C(S,t)$ or an equivalent rotation representation, centerline velocity, cross-sectional angular velocity, resultant section forces and moments, structural kinetic energy, and strain energy. The primary training branch should additionally output the separate powers associated with magnetic, fluid, contact, and boundary actions.

Define the translational Gram matrix

$$\mathbf{G}_a = \int_0^L \mathbf{N}^T \mathbf{W}_a \mathbf{N}\, dS, \tag{2.116}$$

Then the projection of the COMSOL centerline field is

$$\mathbf{a}(t) = \mathbf{G}_a^{-1} \int_0^L \mathbf{N}^T \mathbf{W}_a \left[\mathbf{r}_C(S,t) - \mathbf{r}_0(S)\right] dS. \tag{2.117}$$

Orientation must not be projected linearly by its Euler-angle components. First compute the logarithmic coordinates of the relative rotation

$$\boldsymbol{\theta}_C(S,t) = \log\left(\mathbf{R}_0^T(S)\mathbf{R}_C(S,t)\right)^\vee, \tag{2.118}$$

and then set

$$\mathbf{G}_b = \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\Psi}\, dS, \qquad \mathbf{b}(t) = \mathbf{G}_b^{-1} \int_0^L \boldsymbol{\Psi}^T \mathbf{W}_b \boldsymbol{\theta}_C(S,t)\, dS. \tag{2.119}$$

This gives $\mathbf{q} = \left(\mathbf{a}^T, \mathbf{b}^T\right)^T$. The angular velocity should be checked through $\boldsymbol{\omega}_h = \mathbf{B}_\omega(\mathbf{q})\,\dot{\mathbf{b}}$, rather than by setting $\boldsymbol{\omega} = \boldsymbol{\Psi}\dot{\mathbf{b}}$.

The COMSOL–Galerkin bridge compares the stationary energy/field exports, modal reality on mesh 40/80, the leading 12 frequency clusters, and the mass-metric basis. The maximum relative mesh-frequency difference is 4.15920×10^{-5}, and the maximum off-diagonal entry among the candidate-prefix mass Gram matrices is at most 1.32445×10^{-15}. The three 1 s trajectories do not satisfy the original 0.5% strict perturbation-energy criterion overall, whereas all three satisfy the independent 2% force-v2 criterion on its fixed common support and anchor. The strict energy criterion, the exploratory 1% energy criterion, and the force-v2 criterion address different questions and use different thresholds; their separate values are reported in Section 6.2.

The fluid-loading implementation must match the selected theoretical branch. The theoretical route uses T_A (or its discrete equivalent mass and geometric terms), the pure Froude–Krylov term, and drag. The engineering Morison comparison uses the complete $C_M\mathbf{a}_f - C_A\ddot{\mathbf{r}}$ form but must not activate T_A simultaneously. The same physical action may be counted only once in any data sample.

2.11.2. *Finite-dimensional state, energy, port, and structure-learning outputs.* After the joint Galerkin discretization and the consistent Legendre postprocessing of Section 3, the numerical implementation should output unified finite-dimensional states, energies, ports, and representation labels:

$$\mathcal{O}_h = \left(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}, \mathbf{p}_{rep}, \dot{\mathbf{p}}_{rep}, \mathbf{M}(\mathbf{q}), V_h, E_h, \mathbf{Q}_{diss}^{bulk}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_\partial, \mathbf{c}, \mathbf{G}, representation\right). \tag{2.120}$$

Here, $\mathbf{p}_{rep}$ and $\dot{\mathbf{p}}_{rep}$ must be consistent with the selected Lagrangian representation. The *representation* label must distinguish at least the autonomous-backbone, exact-nonautonomous, and engineering-reduction representations. The strict Kirchhoff branch also stores the discrete constraint $\mathbf{c}(\mathbf{q})$ and its Jacobian $\mathbf{G}(\mathbf{q})$; these two components may be marked as empty in an unconstrained branch.

On this basis, Section 3 constructs two data objects that must not be mixed. The autonomous-backbone data are

$$\mathcal{D}_{sk} = \left(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla_{\mathbf{z}_0} H_{cons}, \mathbf{Q}_{diss}^{bulk}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{sk}, \mathbf{Q}_\partial\right), \qquad \mathbf{z}_0 = (\mathbf{q}, \mathbf{p}_0), \tag{2.121}$$

and the exact nonautonomous data are

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla_{\mathbf{z}} H, \partial_t H, \mathbf{Q}_{diss}^{bulk}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}^{H}, \mathbf{Q}_\partial\right), \qquad \mathbf{z} = (\mathbf{q}, \mathbf{p}). \tag{2.122}$$

If $\dot{\mathbf{p}}_{rep}$ or $\dot{\mathbf{z}}_{rep}$ is obtained by numerical differentiation rather than from a solver residual, analytic equations, or automatic differentiation, the differentiation order, time step, and derivative-error estimate must also be recorded. CHNN derivative supervision, energy-balance supervision, and the subsequent KAM error budget may use only representation-consistent data with traceable errors.

2.11.3. *Training-data hierarchy and model labels.* Each sample must store

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{2.123}$$

Here, K denotes the Kirchhoff-consistent Beam primary training model, WF the strict weak-form model, T the Timoshenko comparison, and S the three-dimensional solid comparison; sk denotes the autonomous-backbone representation, na the exact nonautonomous representation, and red an engineering reduction.

The data set used for the conservative CHNN and the statistical-learning–KAM interface is restricted to

$$\mathcal{D}_K^{cons} = \{(\mathbf{z}_0, \dot{\mathbf{z}}_0, H_{cons}, \nabla H_{cons}) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0\}, \tag{2.124}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \{\left(\mathbf{z}_{rep}, \dot{\mathbf{z}}_{rep}, \mathbf{Q}_{diss}^{bulk}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_{\partial}\right) : model_level = K\}, \tag{2.125}$$

The high-fidelity comparison set is

$$\mathcal{D}_{HF} = \{X_{HF} - X_K : model_level \in \{WF, T, S\}\}. \tag{2.126}$$

Only after closure, power-sign, and regularity tests may $\mathcal{D}_{HF}$ enter a conservative correction, dissipative correction, or general residual, respectively; the entire high-fidelity discrepancy must not be interpreted as a Hamiltonian perturbation. Samples with different *model_level* and *representation* labels must not be mixed directly without labels and pairing.

2.12. **Section summary.** Starting directly from the complete translational–rotational weak form, this section constructed a joint Galerkin approximation using a centerline expansion and an exponential-map representation of cross-sectional orientation. The compatibility conditions for the reference configuration and strains, the initial conditions, the contact active set, and the mutually exclusive accounting of conservative and nonconservative loads were specified explicitly. The base rod, eccentric attachments, and energy-consistent fluid added mass were collected in a velocity-dependent inertial functional, whose Hessian with respect to generalized velocity gives

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}). \tag{2.127}$$

For a general prescribed flow, the generalized momentum is rigorously

$$\mathbf{p} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_A(\mathbf{q}, t), \tag{2.128}$$

Both the geometric inertia and the prescribed-flow cross terms are generated by the complete Euler–Lagrange derivative rather than added empirically. The internal restoring force is obtained from the gradient of the full extension–shear–bending–torsion potential, while magnetic and other distributed couples are projected directly through the rotational test functions. The local linearization eliminates double counting of conservative external loads and explicitly includes boundary follower-load and constraint-tangent contributions.

The resulting finite-dimensional nonlinear dynamical systems are provided in unconstrained and Kirchhoff-constrained forms, together with a projection from COMSOL full-order fields to the joint generalized coordinates and a unified output specification. The energy-consistent fluid branch uses $T_A + \mathbf{f}_{FK} + \mathbf{f}_{drag}$, with the drag further split, under a general background flow, into

$\mathbf{f}_{drag,diss} + \mathbf{f}_{drag,flow}$; the engineering Morison formulation is used only as a mutually exclusive validation branch. Small rotations, constant mass, and frozen added mass are retained only as reduced branches. Euler–Bernoulli serves as the COMSOL Kirchhoff-consistent primary training branch after the slenderness and shear-error tolerances are satisfied. These discrete objects support not only the canonical momentum and Hamiltonian construction of Section 3, but also provide the sole upstream interface to the CHNN training in Section 4 and the structure-preservation and KAM validation in Section 5 through $\mathcal{D}_K^{cons}$, $\mathcal{D}_K^{port}$, $\mathcal{D}_{HF}$, port labels, model-level labels, and constraint labels.

3. Finite-Dimensional Hamiltonian Structure and Canonicalization

3.1. **Finite-Dimensional Phase-Space Variables and Hamiltonian Formulation.**

3.1.1. *Joint Generalized Coordinates.* The joint Galerkin projection of centerline position and cross-sectional orientation developed in Section 2 yields

$$\boxed{\mathbf{q} = \begin{bmatrix} \mathbf{a} \\ \mathbf{b} \end{bmatrix} \in \mathbb{R}^n, \qquad n = n_a + n_b,} \tag{3.1}$$

where $\mathbf{a}$ denotes the generalized coordinates of centerline translation and $\mathbf{b}$ contains the finite-dimensional coefficients of the local rotational coordinates for cross-sectional orientation. The vector $\mathbf{b}$ is an independent configuration variable describing the bending orientation of the cross section, torsion about the tangent, angular velocity, and rotary inertia; it is not an auxiliary variable derived from the centerline displacement. Thus, $\mathbf{q}$ is the complete joint generalized configuration coordinate for translation and cross-sectional orientation.

The discrete dynamics of the finite-shear-stiffness geometrically exact Cosserat parent model can be written as

$$\boxed{\mathbf{M}(\mathbf{q})\,\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}) + \mathbf{g}_f(\mathbf{q}, \dot{\mathbf{q}}, t) + \nabla_{\mathbf{q}} V_h(\mathbf{q}, t) = \mathbf{Q}_{port}.} \tag{3.2}$$

The mass–inertia matrix

$$\mathbf{M}(\mathbf{q}) = \mathbf{M}_{str}(\mathbf{q}) + \sum_e \mathbf{M}_e(\mathbf{q}) + \mathbf{M}_A(\mathbf{q}) \tag{3.3}$$

contains the translational kinetic energy of the rod, cross-sectional rotational kinetic energy, translational–rotational coupling of eccentric attachments, and energy-consistent added mass. The Hamiltonian structure developed below is therefore always based on the complete bending–torsion model; it does not use a constant-mass, small-rotation, centerline-only, or torsion-deleted surrogate.

3.1.2. *Basic Framework of the Hamiltonian Formulation.* The finite-dimensional Hamiltonian formulation follows the chain[17, 29]

$$\mathbf{q} \to \mathbf{p} \to H(\mathbf{q}, \mathbf{p}) \to \begin{cases} \dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \\ \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H + \mathbf{Q}, \end{cases} \to \frac{dH}{dt}. \tag{3.4}$$

This derivation applies to the autonomous conservative backbone and its forced–dissipative extension. A general prescribed flow generates a velocity-linear term in the Lagrangian and consequently shifts the canonical momentum; Section 3.5 gives the complete nonautonomous Hamiltonian formulation.

3.2. **Lagrangian, Momentum, and Hamiltonian of the Autonomous Conservative Backbone.**

3.2.1. *Selection of the Autonomous Potential.* Let the external actuation be controlled by parameters $\boldsymbol{\lambda}(t)$, and choose a reference value $\bar{\boldsymbol{\lambda}}$. Write the discrete potential as

$$V_h\left(\mathbf{q}, t; \boldsymbol{\lambda}(t)\right) = V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) + V_{drv}\left(\mathbf{q}, t; \boldsymbol{\lambda}(t), \bar{\boldsymbol{\lambda}}\right), \tag{3.5}$$

and impose

$$V_{cons}\left(\mathbf{q}; \bar{\boldsymbol{\lambda}}\right) := V_h\left(\mathbf{q}, t; \boldsymbol{\lambda}(t)\right)|_{\boldsymbol{\lambda}(t)\equiv\bar{\boldsymbol{\lambda}}}, \qquad V_{drv} := V_h - V_{cons}. \tag{3.6}$$

After the reference parameters are frozen, no explicit time-dependent term remains; hence

$$\boxed{V_{drv}\left(\mathbf{q}, t; \bar{\boldsymbol{\lambda}}, \bar{\boldsymbol{\lambda}}\right) = 0, \qquad \partial_t V_{cons} = 0.} \tag{3.7}$$

The potential V_{cons} contains the discrete Cosserat strain energy and the conservative magnetic potential, elastic contact potential, and other conservative external potentials selected at the reference operating condition. This definition freezes the actuation parameters rather than arbitrarily selecting a time slice.

3.2.2. *Autonomous-Backbone Lagrangian and Generalized Momentum.* For the autonomous port-closed backbone, the Lagrangian is

$$\boxed{L_{cons}\left(\mathbf{q}, \dot{\mathbf{q}}\right) = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}} - V_{cons}\left(\mathbf{q}\right).} \tag{3.8}$$

The canonical momentum conjugate to $\mathbf{q}$ is defined by[17, 29]

$$\mathbf{p}_0 = \frac{\partial L_{cons}}{\partial \dot{\mathbf{q}}}, \tag{3.9}$$

which gives the relation between canonical momentum and generalized velocity

$$\boxed{\mathbf{p}_0 = \mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}}, \qquad \dot{\mathbf{q}} = \mathbf{M}^{-1}\left(\mathbf{q}\right)\mathbf{p}_0.} \tag{3.10}$$

The subscript 0 distinguishes this quantity from the shifted canonical momentum $\mathbf{p}$ arising under a general prescribed flow in Section 3.5. The $\mathbf{p}_0$ orientation component $\mathbf{p}_{0b}$ is the discrete momentum conjugate to the local coordinate $\mathbf{b}$; it is not, in general, equal to a component of the material angular momentum of any individual cross section.

3.2.3. *Legendre Regularity.* The velocity Hessian is[17, 29]

$$\frac{\partial^2 L_{cons}}{\partial \dot{\mathbf{q}}\, \partial \dot{\mathbf{q}}} = \mathbf{M}\left(\mathbf{q}\right). \tag{3.11}$$

If the joint basis contains no velocity redundancy, the material density, rotary inertia, and added-mass parameters satisfy the required physical positivity conditions, and throughout the configuration domain of interest

$$\boxed{\mathbf{y}^T \mathbf{M}\left(\mathbf{q}\right)\mathbf{y} > 0 \quad \left(\forall \mathbf{y} \neq \mathbf{0}\right),} \tag{3.12}$$

then $\mathbf{M}\left(\mathbf{q}\right) = \mathbf{M}^T\left(\mathbf{q}\right) \succ 0$, and the Legendre map is locally invertible. Geometrically, $\mathbf{M}$ is the mass metric on the tangent space of configuration space, while $\mathbf{M}^{-1}$ is the corresponding dual metric on the cotangent space.

For the fixed N-dimensional reduced manifold $\mathcal{Q}_N$, the configuration coordinate satisfies $q_N \in \mathcal{Q}_N$, the generalized velocity satisfies $\dot{q}_N \in T_{q_N}\mathcal{Q}_N$, and the conjugate momentum satisfies $p_N \in T^*_{q_N}\mathcal{Q}_N$. In the same paired-coordinate trivialization, $\dot{p}_N$ is stored as the momentum-rate component paired with p_N; comparisons across charts additionally require the associated connection or coordinate transformation. The data mapping from COMSOL full-order outputs to this cotangent contract is defined by

$$\boxed{\begin{aligned} q_N &= \Pi_{\mathrm{conf},N}(U), \qquad p_N = B_N(q_N)^T P_{\mathrm{full}}, \\ \dot{q}_N &= M_N(q_N)^{-1} p_N, \quad \dot{p}_N = \dot{B}_N(q_N)^T P_{\mathrm{full}} + B_N(q_N)^T \dot{P}_{\mathrm{full}}. \end{aligned}} \tag{3.13}$$

Here, $B_N(q_N)$ is the current-configuration tangent lift of the fixed translational–$SO(3)$ basis, P_{full} is the full-order momentum, and $M_N = B_N^T E B_N$. If $L_{\mathrm{full}} := \dot{P}_{\mathrm{full}}$ is defined separately and the implementation explicitly accounts for $\dot{B}_N^T P_{\mathrm{full}}$, the last equation may be written as $\dot{p}_N = \dot{B}_N^T P_{\mathrm{full}} + B_N^T L_{\mathrm{full}}$; it cannot in general be reduced to $B_N^T L$. The quantity $p_{\mathrm{mass}} = M_N \dot{q}_{\mathrm{native}}$ is used only as a diagnostic. It equals $B_N^T P_{\mathrm{full}}$ only when the full-order velocity lies in the fixed tangent space and the temporal storage and projection are consistent. The current delivery contains no auditable implementation or raw arrays for $U, B_N, E, P_{\mathrm{full}}$, or the transport term. The existing canonical, force-v2, and G3 numbers therefore remain historical values pending recomputation and do not qualify the contract.

To register notation unambiguously, the magnetic induction is denoted $\mathbf{B}_{mag}$ whenever the magnetic field is meant, while port embedding is denoted by the matrix family $\mathbf{B}_{port}^{(r)} = [\mathbf{0}; \mathbf{I}_r] \in \mathbb{R}^{2r \times r}$; $\mathbf{B}$ in a complex-valued linear layer is local to that algebraic context, and the superscript (r) is omitted when unambiguous. $\mathbf{J}$ is reserved for the symplectic matrix, and section inertia or torsional constants use separate names. The catheter length is L_{cat}, and the load-rate notation is $L_{full} = \dot{P}_{full}$. In dimensions, $\mathbf{q} \in \mathbb{R}^n$ is the full configuration coordinate, $\mathbf{Z} = [\mathbf{Q}; \mathbf{P}] \in \mathbb{R}^{2d}$ is the reduced CHNN state, $\mathbf{B}_N \in \mathbb{R}^{m_f \times N_{\mathrm{diag}}}$, $\mathbf{E} \in \mathbb{R}^{m_f \times m_f}$, and $\mathbf{M}_N = \mathbf{B}_N^T \mathbf{E} \mathbf{B}_N \in \mathbb{R}^{N_{\mathrm{diag}} \times N_{\mathrm{diag}}}$. Here $N_{\mathrm{diag}} = 2$ is diagnostic only; the formal production order remains the undetermined n_{main}. The full-order port uses $r = n$, whereas the reduced CHNN port uses $r = d$. The physical control $\mathbf{u} \in \mathbb{R}^d$ differs from the full-state embedding $\mathbf{U}_{port} = \mathbf{B}_{port}^{(d)} \mathbf{u} \in \mathbb{R}^{2d}$.

3.2.4. *Construction of the Hamiltonian.* The Hamiltonian is defined as

$$H_{cons}(\mathbf{q}, \mathbf{p}_0) = \mathbf{p}_0^T \dot{\mathbf{q}} - L_{cons}(\mathbf{q}, \dot{\mathbf{q}}). \tag{3.14}$$

Substituting $\dot{\mathbf{q}} = \mathbf{M}^{-1} \mathbf{p}_0$ gives

$$\boxed{H_{cons}(\mathbf{q}, \mathbf{p}_0) = \frac{1}{2} \mathbf{p}_0^T \mathbf{M}^{-1}(\mathbf{q}) \mathbf{p}_0 + V_{cons}(\mathbf{q}).} \tag{3.15}$$

This Hamiltonian has the standard form $H = T + V$ and contains both translational and cross-sectional rotational degrees of freedom. Configuration dependence of the mass matrix does not invalidate the Legendre transformation; its contribution enters the geometric inertial terms through the derivative of the Hamiltonian with respect to $\mathbf{q}$.

3.2.5. *Nonlinear Potential and Restoring Force.* The discrete internal potential of the Cosserat rod is determined by the strain-energy functional in Section 2. Define

$$\boxed{\mathbf{f}_{int}(\mathbf{q}) = \nabla_{\mathbf{q}} V_{int,h}(\mathbf{q}), \qquad \mathbf{K}_{tan}(\mathbf{q}) = \frac{\partial \mathbf{f}_{int}}{\partial \mathbf{q}} = \nabla_{\mathbf{q}}^2 V_{int,h}(\mathbf{q}).} \tag{3.16}$$

Relative to a reference configuration $\mathbf{q}_*$, the conservative potential may also be written as the path-independent line integral

$$V_{int,h}(\mathbf{q}) - V_{int,h}(\mathbf{q}_*) = \int_0^1 \mathbf{f}_{int}(\mathbf{q}_* + s(\mathbf{q} - \mathbf{q}_*))^T (\mathbf{q} - \mathbf{q}_*)\, ds. \tag{3.17}$$

Thus, the conservative potential can be represented by the line integral of the restoring force along a path in configuration space. In general, one cannot assume $\mathbf{f}_{int} = \mathbf{K}_{tan}(\mathbf{q})\, \mathbf{q}$. A quadratic approximation is justified only in a linearized neighborhood or when the restoring force genuinely admits that factorization:

$$V_{int,h}(\mathbf{q}_* + \boldsymbol{\eta}) \approx V_{int,h}(\mathbf{q}_*) + \frac{1}{2} \boldsymbol{\eta}^T \mathbf{K}_{tan}(\mathbf{q}_*)\, \boldsymbol{\eta}. \tag{3.18}$$

A global quadratic form is generally invalid because both the potential and the restoring force are geometrically nonlinear.

3.3. **Canonical Hamilton Equations and Geometric Inertia.**

3.3.1. *Standard Symplectic Structure.* Define the phase-space variable and the standard symplectic matrix

$$\mathbf{z}_0 = \begin{bmatrix} \mathbf{q} \\ \mathbf{p}_0 \end{bmatrix}, \qquad \mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}, \qquad \mathbf{J}^T = -\mathbf{J}. \tag{3.19}$$

The autonomous conservative backbone satisfies

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla_{\mathbf{z}_0} H_{cons},} \tag{3.20}$$

that is,

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}, \qquad \dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons}.} \tag{3.21}$$

The corresponding standard Poisson bracket is

$$\{F, G\} = \nabla_{\mathbf{q}} F^T \nabla_{\mathbf{p}_0} G - \nabla_{\mathbf{p}_0} F^T \nabla_{\mathbf{q}} G. \tag{3.22}$$

Because local orientation coordinates and their conjugate momenta are used, the standard canonical bracket applies. If global rotation matrices and material angular momenta were used instead, a Lie–Poisson or constrained symplectic structure would have to be constructed separately.

3.3.2. *Terms Generated by the Configuration-Dependent Mass Matrix.* For the i-th configuration component,

$$\frac{\partial H_{cons}}{\partial q_i} = \frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 + \frac{\partial V_{cons}}{\partial q_i}. \tag{3.23}$$

Therefore, the momentum equation is

$$\boxed{\dot{p}_{0i} = -\frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 - \frac{\partial V_{cons}}{\partial q_i}.} \tag{3.24}$$

Using the derivative identity for an inverse matrix,

$$\frac{\partial \mathbf{M}^{-1}}{\partial q_i} = -\mathbf{M}^{-1}\mathbf{M}_{,i}\mathbf{M}^{-1} \tag{3.25}$$

one obtains

$$-\frac{1}{2}\mathbf{p}_0^T \frac{\partial \mathbf{M}^{-1}}{\partial q_i}\mathbf{p}_0 = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}_{,i}\dot{\mathbf{q}}. \tag{3.26}$$

This term fully represents the contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation. Only when the momentum equation is converted back to a second-order acceleration equation must it be combined with $\dot{\mathbf{p}}_0$, specifically with the contribution $\dot{\mathbf{M}}\dot{\mathbf{q}}$ contained therein.

3.3.3. *Equivalence with the Euler–Lagrange Equations.* From

$$\dot{\mathbf{p}}_0 = \mathbf{M}\ddot{\mathbf{q}} + \dot{\mathbf{M}}\dot{\mathbf{q}} \tag{3.27}$$

and the preceding Hamiltonian momentum equation, one obtains

$$\boxed{\mathbf{M}\ddot{\mathbf{q}} + \mathbf{g}_M(\mathbf{q}, \dot{\mathbf{q}}) + \nabla_{\mathbf{q}} V_{cons}(\mathbf{q}) = \mathbf{0},} \tag{3.28}$$

where

$$\boxed{\mathbf{g}_M = \dot{\mathbf{M}}\dot{\mathbf{q}} - \frac{1}{2}\nabla_{\mathbf{q}}\left(\dot{\mathbf{q}}^T \mathbf{M}\dot{\mathbf{q}}\right).} \tag{3.29}$$

This result agrees with the autonomous conservative Euler–Lagrange equation in Section 2, establishing the equivalence between the geometric-inertia representations in canonical momentum form and in second-order generalized-coordinate form.

3.3.4. *Energy Conservation.* Because $\mathbf{J}$ is skew-symmetric and H_{cons} has no explicit time dependence,

$$\frac{dH_{cons}}{dt} = \nabla H_{cons}^T \mathbf{J} \nabla H_{cons} = 0. \tag{3.30}$$

and hence

$$\boxed{H_{cons}(\mathbf{q}(t), \mathbf{p}_0(t)) = const.} \tag{3.31}$$

3.4. **Forced–Dissipative Hamiltonian Structure.**

3.4.1. *Unified Entry Point for Nonconservative Generalized Forces.* Nonconservative generalized forces enter uniformly through the right-hand side of the Hamiltonian momentum equation:

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}, \qquad \dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{Q}_{port}.} \tag{3.32}$$

Following the mutually exclusive power accounting established in Sections 1 and 2, write

$$\boxed{\mathbf{Q}_{port} = \mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}.} \tag{3.33}$$

where:

- $\mathbf{Q}_{diss}^{bulk}$ contains only passive bulk/drag components whose power with respect to the structural mechanical energy is nonpositive and excludes contact;
- $\mathbf{Q}_{contact}$ contains only elastic contact not represented in the potential together with contact damping, friction, impact, and contact couples;
- $\mathbf{Q}_{in}$ contains magnetic control, actuation, and other nonfluid inputs that have not been included in the potential;
- $\mathbf{Q}_{flow}$ contains background-flow drag input, Froude–Krylov input, and representation-dependent reactive-flow effects;
- $\mathbf{Q}_{\partial}$ separately collects terminal forces, terminal moments, and boundary actuation.

The joint virtual-work projection of these forces and moments was derived in Section 2; the corresponding finite-dimensional generalized forces are used directly here.

3.4.2. *Equivalent Viscous-Damping Special Case.* If, for a particular operating condition, the nonconservative force can be represented equivalently as

$$\mathbf{Q}_{port} = \mathbf{F}_{ext}\left(\mathbf{q}, t\right) - \mathbf{C}\left(\mathbf{q}, \dot{\mathbf{q}}, t\right)\dot{\mathbf{q}}, \qquad \mathbf{C} = \mathbf{C}^T \succcurlyeq 0, \tag{3.34}$$

then $\dot{\mathbf{q}} = \nabla_{\mathbf{p}_0} H_{cons}$ gives

$$\boxed{\dot{\mathbf{p}}_0 = -\nabla_{\mathbf{q}} H_{cons} + \mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}.} \tag{3.35}$$

These equations form a forced–damped Hamiltonian special case on the autonomous backbone. The operator $\mathbf{C}$ may be a state-dependent effective dissipation operator and need not be constant. For quadratic drag, only the passive dissipative component enters $\mathbf{C}\dot{\mathbf{q}}$; the background-flow input remains in $\mathbf{Q}_{flow}$. Coulomb friction, contact switching, and impact may be treated as separate generalized-force ports without requiring a single globally smooth dissipation matrix.

If the magnetic and elastic contact forces have not been included in the potential, one may set

$$\mathbf{F}_{ext} = \mathbf{F}_{mag} + \mathbf{F}_{cont} \tag{3.36}$$

If the same actions have already been generated by the gradients of $V_{mag,cons}$ or $V_{cont,el}$, they must not be counted again in $\mathbf{F}_{ext}$.

3.4.3. *Energy Rate.* For an autonomous Hamiltonian,[17, 20, 21]

$$\frac{dH_{cons}}{dt} = \nabla_{\mathbf{q}} H_{cons}^T \dot{\mathbf{q}} + \nabla_{\mathbf{p}_0} H_{cons}^T \dot{\mathbf{p}}_0. \tag{3.37}$$

Substitution of the forced canonical equations causes the symplectic terms to cancel, yielding

$$\boxed{\frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^T \mathbf{Q}_{port}.} \tag{3.38}$$

Under the equivalent-damping condition of Section 3.4.2,

$$\boxed{\frac{dH_{cons}}{dt} = \nabla_{\mathbf{p}_0} H_{cons}^T \left(\mathbf{F}_{ext} - \mathbf{C}\nabla_{\mathbf{p}_0} H_{cons}\right).} \tag{3.39}$$

where

$$-\nabla_{\mathbf{p}_0} H_{cons}^T \mathbf{C} \nabla_{\mathbf{p}_0} H_{cons} \leq 0, \tag{3.40}$$

The sign of the external input power depends on the operating condition. This energy-rate expression clearly distinguishes external input from passive dissipation and is consistent with the port classification.

3.4.4. *Phase-Space Form of the Complete Forced System.* Define

$$\mathbf{B}_{port} = \begin{bmatrix} \mathbf{0} \\ \mathbf{I} \end{bmatrix}, \tag{3.41}$$

The complete forced–dissipative system on the autonomous backbone is then

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B}_{port}\left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_{\partial}\right).} \tag{3.42}$$

The system is therefore a forced–dissipative Hamiltonian system, or equivalently an open system with a canonical Hamiltonian backbone. It reduces to a classical closed conservative Hamiltonian system only when every port is closed and the Hamiltonian is autonomous. A closed Hamiltonian representation of the dissipative environment would require additional energy-storage or heat-bath degrees of freedom, which are not part of the finite-dimensional variable set constructed in Sections 1 and 2.

3.5. **Canonical Momentum and Hamiltonian under a General Prescribed Flow.**

3.5.1. *General Discrete Lagrangian.* The energy-consistent prescribed-flow formulation in Sections 1 and 2 generates a velocity-linear term. The general discrete Lagrangian is[17, 29]

$$\boxed{L_h\left(\mathbf{q},\dot{\mathbf{q}},t\right)=\frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}}-\dot{\mathbf{q}}^T\mathbf{b}_A\left(\mathbf{q},t\right)+c_A\left(\mathbf{q},t\right)-V_h\left(\mathbf{q},t\right).} \tag{3.43}$$

Here, $\mathbf{M}$ still contains the velocity-quadratic added-mass contribution, while $\mathbf{b}_A$ and c_A represent the cross term and the pure-flow term induced by the prescribed flow. Added inertia already included in $\mathbf{M}_A, \mathbf{b}_A, c_A$ must not be reintroduced on the right-hand side as a complete Morison added-mass generalized force.

3.5.2. *General Canonical Momentum and Hamiltonian.* By definition,

$$\mathbf{p}=\frac{\partial L_h}{\partial\dot{\mathbf{q}}} \tag{3.44}$$

one obtains

$$\boxed{\mathbf{p}=\mathbf{M}\left(\mathbf{q}\right)\dot{\mathbf{q}}-\mathbf{b}_A\left(\mathbf{q},t\right),\qquad\dot{\mathbf{q}}=\mathbf{M}^{-1}\left(\mathbf{q}\right)\left(\mathbf{p}+\mathbf{b}_A\right).} \tag{3.45}$$

When $\mathbf{b}_A=\mathbf{0}$, this relation reduces to $\mathbf{p}_0=\mathbf{M}\dot{\mathbf{q}}$ for the autonomous backbone or the backbone–port representation. When the velocity-linear prescribed-flow term is retained explicitly, the general canonical momentum $\mathbf{p}$ must be used.

The Legendre transformation gives

$$\boxed{H\left(\mathbf{q},\mathbf{p},t\right)=\frac{1}{2}\left(\mathbf{p}+\mathbf{b}_A\right)^T\mathbf{M}^{-1}\left(\mathbf{q}\right)\left(\mathbf{p}+\mathbf{b}_A\right)+V_h\left(\mathbf{q},t\right)-c_A\left(\mathbf{q},t\right).} \tag{3.46}$$

where $-c_A$ follows directly from the Legendre transformation. If $\mathbf{M}=\mathbf{M}\left(\mathbf{q}\right)$ has no explicit time dependence, then

$$\boxed{\nabla_{\mathbf{p}}H=\mathbf{M}^{-1}\left(\mathbf{p}+\mathbf{b}_A\right)=\dot{\mathbf{q}},} \tag{3.47}$$

$$\boxed{\frac{\partial H}{\partial q_i}=-\frac{1}{2}\dot{\mathbf{q}}^T\mathbf{M}_{,i}\dot{\mathbf{q}}+\mathbf{b}_{A,i}^T\dot{\mathbf{q}}+\frac{\partial V_h}{\partial q_i}-\frac{\partial c_A}{\partial q_i},} \tag{3.48}$$

and, at fixed $(\mathbf{q},\mathbf{p})$,

$$\boxed{\left.\frac{\partial H}{\partial t}\right|_{\mathbf{q},\mathbf{p}}=\dot{\mathbf{q}}^T\partial_t\mathbf{b}_A+\partial_t V_h-\partial_t c_A.} \tag{3.49}$$

3.5.3. *Equivalence with the General Euler–Lagrange Equations.* In the absence of external ports, the standard nonautonomous canonical equations still hold:

$$\dot{\mathbf{q}}=\nabla_{\mathbf{p}}H,\qquad\dot{\mathbf{p}}=-\nabla_{\mathbf{q}}H. \tag{3.50}$$

From

$$\dot{\mathbf{p}}=\mathbf{M}\ddot{\mathbf{q}}+\dot{\mathbf{M}}\dot{\mathbf{q}}-\dot{\mathbf{b}}_A \tag{3.51}$$

one recovers

$$\boxed{\mathbf{M}\ddot{\mathbf{q}}+\mathbf{g}_M+\mathbf{g}_f+\nabla_{\mathbf{q}}V_h=\mathbf{0},} \tag{3.52}$$

where

$$\boxed{\mathbf{g}_f=-\dot{\mathbf{b}}_A+\left(\nabla_{\mathbf{q}}\mathbf{b}_A\right)^T\dot{\mathbf{q}}-\nabla_{\mathbf{q}}c_A.} \tag{3.53}$$

This result agrees with the prescribed-flow Euler–Lagrange equation in Section 2. Both $\mathbf{g}_M$ and $\mathbf{g}_f$ are generated automatically by the Legendre transformation and therefore must not be added again as separate generalized forces.

3.5.4. *Exact Representation and Backbone–Port Representation.* When $\mathbf{b}_A, c_A$ are retained, define[17, 29]

$$\mathbf{z} = \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix}, \tag{3.54}$$

The complete forced system is

$$\boxed{\dot{\mathbf{z}} = \mathbf{J}\nabla_{\mathbf{z}} H\left(\mathbf{z}, t\right) + \mathbf{B}_{port}\left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{H} + \mathbf{Q}_{\partial}\right).} \tag{3.55}$$

In the exact representation, $\mathbf{g}_f$ is already contained in the Hamiltonian, so $\mathbf{Q}_{flow}^{H}$ does not include an additional $-\mathbf{g}_f$.

In the backbone–port representation, take

$$\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}} \tag{3.56}$$

and define

$$H_{sk}\left(\mathbf{q}, \mathbf{p}_0, t\right) = \frac{1}{2}\mathbf{p}_0^T \mathbf{M}^{-1}\mathbf{p}_0 + V_h\left(\mathbf{q}, t\right), \tag{3.57}$$

The prescribed-flow cross term is then transferred in its entirety to the port:

$$\boxed{\mathbf{Q}_{flow}^{sk} = \mathbf{Q}_{flow}^{H} - \mathbf{g}_f.} \tag{3.58}$$

The two representations generate the same configuration trajectories, but each must be used as a complete and internally consistent set:

$$\boxed{\left(\mathbf{p}, H, \mathbf{Q}_{flow}^{H}\right) \neq \left(\mathbf{p}_0, H_{sk}, \mathbf{Q}_{flow}^{sk}\right).} \tag{3.59}$$

The two variable sets and generating functions must remain paired. The shifted momentum $\mathbf{p}$ is used with the exact Hamiltonian H, whereas $\mathbf{p}_0$ is used with H_{sk}. The term $-\mathbf{g}_f$ must not be added again to the exact Hamiltonian representation.

3.5.5. *General Energy Rate.* The exact representation satisfies

$$\boxed{\frac{dH}{dt} = \frac{\partial H}{\partial t} + \dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{H} + \mathbf{Q}_{\partial}\right).} \tag{3.60}$$

The backbone–port representation satisfies

$$\boxed{\frac{dH_{sk}}{dt} = \partial_t V_h + \dot{\mathbf{q}}^T\left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{sk} + \mathbf{Q}_{\partial}\right).} \tag{3.61}$$

These two expressions give the energy rates under nonautonomous potentials, prescribed flow, and multiple ports, and remain dynamically equivalent in their respective variable representations.

3.6. **Range of Applicability and the Strict Kirchhoff Constraint Branch.**

3.6.1. *Local Orientation Coordinates.* Exponential coordinates form a local chart on $SO(3)$. To use standard canonical variables within the current chart, require, for every $S \in [0, L]$,

$$\| \mathbf{\Psi}(S)\mathbf{b} \| < \pi. \tag{3.62}$$

This is a coordinate-chart restriction rather than a small-rotation mechanical approximation. When a trajectory crosses the chart boundary, the orientation coordinates must be switched and the conjugate momentum updated consistently through the corresponding cotangent transformation.

3.6.2. *Strict Kirchhoff Primary Branch and Finite-Shear Validation Branch.* The geometrically exact parent model in Section 1 is a finite-shear Cosserat rod. To remain consistent with the Kirchhoff slender-catheter assumptions and to provide Sections 4 and 5 with a nonredundant canonical state, the primary structured-learning and KAM branches preferentially use a locally reduced representation of the strict shear-free constraint. Write the constraint as[17, 18]

$$\mathbf{c}(\mathbf{q}) = \mathbf{0}, \qquad \mathbf{G}(\mathbf{q}) = \frac{\partial \mathbf{c}}{\partial \mathbf{q}}, \tag{3.63}$$

If $\mathbf{G}$ has constant rank throughout the domain of interest, the constraint set is locally a smooth manifold. In the full coordinates, the equations are

$$\boxed{\dot{\mathbf{q}} = \nabla_{\mathbf{p}} H, \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{q}} H + \mathbf{Q}_{port} - \mathbf{G}^T \boldsymbol{\lambda}_c, \qquad \mathbf{c}(\mathbf{q}) = \mathbf{0}.} \tag{3.64}$$

The ideal constraint reaction satisfies $\dot{\mathbf{q}}^T \mathbf{G}^T \boldsymbol{\lambda}_c = 0$. To obtain a canonical system that can be used directly by the CHNN and standard symplectic analysis, introduce independent local coordinates $\mathbf{q} = \boldsymbol{\chi}(\mathbf{y})$ and define

$$\mathbf{C}(\mathbf{y}) = \frac{\partial \boldsymbol{\chi}}{\partial \mathbf{y}}, \qquad \mathbf{M}_r(\mathbf{y}) = \mathbf{C}^T \mathbf{M}(\boldsymbol{\chi}(\mathbf{y}))\, \mathbf{C}, \tag{3.65}$$

When $\mathbf{M}_r \succ 0$, construct the reduced Hamiltonian. The reduced momentum of the autonomous backbone is $\mathbf{p}_r = \mathbf{M}_r \dot{\mathbf{y}}$. If the reduced prescribed-flow model still contains a velocity-linear term, use instead $\mathbf{p}_r = \mathbf{M}_r \dot{\mathbf{y}} - \mathbf{b}_{A,r}$. Sections 4 and 5 use only this independent-coordinate canonical system; the differential–algebraic equations containing the multiplier $\boldsymbol{\lambda}_c$ cannot be treated directly as a standard Hamiltonian system on the full space. The current delivery contains no numerical record of the actual constraint-Jacobian rank, constraint residuals, independent-coordinate instantiation, or evidence that the fixed $N_{\text{diag}} = 2$ pilot belongs to this regular object. This paragraph therefore states a theoretical construction and scope condition, not a numerical certification for the pilot.

The finite-shear-stiffness Cosserat ordinary differential equations are retained for COMSOL comparison, numerical regularization, and constraint-convergence analysis. Data from this regularized branch may replace locally reduced data in CHNN training only when the shear strains, trajectories, energies, and principal frequencies converge to the Kirchhoff results as the shear stiffness increases. This strategy preserves the advantages of the complete parent model while keeping the primary computational object consistent with the original Kirchhoff research objective.

3.7. **Interfaces to CHNN, KAM, and Structure-Preserving Discretization.** Section 3 does not anticipate the neural-network construction or KAM theorem proofs developed in Sections 4 and 5. Instead, based on the HNN, dissipative/port-Hamiltonian network, statistical-learning–KAM, symplectic-algorithm, and parameterization-method literature used there, it specifies the unique and verifiable mathematical objects that connect the preceding and subsequent parts.[19–28]

BDF order 2 with `tsteps` and `tstepsstore=1` is used only to store same-solver-step states and to generate vector-field labels under Equation (3.13). The three-level time-step comparison tests discrete observability and convergence of this data contract, not symplecticity of BDF. Any

symplectic or KAM inference requires a separate symplectic map satisfying the assumptions in Section 5.

3.7.1. *Canonical Nondimensionalization, Complex Coordinates, and the Physical Hamiltonian Baseline.* A subsequent complex representation cannot add $\mathbf{q}$ and $\mathbf{p}$ directly when they have different physical dimensions and scales. Choose a stable reference state $(\mathbf{q}_*, \mathbf{p}_*)$, coordinate and momentum scale matrices $\mathbf{S}_q$ and $\mathbf{S}_p$, a time scale t_*, and an energy scale E_*, and define the action scale $\mathcal{A}_* = E_* t_*$. Require

$$\boxed{\mathbf{S}_q^T \mathbf{S}_p = \mathcal{A}_* \mathbf{I}.} \tag{3.66}$$

Define

$$\mathbf{q} = \mathbf{q}_* + \mathbf{S}_q \mathbf{Q}, \qquad \mathbf{p} = \mathbf{p}_* + \mathbf{S}_p \mathbf{P}, \qquad t = t_* \tau, \qquad H = E_* \widehat{H}. \tag{3.67}$$

The scale-compatibility condition then gives

$$\frac{d\mathbf{Q}}{d\tau} = \nabla_{\mathbf{P}} \widehat{H}, \qquad \frac{d\mathbf{P}}{d\tau} = -\nabla_{\mathbf{Q}} \widehat{H}, \tag{3.68}$$

so the nondimensional variables still satisfy the standard canonical Hamilton equations. Define the complex coordinates[17, 29]

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{3.69}$$

The real and imaginary parts of $\boldsymbol{\xi}$ are canonically paired configuration and momentum variables with compatible scales. A nontrivial real-valued Hamiltonian is generally written as $\widehat{H}\left(\boldsymbol{\xi}, \overline{\boldsymbol{\xi}}\right) \in \mathbb{R}$, rather than as a real-valued holomorphic function of $\boldsymbol{\xi}$ alone.[23, 24]

Section 3 supplies Section 4 with the analytical physical baseline

$$\widehat{H}_{phys} = \widehat{H}_{cons}, \tag{3.70}$$

The real-valued readout in Section 4 should use the physical-residual form

$$\boxed{\widehat{H}_\lambda = \widehat{H}_{phys} + \Delta \widehat{H}_\lambda.} \tag{3.71}$$

The known mass matrix, Cosserat strain energy, and identified conservative magnetic and elastic-contact potentials are not relearned by the network. The term $\Delta \widehat{H}_\lambda$ represents only parameter corrections, conservative corrections induced by Galerkin truncation, and other unmodeled conservative effects. Because a Hamiltonian inferred from vector-field data is identifiable only up to an additive constant, a gauge must be fixed, for example $\widehat{H}_\lambda(\mathbf{z}_*) = 0$ or zero mean over the training domain.[19, 21]

3.7.2. *Backbone–Port Structure and Consistent Data Representations.* In the autonomous-backbone representation with frozen actuation parameters $\boldsymbol{\lambda}(t) \equiv \overline{\boldsymbol{\lambda}}$,

$$V_{drv} = 0, \qquad H_{sk} = H_{cons}, \qquad \partial_t H_{sk} = 0, \tag{3.72}$$

and

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J} \nabla_{\mathbf{z}_0} H_{cons} + \mathbf{B}_{port} \left(\mathbf{Q}_{diss}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in} + \mathbf{Q}_{flow}^{sk} + \mathbf{Q}_{\partial} \right).} \tag{3.73}$$

Section 4 should parameterize or supervise the ports separately rather than allowing a single unstructured residual to absorb every physical contribution:

$$\mathbf{Q}_{port,\theta} = \mathbf{Q}_{diss,\mu}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{\partial} + \mathbf{r}_\eta. \tag{3.74}$$

Here, $\mathbf{Q}_{diss,\mu}^{bulk}$ represents only the passive bulk/drag component with nonpositive power and excludes contact; $\mathbf{Q}_{contact}$ remains a separate contact port; magnetic control, background-flow

input, and boundary actuation remain separate ports; and $\mathbf{r}_\eta$ represents only the residual remaining after the known model and known ports have been removed. This hierarchy is consistent with the structured modeling principles of dissipative SymODEN and port-Hamiltonian neural networks.[20, 21]

For nonfrozen or pulsatile operating conditions, $H_{sk} = H_{cons} + V_{drv}$. The following two representations are mutually exclusive:

1. If the complete H_{sk} is used as the generator, the same conservative actuation force must not also enter $\mathbf{Q}_{in}$;
2. If the autonomous H_{cons} is used as the generator, set $\mathbf{Q}_{drv} = -\nabla_{\mathbf{q}} V_{drv}$ and include it exactly once in $\mathbf{Q}_{in}$.

Under a general prescribed flow, use $\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A$ and the corresponding nonautonomous Hamiltonian; the autonomous backbone uses $\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}$. The two data types must be distinguished by a *representation* label. The backbone and nonautonomous data supplied to Section 4 are, respectively,

$$\mathcal{D}_{sk} = \left(\mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \dot{\mathbf{z}}_0, \widehat{H}_{cons}, \nabla \widehat{H}_{cons}, \mathbf{Q}^{bulk}_{diss}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}^{sk}_{flow}, \mathbf{Q}_{\partial}\right), \tag{3.75}$$

$$\mathcal{D}_{na} = \left(\mathbf{z}, \dot{\mathbf{z}}, t, H, \nabla H, \partial_t H, \mathbf{Q}^{bulk}_{diss}, \mathbf{Q}_{contact}, \mathbf{Q}_{in}, \mathbf{Q}^{H}_{flow}, \mathbf{Q}_{\partial}\right). \tag{3.76}$$

The statistical-learning–KAM interface preferentially uses derivative-supervised autonomous conservative samples and does not interpret the sum of trajectory loss, dissipation loss, and regularization terms directly as the KAM perturbation size.[22]

3.7.3. *Stable Equilibrium, Modal Canonicalization, and the Integrable Reference System.* The subsequent KAM analysis starts from the autonomous conservative backbone $H_{cons}(\mathbf{q}, \mathbf{p}_0)$, without assuming that it is globally integrable. Choose an equilibrium $(\mathbf{q}_*, \mathbf{0})$ satisfying

$$\nabla_{\mathbf{q}} V_{cons}(\mathbf{q}_*) = \mathbf{0}, \qquad \mathbf{M}_* = \mathbf{M}(\mathbf{q}_*) \succ 0, \qquad \mathbf{K}_* = \nabla^2_{\mathbf{q}} V_{cons}(\mathbf{q}_*) \succ 0. \tag{3.77}$$

Solve the generalized eigenvalue problem

$$\mathbf{K}_* \boldsymbol{\phi}_j = \omega_j^2 \mathbf{M}_* \boldsymbol{\phi}_j, \tag{3.78}$$

and choose the modal matrix $\boldsymbol{\Phi}$ such that

$$\boldsymbol{\Phi}^T \mathbf{M}_* \boldsymbol{\Phi} = \mathbf{I}, \qquad \boldsymbol{\Phi}^T \mathbf{K}_* \boldsymbol{\Phi} = \boldsymbol{\Omega}^2. \tag{3.79}$$

The linear canonical transformation

$$\mathbf{q} - \mathbf{q}_* = \boldsymbol{\Phi} \mathbf{Q}_m, \qquad \mathbf{p}_0 = \boldsymbol{\Phi}^{-T} \mathbf{P}_m \tag{3.80}$$

reduces the quadratic Hamiltonian to

$$H_2 = \frac{1}{2}\left(\mathbf{P}_m^T \mathbf{P}_m + \mathbf{Q}_m^T \boldsymbol{\Omega}^2 \mathbf{Q}_m\right) = \sum_{j=1}^{n} \omega_j I_j, \tag{3.81}$$

where

$$x_j = \sqrt{\omega_j} Q_{m,j}, \qquad y_j = \frac{P_{m,j}}{\sqrt{\omega_j}}, \qquad I_j = \frac{x_j^2 + y_j^2}{2}. \tag{3.82}$$

With quadratic terms alone, the frequencies are independent of the actions, and hence

$$\frac{\partial^2 H_2}{\partial \mathbf{I}^2} = \mathbf{0}, \tag{3.83}$$

Thus the linear harmonic oscillator is isochronously degenerate. The condition $\mathbf{K}_* \succ 0$ cannot replace the Kolmogorov nondegeneracy condition; nondegeneracy must be supplied by geometric or material nonlinearities, conservative magnetic potentials, or other higher-order conservative terms.

If H_{cons} is analytic in the neighborhood of interest, or has the regularity required by the selected finitely differentiable KAM theorem, and the linear frequencies avoid low-order resonances to the required order, a finite-order Birkhoff normal form can be constructed:[18, 25, 29]

$$H_{cons} \circ \mathbf{\Psi} = H_* + \boldsymbol{\omega} \cdot \mathbf{I} + Z_4(\mathbf{I}) + \cdots + Z_r(\mathbf{I}) + R_{r+1}(\mathbf{I}, \boldsymbol{\theta}). \tag{3.84}$$

Define

$$\boxed{H_0(\mathbf{I}) = H_* + \boldsymbol{\omega} \cdot \mathbf{I} + Z_4(\mathbf{I}) + \cdots + Z_r(\mathbf{I}), \qquad \varepsilon H_1 = R_{r+1}.} \tag{3.85}$$

The lowest nontrivial case commonly has

$$Z_4(\mathbf{I}) = \frac{1}{2}\mathbf{I}^T \mathbf{A} \mathbf{I}, \tag{3.86}$$

The Kolmogorov nondegeneracy condition over the action domain under consideration then requires

$$\boxed{\det\left(\frac{\partial^2 H_0}{\partial \mathbf{I}^2}\right) \neq 0,} \tag{3.87}$$

Near the origin, this reduces to $\det \mathbf{A} \neq 0$. If this matrix is singular, classical Kolmogorov KAM cannot be claimed; one must increase the normal-form order, choose another reference family, or invoke and verify an isoenergetic or degenerate KAM theorem.

On a fixed complex neighborhood or Sobolev domain $\mathcal{D}$, with a common function norm $\| \cdot \|_{\mathcal{X}(\mathcal{D})}$, write the nondimensional Hamiltonian as

$$\widehat{H}_{cons} = \widehat{H}_0 + \varepsilon P, \qquad \| P \|_{\mathcal{X}(\mathcal{D})} = 1, \tag{3.88}$$

and define

$$\boxed{\varepsilon = \| \widehat{H}_{cons} - \widehat{H}_0 \|_{\mathcal{X}(\mathcal{D})}.} \tag{3.89}$$

The frequency vector must also satisfy the corresponding Diophantine condition. If a low-order resonance is present, the analysis must use a resonant normal form and an appropriate theorem; the resonant term cannot simply be absorbed into an ordinary small perturbation.

3.7.4. *Learning Error, Discretization Error, and the KAM Smallness Budget.* To place the learning error from Section 4 and the symplectic-discretization error from Section 5 in one mathematical chain, all errors must be measured on the same domain and in compatible function norms. At the continuous Hamiltonian level, first define

$$\delta_{model} = \| H_{cons} - H_0 \|_{\mathcal{X}}, \qquad \delta_{learn} = \| H_\lambda - H_{cons} \|_{\mathcal{X}}, \tag{3.90}$$

so that

$$\boxed{\| H_\lambda - H_0 \|_{\mathcal{X}} \leq \delta_{model} + \delta_{learn}.} \tag{3.91}$$

At the discrete level, do not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Let Φ_h^H denote the exact flow of Hamiltonian H over the time interval h, let $F_{\lambda,h}$ denote the conservative symplectic one-step map actually used in Section 5, and set the unperturbed reference map to $F_{0,h} = \Phi_h^{H_0}$. On a fixed compact domain, define in the C^r or analytic norm appropriate to the selected symplectic-map KAM theorem[18, 25]

$$\delta_{alg} = \| F_{\lambda,h} - \Phi_h^{H_\lambda} \|, \qquad \delta_{map} = \| F_{\lambda,h} - F_{0,h} \|. \tag{3.92}$$

The triangle inequality gives

$$\boxed{\delta_{map} \leq \delta_{alg} + \| \Phi_h^{H_\lambda} - \Phi_h^{H_{cons}} \| + \| \Phi_h^{H_{cons}} - \Phi_h^{H_0} \|.} \tag{3.93}$$

If the vector field depends locally Lipschitz-continuously on Hamiltonian perturbations and the flow remains in the same domain over the time interval considered, the last two terms can be bounded by $C_{learn}(h)\delta_{learn}$ and $C_{model}(h)\delta_{model}$, respectively. Hence,

$$\boxed{\delta_{map} \leq \delta_{alg} + C_{learn}(h)\delta_{learn} + C_{model}(h)\delta_{model}.} \tag{3.94}$$

If backward error analysis additionally constructs a modified Hamiltonian for a specified step size and finite time scale, an equivalent modified-Hamiltonian budget may be used. The domain of validity, truncation order, and remainder bound must accompany this object; global exact existence cannot be inferred solely from the statement that the integrator is symplectic.

Persistence of invariant tori can be concluded only when the continuous or discrete budget is below the concrete threshold of the selected KAM theorem, which depends on the analytic domain, Diophantine constants, nondegeneracy margin, dimension, and step size. The statistical-learning result of Chen–Matsubara–Yaguchi directly controls only the conservative HNN branch under its assumptions. Transferring a comparable probabilistic bound to the CHNN requires a real-valued readout and a corresponding complexity analysis of the function class.[22]

Quadratic drag, contact damping and friction, nonpotential magnetic actuation, background-flow input, boundary input, and contact activation or release are not included in the preceding classical autonomous-Hamiltonian perturbation. Their contributions to the complete safe operating domain must instead be treated through energy balance, port gains, structural defects, Lyapunov indicators, or a specialized theory of dissipative invariant tori.

3.7.5. *Symplectic Maps, Birkhoff Structure, and the Torus Equation.* For the autonomous conservative branch, a structure-preserving integrator produces the one-step map

$$F_h^{cons} \neq \Phi_h^{cons} \text{ in general}, \qquad \delta_{alg} := \|F_h^{cons} - \Phi_h^{cons}\|. \tag{3.95}$$

Here F_h is used below as shorthand for the actual numerical conservative map F_h^{cons}; Φ_h^{cons} remains the exact flow of the continuous Hamiltonian, and δ_{alg} is their algorithmic difference, so a general integrator must not be identified with the exact flow. Section 5 first verifies whether F_h is an exact symplectic map and then applies the KAM conditions for symplectic maps and symplectic algorithms due to Shang together with the parameterization method of Haro et al.[25–28] The torus embedding $K : \mathbb{T}^d \to \mathcal{M}$ satisfies[28]

$$\boxed{F_h\left(K\left(\boldsymbol{\theta}\right)\right) = K\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right).} \tag{3.96}$$

The invariance residual of a numerical approximation K_N is

$$E_N\left(\boldsymbol{\theta}\right) = F_h\left(K_N\left(\boldsymbol{\theta}\right)\right) - K_N\left(\boldsymbol{\theta} + \boldsymbol{\omega}\right). \tag{3.97}$$

For the complete data vector field, the known conservative backbone and known ports must first be removed:

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla H_\lambda - \mathbf{B}_{port}\left(\mathbf{Q}_{diss,\mu}^{bulk} + \mathbf{Q}_{contact} + \mathbf{Q}_{in,\nu} + \mathbf{Q}_{flow,\kappa} + \mathbf{Q}_{\partial}\right).} \tag{3.98}$$

The "self-adjoint Birkhoff-system representation" used here and the "Hamiltonian Birkhoff normal form" used to construct H_0 in Section 3.7.3 are distinct concepts. The former concerns the variational self-adjointness and $K\left(\mathbf{z}, t\right)$-symplectic structure of a general first-order system, whereas the latter is the Hamiltonian normal form near an elliptic equilibrium. Section 5 tests only whether X_{res} admits a regular self-adjoint Birkhoff-system representation. If the closure, time-compatibility, or nondegeneracy conditions fail, X_{res} remains a general residual and the complete open system cannot be called a strict $K\left(\mathbf{z}, t\right)$-symplectic system. Explicit time dependence alone also does not necessarily imply a nonstandard symplectic structure: in appropriate canonical variables and extended phase space, the exact nonautonomous Hamiltonian branch should still be treated preferentially with standard symplectic theory.[18, 25]

3.7.6. *Structural Decomposition of the Difference between the Primary COMSOL Training Model and High-Fidelity Models.* Section 3 treats the Kirchhoff-consistent COMSOL branch as the primary training dynamics X_K, and the Weak Form, Timoshenko, or solid model as the high-fidelity reference X_{HF}. Define the paired-condition model discrepancy

$$\Delta X_{HF} = X_{HF} - X_K. \tag{3.99}$$

This discrepancy cannot be inserted wholesale into the Hamiltonian. First test its conservative closure in canonical variables. If a real-valued ΔH_{HF} exists such that

$$\Delta X_{HF}^{cons} = \mathbf{J}\nabla\Delta H_{HF}, \tag{3.100}$$

then that component may enter the conservative correction head in Section 4. If a momentum port satisfies

$$\nabla_{\mathbf{P}}\widehat{H}^T\Delta\mathbf{Q}_{HF}^{diss,bulk} \le 0, \tag{3.101}$$

it may enter the bulk/drag dissipative correction head. Contact-model discrepancy remains in the separate $\Delta\mathbf{Q}_{HF}^{contact}$ channel and must not be merged into that head. Components that do not satisfy closure, power-sign, or smoothness conditions remain in the general model residual $\mathbf{r}_{HF}$. Thus,

$$\boxed{\Delta X_{HF} = \mathbf{J}\nabla\Delta H_{HF} + \mathbf{B}_{port}\Delta\mathbf{Q}_{HF}^{diss,bulk} + \mathbf{B}_{port}\Delta\mathbf{Q}_{HF}^{contact} + \mathbf{B}_{port}\Delta\mathbf{Q}_{HF}^{port} + \mathbf{r}_{HF}.} \tag{3.102}$$

This decomposition is a structural projection to be identified and validated, not an identity that holds automatically for an arbitrary high-fidelity discrepancy. The model error used in KAM analysis consists only of the conservatively closed component and its bound in the common function norm. Shear dissipation, contact switching, and local solid effects must not be included in H_1 without proof.

3.7.7. *Interfaces for Structured Learning and Stability Analysis.* The preceding derivations yield three interfaces required by the subsequent structured-learning and stability analyses:

$$\begin{aligned}\mathcal{I}_{CHNN} = \big(&\mathbf{Q}, \mathbf{P}, \boldsymbol{\xi}, \widehat{H}_{phys}, \nabla\widehat{H}_{phys}, \mathbf{Q}_{diss}^{bulk}, \mathbf{Q}_{contact},\\ &\mathbf{Q}_{in}, \mathbf{Q}_{flow}, \mathbf{Q}_{\partial}, model_level, representation\big),\end{aligned} \tag{3.103}$$

$$\mathcal{I}_{KAM} = \left(H_0, H_1, \varepsilon, \boldsymbol{\omega}\left(\mathbf{I}\right), \partial_{\mathbf{I}}\boldsymbol{\omega}, \gamma, \tau, \mathcal{D}, \mathcal{X}\right), \tag{3.104}$$

$$\mathcal{I}_{geom} = \left(\mathbf{J}, F_{0,h}, F_{\lambda,h}, X_{res}, E_N, \delta_{model}, \delta_{learn}, \delta_{alg}, \delta_{map}\right). \tag{3.105}$$

Each object is traceable to the physical model and Galerkin outputs in Sections 1 and 2 and to the canonicalization performed in this section. If an object has not yet been constructed or verified numerically, subsequent sections may list it only as a condition to be tested and must not present the conditional interface as an established theorem conclusion.

3.8. **Section Summary.** Using the complete joint coordinates for translation and cross-sectional orientation, the canonical momentum and Hamiltonian of the autonomous conservative backbone are

$$\boxed{\mathbf{p}_0 = \mathbf{M}\dot{\mathbf{q}}, \qquad H_{cons} = \frac{1}{2}\mathbf{p}_0^T\mathbf{M}^{-1}\mathbf{p}_0 + V_{cons},} \tag{3.106}$$

and the forced–dissipative canonical equations are

$$\boxed{\dot{\mathbf{z}}_0 = \mathbf{J}\nabla H_{cons} + \mathbf{B}_{port}\mathbf{Q}_{port}, \qquad \frac{dH_{cons}}{dt} = \dot{\mathbf{q}}^T\mathbf{Q}_{port}.} \tag{3.107}$$

The contribution of the configuration-dependent mass matrix in the Hamiltonian momentum equation is equivalent to the geometric-inertia vector $\mathbf{g}_M$ in Section 2. A nonlinear conservative

potential can be represented as a path-independent line integral of the restoring force; under equivalent viscous damping, the energy rate reduces to the sum of external-input power and dissipative power.

Under a general prescribed flow, the canonical momentum and Hamiltonian become

$$\boxed{\mathbf{p} = \mathbf{M}\dot{\mathbf{q}} - \mathbf{b}_A, \qquad H = \frac{1}{2}\left(\mathbf{p} + \mathbf{b}_A\right)^T \mathbf{M}^{-1}\left(\mathbf{p} + \mathbf{b}_A\right) + V_h - c_A.} \tag{3.108}$$

This general representation retains cross-sectional orientation, independent torsion, rotary inertia, added mass, and external ports while unifying the autonomous backbone, prescribed-flow dynamics, and forced–dissipative dynamics. Canonical nondimensionalization, complex coordinates, a physical-residual Hamiltonian, Birkhoff normal form, a unified KAM error budget, and the three interfaces $\mathcal{I}_{CHNN}$, $\mathcal{I}_{KAM}$, and $\mathcal{I}_{geom}$ then enable Section 3 to provide the subsequent analyses with mutually consistent and traceable mathematical objects for structured CHNN learning, symplectic discretization, parameterized-torus validation, and safe-domain computation.

4. Structured Hamiltonian Neural Networks with Complex-Valued Representations

4.1. **Learning Object and Upstream Interfaces.** This section constructs a Hamiltonian neural network with complex-valued representations on the reduced canonical finite-dimensional system and connects it to KAM stability analysis. To keep the physical objects unambiguous, a single complex-valued scalar is not used to generate conservative, dissipative, and driven dynamics simultaneously. Section 3 has already provided the independent canonical variables after local reduction of the strict Kirchhoff constraint, canonical nondimensionalization, the physical Hamiltonian baseline, and mutually exclusive port accounting. The CHNN is defined only on these objects.

Let the reduced nondimensional canonical state be

$$\mathbf{Z} = \begin{bmatrix} \mathbf{Q} \\ \mathbf{P} \end{bmatrix} \in \mathbb{R}^{2d}, \qquad \mathbf{J} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{I} & \mathbf{0} \end{bmatrix}, \tag{4.1}$$

and define the complex coordinate

$$\boxed{\boldsymbol{\xi} = \frac{\mathbf{Q} + i\mathbf{P}}{\sqrt{2}}, \qquad \overline{\boldsymbol{\xi}} = \frac{\mathbf{Q} - i\mathbf{P}}{\sqrt{2}}.} \tag{4.2}$$

Here, $\mathbf{Q}$ and $\mathbf{P}$ have undergone a canonical scaling satisfying $\mathbf{S}_q^T \mathbf{S}_p = E_{ref} t_{ref} \mathbf{I}$, so they may be combined as the real and imaginary parts of a complex variable. The dimensional variables $\mathbf{q}$ and $\mathbf{p}$ are not added directly before canonical nondimensionalization.

The COMSOL data also carry

$$model_level \in \{K, WF, T, S\}, \qquad representation \in \{sk, na, red\}, \tag{4.3}$$

which identify, respectively, the Kirchhoff-consistent Beam model, strict weak-form model, Timoshenko model, and solid model, together with the autonomous-backbone, exact nonautonomous, and engineering-reduction representations. Data with different labels are not mixed without a documented compatibility analysis.

4.2. **Conservative Hamiltonian Head and Complex-Valued Internal Representation.**

4.2.1. *Real-Valued Physical Hamiltonian.* A standard HNN generates the conservative vector field from the gradient of a real-valued scalar Hamiltonian.[19] We use

$$\boxed{\widehat{H}_\lambda\left(\mathbf{Z}\right) = \widehat{H}_{phys}\left(\mathbf{Z}\right) + \Delta\widehat{H}_\lambda\left(\mathbf{Z}\right) \in \mathbb{R},} \tag{4.4}$$

where $\widehat{H}_{phys}$ is supplied by the mass matrix, Cosserat/Kirchhoff strain energy, identified conservative magnetic potential, and elastic contact potential for a fixed active set from Section 3. The network learns only material-parameter discrepancies, conservative corrections caused by

Galerkin truncation, and other conservatively closed model discrepancies. To remove the additive gauge freedom, impose

$$\widehat{H}_\lambda\left(\mathbf{Z}_{ref}\right)=0 \tag{4.5}$$

or require zero mean over the training domain.

4.2.2. *Complex-Valued Backbone and Real-Valued Readout.* Let

$$\mathbf{h}_\theta=\mathcal{N}_\theta^{\mathbb{C}}\left(\boldsymbol{\xi},\overline{\boldsymbol{\xi}}\right),\qquad \Delta\widehat{H}_\lambda=\mathcal{R}_{H,\lambda}\left(\mathbf{h}_\theta\right)\in\mathbb{R}. \tag{4.6}$$

A complex linear layer represents amplitude scaling and phase rotation simultaneously through its real–imaginary block structure. For noncircular phase-space data, both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ are retained as inputs, or an equivalent augmented representation is used. A practical CVNN may be treated as a real-differentiable mapping $\mathbb{R}^{2d}\to\mathbb{R}$; global holomorphy of the network is not required.[23, 24] A nonconstant real-valued Hamiltonian generally cannot be a purely holomorphic function on a connected open set, so the physical readout may depend on both $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$.

The primary theoretical branch uses activation functions sufficiently smooth in the real and imaginary parts, for example

$$\phi_{split}(z)=\tanh(Rez)+i\tanh(Imz). \tag{4.7}$$

modReLU, CReLU, and zReLU may be included in engineering ablations, but their piecewise smoothness or nondifferentiability at boundaries prevents their direct use in statistical-learning–KAM theorems requiring higher-order smoothness. Complex weights use complex Glorot or semi-unitary initialization; the Hamiltonian backbone does not use BatchNorm based on batch statistics by default.[23, 24]

4.2.3. *Widely Linear Complex Layers and Real-Affine Completeness.* To retain an amplitude–phase inductive bias while preserving expressive compatibility with general real networks, the complex backbone uses a widely linear layer. Let

$$\mathbf{z}_{out}=\mathbf{W}\mathbf{z}+\mathbf{V}\overline{\mathbf{z}}+\mathbf{b} \tag{4.8}$$

where $\mathbf{z}=\mathbf{x}+i\mathbf{y}$, $\mathbf{W}=\mathbf{A}+i\mathbf{B}$, $\mathbf{V}=\mathbf{C}+i\mathbf{D}$, and $\mathbf{b}=\mathbf{b}_R+i\mathbf{b}_I$. Then

$$\mathbf{x}_{out}=(\mathbf{A}+\mathbf{C})\,\mathbf{x}+(-\mathbf{B}+\mathbf{D})\,\mathbf{y}+\mathbf{b}_R \tag{4.9}$$

$$\mathbf{y}_{out}=(\mathbf{B}+\mathbf{D})\,\mathbf{x}+(\mathbf{A}-\mathbf{C})\,\mathbf{y}+\mathbf{b}_I \tag{4.10}$$

Proposition 4.1 (Real-affine completeness)**.** *For any real affine mapping*

$$\mathbf{x}_{out}=\mathbf{M}_{11}\mathbf{x}+\mathbf{M}_{12}\mathbf{y}+\mathbf{c}_1 \tag{4.11}$$

$$\mathbf{y}_{out}=\mathbf{M}_{21}\mathbf{x}+\mathbf{M}_{22}\mathbf{y}+\mathbf{c}_2 \tag{4.12}$$

there exist unique $\mathbf{W},\mathbf{V},\mathbf{b}$ *for which the widely linear complex layer is exactly equivalent, with the explicit construction*

$$\mathbf{A}=\frac{\mathbf{M}_{11}+\mathbf{M}_{22}}{2},\qquad \mathbf{C}=\frac{\mathbf{M}_{11}-\mathbf{M}_{22}}{2} \tag{4.13}$$

$$\mathbf{B}=\frac{\mathbf{M}_{21}-\mathbf{M}_{12}}{2},\qquad \mathbf{D}=\frac{\mathbf{M}_{21}+\mathbf{M}_{12}}{2} \tag{4.14}$$

$$\mathbf{b}=\mathbf{c}_1+i\mathbf{c}_2 \tag{4.15}$$

Proof. Substituting the stated construction into the real and imaginary equations recovers the four blocks $\mathbf{M}_{11},\mathbf{M}_{12},\mathbf{M}_{21},\mathbf{M}_{22}$. Conversely, solving the four linear block equations gives uniqueness. □

When $\mathbf{V} = \mathbf{0}$, the real block matrix associated with a standard complex-linear layer must commute with the standard complex structure and is therefore a proper subclass of general real linear mappings. We use standard complex-linear layers as a stronger amplitude–phase structural bias and widely linear layers as the default theoretical model when general real-affine expressivity must be retained. This result establishes only a function-class inclusion and does not by itself imply fewer parameters or higher sample efficiency for the complex-valued model.

4.2.4. *Embedding of Real MLPs and Conditional Inheritance of Sobolev Approximation.*

Theorem 4.1 (Layerwise embedding of the real MLP class)**.** *Let a real-valued multilayer perceptron take* $(\mathbf{Q}, \mathbf{P})$ *as input and use a smooth hidden-layer activation satisfying* $\sigma(0) = 0$*. Then there exists a complex network taking* $\boldsymbol{\xi} = (\mathbf{Q} + i\mathbf{P})/\sqrt{2}$ *and* $\overline{\boldsymbol{\xi}}$ *as inputs and using widely linear layers and split activations, whose real-valued output is identical to that of the real MLP.*

Proof. The first layer treats $(Re\boldsymbol{\xi}, Im\boldsymbol{\xi})$ as the realified input and uses Proposition 4.1 to realize the target real affine mapping while setting the target imaginary preactivation to zero. Since $\sigma(0) = 0$, the imaginary channel remains zero after the split activation. Starting from the second layer, the block-diagonal special case of Proposition 4.1 realizes each affine layer of the original real network. The final real-valued readout is therefore identical to that of the original MLP. □

Corollary 4.1 (Conditional inheritance of Sobolev approximation)**.** *Let* $\Omega \subset \mathbb{R}^{2d}$ *be a bounded Lipschitz domain,* $s \in \mathbb{N}$*, and* $1 \le r < \infty$*. If the class of real MLPs with the chosen activation is dense in* $W^{s,r}(\Omega)$*, then the class of real-valued readouts of the preceding widely linear complex networks is also dense in* $W^{s,r}(\Omega)$*.*

This corollary transfers an existing real-network density result and does not remove the original assumptions on the activation, domain, network width, or derivative approximation.[35] The standard complex-linear subclass is constrained by its block structure and does not automatically inherit the same Sobolev-density conclusion.

The theoretical advantages established by Zhang et al. for complex-reaction networks depend on their specific complex reaction structure, activation, and problem setting.[34] They cannot be generalized directly into the claim that an arbitrary complex-valued CHNN must outperform a real-valued HNN. Here, that work motivates only the possibility of a task-relevant inductive bias from complex structure; actual sample-efficiency and error advantages are evaluated through parameter-matched ablation studies.

4.2.5. *Physical Limits of the Complex Representation.* Consider the complex scalar representation

$$\widetilde{\mathcal{H}} = H + iG \tag{4.16}$$

This is retained only as a special representation layer. A residual vector field may be represented by a real auxiliary potential G only if the corresponding one-form is closed and locally integrable. General quadratic drag, contact friction, magnetic control input, and background-flow effects do not satisfy this condition. The imaginary part is not interpreted directly as physical dissipation; physical energy is always given by the real-valued $\widehat{H}_\lambda$.

4.3. **Open-Port Dynamics and Branch Networks.** The metriplectic framework of Coquinot–Morrison requires a Poisson bracket, a symmetric positive-semidefinite metric bracket, energy and entropy variables, and the corresponding degeneracy conditions and entropy-production relation.[30] The current catheter model does not establish an independent thermodynamic entropy field or these degeneracy structures. Therefore, the open-port model is not formulated metriplectically; that framework instead delineates the conditions required by future thermo-viscoelastic extensions and a rigorous entropy-production mechanism.

The complete structured model on the autonomous backbone is

$$\boxed{\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_\lambda(\mathbf{Z}) + \mathbf{B}_{port}\left(\widehat{\mathbf{Q}}^{bulk}_{diss,\mu} + \widehat{\mathbf{Q}}_{contact} + \widehat{\mathbf{Q}}_{in,\nu} + \widehat{\mathbf{Q}}_{flow,\kappa} + \widehat{\mathbf{Q}}_{\partial} + \widehat{\mathbf{r}}_\eta\right),} \tag{4.17}$$

where

$$\mathbf{B}_{port} = \begin{bmatrix}\mathbf{0}\\ \mathbf{I}\end{bmatrix}. \tag{4.18}$$

Here $\mathbf{B}_{port} \in \mathbb{R}^{2d\times d}$ embeds a physical port input $\mathbf{u} \in \mathbb{R}^d$ only into the momentum-update channel. The full-state port vector $\mathbf{U}_{port} = \mathbf{B}_{port}\mathbf{u} \in \mathbb{R}^{2d}$ must not be conflated with the physical input $\mathbf{u}$.

The present work instantiates neither a sampled-data controller nor an online adaptive law, so these equations do not support a sampled-control stability claim. The BDF data-output interval Δt_{data}, a future control-sampling period Δt_{ctrl}, and the Birkhoff numerical-map step h_{Birk} are three independent quantities and are not assumed equal. Any future adaptive extension must register the regressor $\mathbf{Y}(\mathbf{Z}, t) \in \mathbb{R}^{d\times r}$, parameter vector $\widehat{\boldsymbol{\theta}} \in \mathbb{R}^r$, gain dimensions, transpose or outer-product convention, and parameter-projection constraint together. Its output must first form $\mathbf{u} \in \mathbb{R}^d$ and then enter the momentum channel through $\mathbf{B}_{port}\mathbf{u}$; it cannot be injected directly into $\dot{\mathbf{Q}}$. No adaptive-stability conclusion is verifiable until those objects and their implementation are supplied.

The branches have distinct physical meanings.

1. $\widehat{\mathbf{Q}}^{bulk}_{diss,\mu}$ represents only the passive bulk/drag component, excluding contact, whose power with respect to the structural mechanical energy is nonpositive. For the equivalent-viscous subclass, one may take

$$\widehat{\mathbf{Q}}^{bulk}_{diss,\mu} = -\mathbf{D}_\mu(\mathbf{Z})\nabla_{\mathbf{P}}\widehat{H}_\lambda, \qquad \mathbf{D}_\mu = \mathbf{L}_\mu\mathbf{L}_\mu^T \succcurlyeq 0, \tag{4.19}$$

   - so that $\nabla_{\mathbf{P}}\widehat{H}_\lambda^T\widehat{\mathbf{Q}}^{bulk}_{diss,\mu} \le 0$. Elastic contact not represented in the potential, contact switching, Coulomb friction, and impact enter the separate nonpotential $\widehat{\mathbf{Q}}_{contact}$ port; if elastic contact is already in the potential, it is not counted again, and contact damping or friction does not enter the bulk dissipation head.[20]
2. $\widehat{\mathbf{Q}}_{in,\nu}$ represents magnetic control and other nonfluid inputs not included in the potential. Magnetic actions derivable from a potential at fixed magnetic parameters enter $\widehat{H}_{phys}$. A time-dependent conservative magnetic potential may either enter the nonautonomous Hamiltonian or enter the input port once as $-\nabla V_{drv}$; the two representations are mutually exclusive.
3. $\widehat{\mathbf{Q}}_{flow,\kappa}$ represents the Froude–Krylov input, background-flow drag input, and reactive effects induced by the prescribed flow. The background-flow power is generally sign-indefinite and cannot be absorbed wholesale into a dissipation matrix.
4. $\widehat{\mathbf{Q}}_{\partial}$ is supplied by the known terminal force, terminal moment, and boundary actuation and is not relearned by the network in principle.
5. $\widehat{\mathbf{r}}_\eta$ represents only the residual remaining after the known physical backbone and all known ports have been removed.

This decomposition follows the separation of Hamiltonian, dissipation, and control in Dissipative SymODEN and the treatment of explicit time-dependent forcing and identifiability in port-Hamiltonian neural networks.[20, 21]

Numerical studies of pHNNs show that simultaneously identifying the Hamiltonian, time-dependent forcing, and damping from state trajectories alone can cause information leakage among branches, demonstrating that identifiability is not guaranteed by the architecture.[21] We therefore fix the physical Hamiltonian and known ports established in Sections 1 to 3; the network is allowed to learn only conservative corrections, sign-constrained dissipative corrections, and the residual after known terms have been removed. Staged training, port-power supervision, and sparsity or magnitude regularization reduce mutual compensation among branches. These

measures are structural constraints designed to mitigate leakage, but uniqueness and stability must still be tested through paired operating conditions and ablation experiments.

4.4. **COMSOL Data Hierarchy and Training Objects.**

4.4.1. *Kirchhoff-Consistent Primary Training Set.* The primary COMSOL training model uses the three-dimensional Euler–Bernoulli Beam interface with geometric nonlinearity, dynamic inertia, and torsional degrees of freedom. In COMSOL 6.3, the Beam interface applies the Euler–Bernoulli formulation to slender beams and neglects transverse shear while retaining three-dimensional displacement, cross-sectional rotation, and torque; geometric nonlinearity is represented in a corotational large-rotation, small-strain form.[4–6] The model is therefore consistent with the strict Kirchhoff branch in the shear-free assumption but is not unconditionally identical to the geometrically exact Cosserat equations.

The conservative training set is restricted to

$$\mathcal{D}_K^{cons} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \widehat{H}_{phys,i} \right) : model_level = K,\ representation = sk,\ \mathbf{Q}_{port} = 0 \right\}. \tag{4.20}$$

The open-port training set is

$$\mathcal{D}_K^{port} = \left\{ \left(\mathbf{Z}_i, \dot{\mathbf{Z}}_i, \mathbf{Q}_{diss,i}^{bulk}, \mathbf{Q}_{contact,i}, \mathbf{Q}_{in,i}, \mathbf{Q}_{flow,i}, \mathbf{Q}_{\partial,i} \right) \right\}. \tag{4.21}$$

Samples used for probabilistic transfer to KAM theory may only come from $\mathcal{D}_K^{cons}$. The total training loss of the open-port model cannot be interpreted as a Hamiltonian perturbation.

The P1-series pilots use only three small trajectories—mode1, mode2, and mixed12—in the fixed $N = 2$ basis, with leave-one-whole-trajectory-out partitioning rather than random point splitting across a trajectory. P1 contains 3 folds×4 models×5 seeds, or 60 derivative-training runs; P1.5 contains 45 residual and 15 physics-only runs; and P1.6 contains 30 constrained and 30 unconstrained runs. The five seeds are algorithmic repetitions on the same physical trajectories and quantify optimization variability; they are not five independent physical samples. These pilots compare structural assumptions and expose failure modes but do not replace the planned 96-trajectory data set or full-scale P2 training.

All preceding references to $N = 2$ in the pilot denote the $N_{\text{diag}} = 2$ diagnostic basis, not the formal production order n_{main} or a modal-sufficiency conclusion.

4.4.2. *Strict Weak-Form and High-Fidelity References.* When a term-by-term reproduction of the coupled weak form is required, Weak Form PDE or Weak Contribution is used to enter test functions, boundary contributions, and auxiliary variables.[7, 8] Timoshenko Beam and three-dimensional Solid Mechanics models are used to assess shear, local contact, and cross-sectional effects. For paired operating conditions, define

$$\Delta X_{HF} = X_{HF} - X_K. \tag{4.22}$$

Only the component that passes the closure test enters the conservative correction $\mathbf{J}\nabla\Delta H_{HF}$; a component with nonpositive power enters the dissipative correction; and all remaining components remain general residuals. The complete ΔX_{HF} must not be interpreted as a Hamiltonian perturbation or as H_1 in KAM theory.

4.5. **Loss Functions and Staged Training.**

4.5.1. *Conservative Derivative Supervision.* For $\mathcal{D}_K^{cons}$, define

$$\mathcal{L}_H = \frac{1}{N_c} \sum_{i=1}^{N_c} \left\| \dot{\mathbf{Z}}_i - \mathbf{J}\nabla \widehat{H}_\lambda \left(\mathbf{Z}_i \right) \right\|_2^2. \tag{4.23}$$

If the physical Hamiltonian is known, add supervision of the energy correction

$$\mathcal{L}_{\Delta H} = \frac{1}{N_c} \sum_{i=1}^{N_c} \left| \Delta \widehat{H}_\lambda \left(\mathbf{Z}_i \right) - \Delta \widehat{H}_{ref,i} \right|^2. \tag{4.24}$$

4.5.2. *Port and Energy-Balance Supervision.* Define the port residual

$$\mathcal{L}_{port} = \frac{1}{N}\sum_{i=1}^{N} \parallel \dot{\mathbf{Z}}_i - \mathbf{J}\nabla\widehat{H}_\lambda - \mathbf{B}_{port}\left(\widehat{\mathbf{Q}}^{bulk}_{diss,\mu} + \widehat{\mathbf{Q}}_{contact} + \widehat{\mathbf{Q}}_{in,\nu} + \widehat{\mathbf{Q}}_{flow,\kappa} + \mathbf{Q}_\partial + \widehat{\mathbf{r}}_\eta\right) \parallel_2^2 . \tag{4.25}$$

The energy-balance loss is

$$\mathcal{L}_E = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{d\widehat{H}_\lambda}{d\tau} - \nabla_{\mathbf{P}}\widehat{H}_\lambda^T\left(\widehat{\mathbf{Q}}_{diss,\mu} + \widehat{\mathbf{Q}}_{in,\nu} + \widehat{\mathbf{Q}}_{flow,\kappa} + \mathbf{Q}_\partial + \widehat{\mathbf{r}}_\eta\right) - \partial_\tau\widehat{H}_\lambda\right|^2 . \tag{4.26}$$

The dissipation-sign constraint is imposed only on $\widehat{\mathbf{Q}}_{diss,\mu}$:

$$\mathcal{L}_{pass} = \frac{1}{N}\sum_{i=1}^{N}\left[\max\left\{0, \nabla_{\mathbf{P}}\widehat{H}_\lambda^T\widehat{\mathbf{Q}}_{diss,\mu}\right\}\right]^2 . \tag{4.27}$$

4.5.3. *Trajectory, Gauge, and Symplectic-Structure Losses.* The trajectory loss is

$$\mathcal{L}_{traj} = \frac{1}{N}\sum_{i=1}^{N} \parallel \mathbf{Z}_{i,n+1} - Integrate\left(X_\theta, \mathbf{Z}_{i,n}, \Delta\tau\right) \parallel_2^2 . \tag{4.28}$$

The additive-gauge loss is

$$\mathcal{L}_{gauge} = \left|\widehat{H}_\lambda\left(\mathbf{Z}_{ref}\right)\right|^2 . \tag{4.29}$$

The symplectic-structure loss is applied only to the conservative submap $F^{cons}_{h,\lambda}$:

$$\mathcal{L}_{symp} = \frac{1}{N}\sum_{i=1}^{N} \parallel \left(DF^{cons}_{h,\lambda}\right)^T\mathbf{J}DF^{cons}_{h,\lambda} - \mathbf{J} \parallel_F^2 . \tag{4.30}$$

The complete dissipative map is generally not symplectic. The loss must therefore not be imposed on the complete open map and then interpreted as evidence of a strictly symplectic flow.

Training proceeds in three stages: first fit $\widehat{H}_\lambda$; then identify the ports while fixing or only weakly updating the Hamiltonian head; and finally perform joint fine-tuning. The statistical-learning–KAM analysis reads only the conservative derivative error associated with $\mathcal{L}_H$ and its function-space generalization bound.

The P1/P1.5/P1.6 runs instantiate only the conservative derivative-only pilot and structural ablations of this workflow. P1.5 tests the conditional benefit and computational cost of physics-only, Real-HNN residual, and CHNN residual variants; P1.6 imposes $f(0) = 0$ and the zero-point Jacobian structure as a hard-constraint ablation. Their results are reported in Section 6.3, whereas open-port joint fine-tuning, high-order function-space certificates, and transfer to KAM remain to be examined in the full-scale P2 study.

4.5.4. *CHNN Implementation and Structural Validation.* To implement the preceding mathematical structure, we developed an independent CHNN software system aligned with the unified interfaces. Rather than directly assembling existing HNN, CVNN, and KAM–HNN reproduction projects, the implementation retains unit-tested complex-valued operators and reorganizes them according to the canonical state, gray-box Hamiltonian, and mutually exclusive port interfaces used here. The primary software model is

$$H_\theta\left(\mathbf{x}, \boldsymbol{\mu}\right) = H_{phys}\left(\mathbf{x}, \boldsymbol{\mu}\right) + \Delta H_\theta\left(\mathbf{x}, \boldsymbol{\mu}\right), \tag{4.31}$$

$$\dot{\mathbf{x}} = \mathbf{J}\nabla_{\mathbf{x}}H_\theta + \mathbf{B}_{port}\left(\mathbf{Q}^{bulk}_{diss} + \mathbf{Q}_{in} + \mathbf{Q}_{flow} + \mathbf{Q}_\partial + \mathbf{Q}_{contact} + \mathbf{r}_\eta\right), \qquad \mathbf{B}_{port} = \begin{bmatrix}\mathbf{0}\\ \mathbf{I}\end{bmatrix}. \tag{4.32}$$

The passive dissipation head uses

$$\mathbf{Q}_{diss}^{bulk} = -\mathbf{D}_\theta \nabla_{\mathbf{p}} H_\theta, \qquad \mathbf{D}_\theta = \mathbf{L}_\theta \mathbf{L}_\theta^T \succcurlyeq \mathbf{0}, \tag{4.33}$$

which guarantees at the code-structure level that

$$(\nabla_{\mathbf{p}} H_\theta)^T \mathbf{Q}_{diss}^{bulk} \leq 0. \tag{4.34}$$

The implementation corresponds to the theory as follows. The canonical-scaling module first transforms the dimensional pair $(\mathbf{q}, \mathbf{p})$ into the nondimensional canonical pair $(\mathbf{Q}, \mathbf{P})$ and then constructs the complex coordinates. The complex backbone provides both standard complex-linear layers and widely linear layers of the form $\mathbf{W}\boldsymbol{\xi} + \mathbf{V}\overline{\boldsymbol{\xi}} + \mathbf{b}$. The Hamiltonian head outputs only a real scalar and uses the gray-box structure $H_{phys} + \Delta H_\theta$. Dissipation, magnetic input, flow, boundary, contact, and residual error are represented by independent generalized-force heads and enter only the momentum equation through $\mathbf{B}$. The data interface uses `representation` to distinguish $\mathbf{p}_0 = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}}$ from $\mathbf{p} = \mathbf{M}(\mathbf{q})\,\dot{\mathbf{q}} - \mathbf{b}_A$, uses `model_level` to distinguish the Kirchhoff Beam, strict weak-form, and high-fidelity reference models, and uses `conservative_mask` to isolate conservative samples accessible to the statistical-learning–KAM analysis. The training pipeline implements the three stages "conservative Hamiltonian–ports–joint fine-tuning" and provides a real-valued HNN and a black-box vector-field network as comparison baselines.

The software passed the following structural-consistency tests:

1. equivalent implementation of an arbitrary real block-linear mapping by the widely linear layer;
2. round-trip consistency of canonical scaling and recovery of the Galerkin basis subspace for prescribed basis functions;
3. the total Hamiltonian satisfies $H_\theta = H_{phys} + \Delta H_\theta$;
4. the continuous conservative field satisfies $\nabla H_\theta^T \mathbf{J} \nabla H_\theta = 0$;
5. passive dissipative power is nonpositive;
6. every external port has zero direct leakage into the $\dot{\mathbf{q}}$ channel;
7. inconsistent momentum-representation labels, model-level labels, or projected-data formats are rejected, and automatic-differentiation backpropagation succeeds for every loss term.

In a unified software environment, Python source compilation, editable installation, and 13 component and interface tests all passed, and the three-stage training pipeline executed end to end on synthetic data. These results verify consistency between the implementation and the stated structural identities; their evidential scope is limited to implementation correctness. They do not replace identification accuracy on catheter COMSOL data, function-class complexity constants, higher-order error certificates, or KAM invariant-torus certification. The physical baseline Hamiltonian $\widehat{H}_{phys}$, configuration-dependent mass–inertia matrix, exact nonautonomous terms, and COMSOL–Galerkin projection for a specific catheter will be instantiated from the actual model and data in the numerical experiments.

4.6. **From CHNN Learning Error to the KAM Smallness Condition.**

4.6.1. *The Integrable Reference System Is Not an Automatic Consequence of the Network Real Part.* Section 3 constructs, through a stable equilibrium, modal canonicalization, and a Hamiltonian Birkhoff normal form,

$$H_{cons} = H_0 + \varepsilon H_1. \tag{4.35}$$

The Hamiltonian H_0 must be verified to be integrable and nondegenerate. A real-valued H_λ is not integrable merely because it is real. The linear harmonic oscillator satisfies $\partial_{\mathbf{I}}^2 H_2 = 0$, so nondegeneracy must arise from higher-order conservative terms.

4.6.2. *Conservative Training Data, Function Class, and Additive Gauge.* The following statistical-learning statements apply only to the conservative training set $\mathcal{D}_K^{cons}$. Let $\Omega \subset \mathbb{R}^{2d}$ be a bounded connected Lipschitz domain, let the true conservative Hamiltonian be $H_\star$, and let the samples $\mathbf{Z}_1, \ldots, \mathbf{Z}_N$ be independent and identically distributed according to a probability measure $\mathbb{P}_Z$. Assume noiseless derivative labels satisfy

$$\dot{\mathbf{Z}}_i = \mathbf{J}\nabla H_\star\left(\mathbf{Z}_i\right). \tag{4.36}$$

Independence, a full-dimensional phase-space density lower bound, and the uniform regularity conditions are assumptions rather than established facts for the current pilot; the three fixed P1/P1.5/P1.6 trajectories with time-correlated sampling cannot be inserted into this theorem as a generalization guarantee.

Realify all complex parameters of the Hamiltonian head as $\boldsymbol{\vartheta} \in \Theta = [-B, B]^P$, and denote the learning error by

$$e_{\boldsymbol{\vartheta}} = \widehat{H}_{\boldsymbol{\vartheta}} - H_\star. \tag{4.37}$$

State derivatives alone do not identify the absolute constant of the Hamiltonian. Therefore, impose on the theoretical domain

$$\int_\Omega e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \, d\mathbf{Z} = 0, \tag{4.38}$$

or use a known energy anchor to select the same representative of the additive equivalence class. Choose a statistical exponent $r > 2d$ and define the scalar gradient loss

$$g_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) = \parallel \dot{\mathbf{Z}} - \mathbf{J}\nabla \widehat{H}_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r = \parallel \nabla e_{\boldsymbol{\vartheta}}\left(\mathbf{Z}\right) \parallel_{\ell^r}^r . \tag{4.39}$$

To apply a uniform generalization bound for bounded losses, also assume $0 \leq g_{\boldsymbol{\vartheta}} \leq c$. This boundedness is an independent regularity assumption and does not follow automatically from a small empirical loss.

4.6.3. *Covering Numbers and Rademacher Complexity.* For a sample $S = (\mathbf{Z}_1, \ldots, \mathbf{Z}_N)$, define the normalized empirical metric

$$d_S(g, g\prime) = \left(\frac{1}{N} \sum_{i=1}^{N} \left| g\left(\mathbf{Z}_i\right) - g\prime\left(\mathbf{Z}_i\right) \right|^2 \right)^{1/2} . \tag{4.40}$$

Assumption 4.1 (Samplewise Lipschitz continuity from parameters to loss)**.** For a given sample S, there exists a finite constant L_S such that every $\boldsymbol{\vartheta}, \boldsymbol{\vartheta}\prime \in \Theta$ satisfies

$$d_S\left(g_{\boldsymbol{\vartheta}}, g_{\boldsymbol{\vartheta}\prime}\right) \leq L_S \parallel \boldsymbol{\vartheta} - \boldsymbol{\vartheta}\prime \parallel_2 . \tag{4.41}$$

When the input domain and parameter set are bounded, the network depth is finite, and the first and second derivatives of the activation are bounded on the relevant range, this assumption can hold on a compact parameter set. In general, L_S grows with depth, width, weight bounds, and derivative bounds.

Lemma 4.1 (Finite-parameter covering number)**.** *Under Assumption 4.1, let* $\mathcal{G} = \{g_{\boldsymbol{\vartheta}} : \boldsymbol{\vartheta} \in \Theta\}$*, and let* $\mathcal{G}_S$ *denote the restriction of this function class to the sample* S*. Then*

$$\mathcal{N}\left(\varepsilon, \mathcal{G}_S, d_S\right) \leq \left(1 + \frac{2B\sqrt{P}\, L_S}{\varepsilon} \right)^P \tag{4.42}$$

Proof. On the parameter cube $[-B, B]^P$, place a grid of Euclidean radius ε / L_S, and use Assumption 4.1 to push the parameter covering forward to a covering of the loss functions. □

Because the parameter set is compact and the loss is continuous in the parameters, the supremum is attained. Define the empirical Rademacher complexity by

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) = \mathbb{E}_{\boldsymbol{\sigma}}\left[\max_{g\in\mathcal{G}} \frac{1}{N}\sum_{i=1}^{N} \sigma_i g(\mathbf{Z}_i)\right] \tag{4.43}$$

Let

$$R_S = \max_{g\in\mathcal{G}} d_S(g,0) \tag{4.44}$$

Dudley's entropy integral gives[36]

$$\widehat{\mathfrak{R}}_S(\mathcal{G}) \le \inf_{0<\alpha<R_S}\left[4\alpha + \frac{12}{\sqrt{N}}\int_{\alpha}^{R_S}\sqrt{\log \mathcal{N}(\varepsilon, \mathcal{G}_S, d_S)}\, d\varepsilon\right] \tag{4.45}$$

A purely complex-linear network is a structurally constrained subclass of the corresponding real network. A widely linear network must instead be counted using its actual number P of real parameters and its actual Lipschitz constant L_S; "one complex parameter" must not be incorrectly counted as one real degree of freedom.

4.6.4. *From Gradient Risk to Uniform Hamiltonian Error.* Assume that the loss class is fixed before sampling. A standard uniform Rademacher generalization bound then implies that, with probability at least $1-\delta$, simultaneously for every $\boldsymbol{\vartheta}\in\Theta$,

$$\mathbb{E}_{\mathbb{P}_Z} g_{\boldsymbol{\vartheta}} \le \widehat{\mathcal{L}}_r(\boldsymbol{\vartheta}) + 2\widehat{\mathfrak{R}}_S(\mathcal{G}) + 3c\sqrt{\frac{\log(2/\delta)}{2N}} =: B_{N,\delta}(\boldsymbol{\vartheta}). \tag{4.46}$$

Assume further that $\mathbb{P}_Z$ has a density f_Z with respect to Lebesgue measure and that throughout Ω,

$$f_Z(\mathbf{Z}) \ge m_Z > 0. \tag{4.47}$$

Then, on the preceding uniform event,

$$\| \nabla e_{\boldsymbol{\vartheta}} \|_{L^r(\Omega;\ell^r)} \le \left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z}\right)^{1/r}. \tag{4.48}$$

Using the additive gauge, the Poincaré inequality, and the Morrey–Sobolev embedding for $r > 2d$,[37] one obtains:

Theorem 4.2 (Uniform generalization bound for the Hamiltonian)**.** *Under the preceding assumptions, with probability at least $1-\delta$, simultaneously for every $\boldsymbol{\vartheta}\in\Theta$,*

$$\boxed{\| \widehat{H}_{\boldsymbol{\vartheta}} - H_\star \|_{L^\infty(\Omega)} \le C_{\Omega,r}(1+C_P)\left(\frac{B_{N,\delta}(\boldsymbol{\vartheta})}{m_Z}\right)^{1/r}.} \tag{4.49}$$

where C_P is the Poincaré constant. The overall $1/r$ power arises from converting the r-th-power risk back to a gradient norm and cannot be absorbed into a constant independent of the training error and sample size.

4.6.5. *Higher-Order Sobolev or Analytic Error Certificate.* Theorem 4.2 provides only a C^0-level uniform error. A high-frequency, small-amplitude function may converge to zero in C^0 while its higher derivatives diverge. Therefore, neither a first-derivative risk nor proximity of Hamiltonian values alone is sufficient to claim the KAM smallness condition.

Assume that Section 3 has constructed an integrable reference and coordinate transformation on the local action–angle domain

$$\mathcal{A} = \mathbb{T}^M \times U \tag{4.50}$$

If the original training data are represented in physical canonical coordinates, the learning error must be pulled back to $\mathcal{A}$, and bounds on the higher derivatives of the coordinate transformation and its inverse must be included in the constants. Choose integers s, ℓ and a statistical exponent r satisfying

$$s > \ell + \frac{2M}{r}. \tag{4.51}$$

Let the training algorithm output $\widehat{\boldsymbol{\vartheta}}$. In addition to the first-order statistical certificate from Theorem 4.2, require a higher-order certificate for this specific output: on an event of probability at least $1-\eta$,

$$\sum_{2\le|\alpha|\le s} \| D^{\alpha}\left(\widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_{\star}\right) \|^{r}_{L^{r}(\mathcal{A})} \le C_{s,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right)^{r}. \tag{4.52}$$

The certificate must control higher derivatives of the true learning error, not merely the magnitude of the network's own derivatives. Periodic Sobolev embedding then gives, on the intersection of the two events,

$$\| \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} - H_{\star} \|_{C^{\ell}(\mathcal{A})} \le \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right), \tag{4.53}$$

where one may take

$$\mathfrak{C}_{N,\delta,\eta} = C_{s,r,\mathcal{A}}\left[(C_P^{r}+1)\,\frac{B_{N,\delta}\left(\widehat{\boldsymbol{\vartheta}}\right)}{m_Z} + C_{s,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right)^{r}\right]^{1/r}. \tag{4.54}$$

If the higher-order certificate is obtained by deterministic interval validation or interpolation with a rigorous remainder, set $\eta = 0$. If it is itself probabilistic, independence from the generalization event is not assumed, and the union bound yields only a joint probability of at least $1-\delta-\eta$.

If an analytic KAM theorem is selected, the preceding finitely smooth certificate is still insufficient; the analytic supremum norm of the learning error on a specified complex neighborhood must be controlled separately. Explicit dependence of a widely linear network on $\boldsymbol{\xi}$ and $\overline{\boldsymbol{\xi}}$ does not prevent the real-variable Hamiltonian from admitting a real-analytic extension, but smallness in a real-domain C^{ℓ} norm does not automatically imply smallness in a complex-neighborhood analytic norm.

4.6.6. *Accounting for the Integrable Reference, Pre-Existing Physical Perturbation, and Learning Error.* Fix the integrable reference Hamiltonian $h(\mathbf{I})$ constructed and verified in Section 3. Write the true conservative system and the learned system as

$$H_{\star} = h + P_0, \qquad \widehat{H}_{\widehat{\boldsymbol{\vartheta}}} = H_{\star} + e_{\widehat{\boldsymbol{\vartheta}}} = h + P_{\widehat{\boldsymbol{\vartheta}}}, \tag{4.55}$$

where

$$P_{\widehat{\boldsymbol{\vartheta}}} = P_0 + e_{\widehat{\boldsymbol{\vartheta}}}. \tag{4.56}$$

Therefore, in the common $C^{\ell}(\mathcal{A})$ norm,

$$\| P_{\widehat{\boldsymbol{\vartheta}}} \|_{C^{\ell}} \le \| P_0 \|_{C^{\ell}} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right). \tag{4.57}$$

If the true system is already only nearly integrable, P_0 consumes part of the KAM perturbation budget; the entire threshold cannot be assigned to the learning error. Coordinate-transformation, Galerkin-truncation, parameter-identification, and model-closure errors may be added through the triangle inequality only after they have been controlled in the same function-space norm used by the KAM theorem. General dissipation, friction, background-flow input, and nonpotential magnetic control are not Hamiltonian perturbations here and remain in the open-port branches of Section 4.

4.6.7. *Probabilistic Transfer from the Learning-Error Certificate to the KAM Condition.* Select an external KAM theorem in advance. Let $\mathsf{A}_{KAM}(h, P)$ denote all of its structural assumptions other than perturbation smallness, including the action–angle domain, regularity order, type of nondegeneracy, Diophantine condition, distance to the boundary, and admissible perturbation class. Let ε_{KAM} be its smallness threshold and let $\mathcal{K}(h + P)$ denote the torus-persistence conclusion supplied by that theorem.

Proposition 4.2 (Probabilistic transfer)**.** *Under the preceding assumptions, with probability at least* $1 - \delta - \eta$*,*

$$\boxed{\mathsf{A}_{KAM}\left(h, P_{\widehat{\boldsymbol{\vartheta}}}\right) \ \wedge \ \left(\| P_0 \|_{C^{\ell}(\mathcal{A})} + \mathfrak{C}_{N,\delta,\eta}\left(\widehat{\boldsymbol{\vartheta}}\right) < \varepsilon_{KAM}\right) \ \Rightarrow \ \mathcal{K}\left(\widehat{H}_{\widehat{\boldsymbol{\vartheta}}}\right).} \tag{4.58}$$

Proof. On the intersection of the uniform-generalization event and the higher-order-certificate event, Section 4.6.5 gives a C^{ℓ} bound for the learning error. The triangle inequality in Section 4.6.6 then makes the total Hamiltonian perturbation smaller than ε_{KAM}. If all remaining conditions in A_{KAM} also hold, the selected KAM theorem applies. By the union bound, the probability of the joint event is at least $1 - \delta - \eta$. □

This proposition establishes only the probabilistic transfer from a learning-error certificate to the small-perturbation condition of an external KAM theorem. The type, regularity, measure estimate, and frequency properties of the persistent tori are determined entirely by the selected external theorem and cannot be inferred from the training loss alone.

4.6.8. *Continuous and Discrete Error Budgets.* On a common analytic domain $\mathcal{D}_{\rho}$, define

$$\varepsilon_{model} = \| X_{cons} - X_0 \|_{\rho}, \qquad \varepsilon_{learn} = \| X_{\lambda} - X_{cons} \|_{\rho}. \tag{4.59}$$

Let Φ_h^0 be the exact map of X_0 over a time step h, and let $\Psi_{h,\lambda}$ be an order-s_a symplectic algorithm applied to the learned Hamiltonian. If the vector fields and their first derivatives are uniformly bounded and the trajectories remain in a smaller analytic domain $\mathcal{D}_{\rho\prime}$, then there exists a constant $C_{\rho,\rho\prime}$ such that

$$\boxed{\| \Psi_{h,\lambda} - \Phi_h^0 \|_{\rho\prime} \leq C_{\rho,\rho\prime} h \left(\varepsilon_{model} + \varepsilon_{learn} + h^{s_a}\right) =: \varepsilon_{map}.} \tag{4.60}$$

Here, h^{s_a} arises from the local algorithmic error. This budget does not assume unconditionally that a globally exact "discrete Hamiltonian" exists. Only when

$$\varepsilon_{map} < \delta_{KAM}\left(\rho\prime, \gamma, \tau, \text{nondegeneracy margin}, h\right) \tag{4.61}$$

and the analyticity, Diophantine nonresonance, and step-size nonresonance conditions are satisfied may a specific symplectic-map KAM theorem be invoked.[25–27, 33]

4.7. **Three Discrete-Dynamics Branches.**

1. **Autonomous conservative branch.** Apply a generating-function method, implicit midpoint rule, or another algorithm proved to be symplectic to $\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_{\lambda}$, yielding F_h^{cons}. This map is used for Shang's KAM theory for symplectic algorithms and for validation by the Haro parameterization method.
2. **Exact nonautonomous Hamiltonian branch.** Apply the standard nonautonomous canonical equations to $H(\mathbf{Z}, \tau)$, or introduce the extended variables (τ, p_{τ}) and $\mathcal{H} = H + p_{\tau}$ and use a symplectic algorithm in extended phase space. Explicit time dependence alone does not require a nonstandard $K(\mathbf{z}, t)$-symplectic structure.
3. **Open-port branch.** Integrate the complete forced–dissipative system consistently while monitoring its energy balance. Its map is generally neither a standard symplectic map nor automatically a Birkhoff system. It enters the $K(\mathbf{z}, t)$-symplectic branch of Section 5 only if the residual vector field remaining after removal of the known backbone and ports passes the self-adjoint representation test.

4.8. **Output Objects for Invariant-Torus Analysis.** The objects required by the invariant-torus analysis in Section 5 are therefore

$$\mathcal{O}_{4\to5} = (F_h^{cons}, \varepsilon_{map}, X_{res}, \mathcal{D}, \gamma, \tau, \text{nondegeneracy margin}, model_level, representation)\,, \tag{4.62}$$

where

$$\boxed{X_{res} = X_{data} - \mathbf{J}\nabla\widehat{H}_\lambda - \mathbf{B}_{port}\left(\widehat{\mathbf{Q}}_{diss,\mu}^{bulk} + \widehat{\mathbf{Q}}_{contact} + \widehat{\mathbf{Q}}_{in,\nu} + \widehat{\mathbf{Q}}_{flow,\kappa} + \mathbf{Q}_\partial\right).} \tag{4.63}$$

If any quantity has not yet been constructed or verified numerically, subsequent sections list it only as a condition to be tested rather than as an established conclusion.

5. Symplectic Discretization, Birkhoff Structures, and Invariant-Torus Validation

5.1. **Analysis branches.** A fixed-dimensional Galerkin truncation has already been completed before the KAM analysis. The principal theorem should therefore use a finite-dimensional Hamiltonian-flow theorem or a finite-dimensional symplectic-map theorem. Kuksin's infinite-dimensional KAM theory[32] is not used as a direct theorem for the fixed truncated system; it is retained only as a theoretical extension for discussions of the limit of infinitely many modes, the PDE limit, or consistency of the truncation. Moser's invariant-curve theorem for area-preserving annulus maps[31] primarily concerns two-dimensional twist maps. It may be used for low-dimensional sections and methodological comparison, but it cannot replace the Pöschel/Shang conditions required for multi-degree-of-freedom continuous Hamiltonian systems or high-dimensional symplectic maps.[26, 27, 33]

Rather than describing every dynamical branch as a $K(\mathbf{z}, t)$-symplectic system, this section distinguishes among:

- standard $\mathbf{J}$-symplectic discretization of an autonomous conservative Hamiltonian system;
- standard nonautonomous or extended-phase-space symplectic discretization of an exact nonautonomous Hamiltonian system;
- time-dependent $\mathbf{K}(\mathbf{z}, t)$-symplectic discretization of a Birkhoff system that passes the variational self-adjointness test.

The Hamiltonian Birkhoff normal form is used in Sections 3 and 4 to construct $H_0 + \varepsilon H_1$. The self-adjoint Birkhoff representation considered here is used to test the structure of a general first-order system. The names are similar, but the mathematical objects are different.

5.2. **Standard symplectic branch and generating functions.** For the autonomous conservative CHNN

$$\dot{\mathbf{Z}} = \mathbf{J}\nabla\widehat{H}_\lambda(\mathbf{Z})\,, \tag{5.1}$$

the exact flow satisfies

$$(D\Phi_t)^T\,\mathbf{J}\,(D\Phi_t) = \mathbf{J}. \tag{5.2}$$

If the one-step map F_h^{cons} is an exact symplectic map sufficiently close to the identity and the relevant Jacobian block of the chosen projection is nondegenerate, a local generating function exists. Retaining the leading term of a center-type generating function gives the implicit midpoint rule

$$\frac{\mathbf{Z}_{n+1} - \mathbf{Z}_n}{h} = \mathbf{J}\nabla\widehat{H}_\lambda\left(\frac{\mathbf{Z}_{n+1} + \mathbf{Z}_n}{2}\right), \tag{5.3}$$

which is a second-order symplectic method. Higher-order methods may be constructed through generating-function expansions, composition methods, or other proven symplectic algorithms[25].

For a nonautonomous Hamiltonian $H(\mathbf{Z}, t)$, introduce the extended phase space

$$\mathcal{H}(\mathbf{Z}, t, p_t) = H(\mathbf{Z}, t) + p_t, \tag{5.4}$$

and integrate with respect to the extended canonical symplectic form. A transition to the formulation of the next subsection is needed only when the variables are noncanonical or when the system possesses a special Birkhoff representation.

The order-2 BDF scheme used for canonical data processing is not a symplectic stepper of the type defined in this section. It supplies only an observable `tsteps` sequence for COMSOL solver-step state sampling and canonical vector-field label generation. Three-level observed convergence and satisfaction of the predefined G0–G6 thresholds show only that the data contract is suitable for subsequent amplitude-sentinel experiments. Neither BDF energy drift nor second-order convergence is used to infer symplecticity; a subsequent KAM computation must use a separately proven symplectic map and quantify its discretization defect.

5.3. **Self-adjoint representation of a Birkhoff system.** Let $\mathbf{z} \in \mathbb{R}^{2n}$, let $F_i(\mathbf{z}, t)$ be Birkhoff functions, let $B(\mathbf{z}, t)$ be the Birkhoffian, and define

$$K_{ij}(\mathbf{z}, t) = \frac{\partial F_j}{\partial z_i} - \frac{\partial F_i}{\partial z_j}. \tag{5.5}$$

The Birkhoff equations are

$$\boxed{K_{ij}(\mathbf{z}, t)\,\dot{z}_j - \frac{\partial B}{\partial z_i} - \frac{\partial F_i}{\partial t} = 0.} \tag{5.6}$$

Consider a general system written as

$$K_{ij}\dot{z}_j + D_i = 0. \tag{5.7}$$

On a local star-shaped domain, if $\mathbf{K}$ is nondegenerate and satisfies skew symmetry, the spatial closure condition, and the temporal compatibility condition, then the system admits a regular self-adjoint Birkhoff representation. Equivalently, functions F_i and B can be found such that the preceding equation holds[18, 25]. Define the time-dependent two-form

$$\Omega_K = \frac{1}{2} K_{ij}(\mathbf{z}, t)\; dz_i \wedge dz_j. \tag{5.8}$$

Its phase flow satisfies the time-dependent structure-preservation relation

$$\left(D\Phi_{t,t_0}\right)^T \mathbf{K}\left(\Phi_{t,t_0}, t\right) D\Phi_{t,t_0} = \mathbf{K}(\mathbf{z}_0, t_0). \tag{5.9}$$

When $\mathbf{K} = \mathbf{J}^{-1}$ is constant, this condition reduces to the canonical symplectic condition.

5.4. **Identification of a Birkhoff structure for the residual vector field.** The analysis is applied only to X_{res}, as defined in Section 4. Parameterize the one-form directly as

$$\mathcal{F}_\rho = \sum_{i=1}^{2n} F_{\rho,i}(\mathbf{z}, t)\; dz_i, \qquad K_{\rho,ij} = \partial_{z_i} F_{\rho,j} - \partial_{z_j} F_{\rho,i}, \tag{5.10}$$

introduce B_β, and define the candidate vector field

$$\widetilde{X}_{\rho,\beta} = \mathbf{K}_\rho^{-1}\left(\nabla B_\beta + \partial_t \mathbf{F}_\rho\right). \tag{5.11}$$

The training loss is

$$\mathcal{L}_{Birk} = \frac{1}{N}\sum_{\ell=1}^{N} \;\| \mathbf{K}_\rho X_{res} - \nabla B_\beta - \partial_t \mathbf{F}_\rho \;\|_2^2 + \mu_{nd} \frac{1}{N}\sum_{\ell=1}^{N} \left[\max\{0, \kappa - \sigma_{min}(\mathbf{K}_\rho)\}\right]^2. \tag{5.12}$$

On the domain $\mathcal{D}$, define the defect of the self-adjoint Birkhoff representation by

$$\boxed{\delta_{Birk} = \inf_{F,B,\ \inf_{\mathcal{D}} \sigma_{min}(K_F) \geq \kappa} \| K_F X_{res} - \nabla B - \partial_t F \ \|_{\mathcal{X}(\mathcal{D})} .} \tag{5.13}$$

If $\delta_{Birk} = 0$, an exact representation is obtained. If $\delta_{Birk} > 0$, only the projected vector field $\widetilde{X}$ has an exact K-symplectic structure, and the remaining error must be controlled separately. The complete open system containing a nonstructured residual cannot therefore be described as a strictly $K(\mathbf{z}, t)$-symplectic system.

5.5. **A $K(\mathbf{z}, t)$-symplectic discretization for Birkhoff systems.** Under regularity, self-adjointness, analyticity, and nondegeneracy conditions, the Birkhoff phase flow admits a gradient map and a scalar generating function under an appropriate local projection. If the generating function is analytic in the step size, it can be expanded as

$$S(\mathbf{Z}, h) = \sum_{r=1}^{\infty} S_r(\mathbf{Z}) h^r. \tag{5.14}$$

Truncation at order m gives an implicit order-m approximately $K(\mathbf{z}, t)$-symplectic method satisfying

$$(D\Psi_h)^T \mathbf{K}(\Psi_h(\mathbf{z}), t + h) D\Psi_h = \mathbf{K}(\mathbf{z}, t) + O\left(h^{m+1}\right). \tag{5.15}$$

This result guarantees only approximate structure preservation and the stated order when its hypotheses hold; it is exactly K-symplectic only when the remainder vanishes on the domain of interest, and it does not by itself imply persistence of classical Hamiltonian KAM tori. If a nonstructured residual is superposed, the resulting structural defect must enter the error budget.

5.6. **Invariant-torus computation by the parameterization method.** For the exact symplectic one-step map F_h^{cons} of the autonomous conservative branch, seek an embedding

$$K : \mathbb{T}^d \to \mathcal{M} \tag{5.16}$$

satisfying

$$\boxed{F_h^{cons}(K(\boldsymbol{\theta})) = K(\boldsymbol{\theta} + \boldsymbol{\omega}).} \tag{5.17}$$

Use the Fourier truncation

$$K_N(\boldsymbol{\theta}) = \sum_{|k| \leq N} \widehat{K}_k e^{2\pi i k \cdot \boldsymbol{\theta}}, \qquad \widehat{K}_{-k} = \overline{\widehat{K}_k}, \tag{5.18}$$

and define the invariance residual

$$E_N(\boldsymbol{\theta}) = F_h^{cons}(K_N(\boldsymbol{\theta})) - K_N(\boldsymbol{\theta} + \boldsymbol{\omega}). \tag{5.19}$$

FFT analysis, Poincaré sections, and long trajectories are used only to initialize K_N and $\boldsymbol{\omega}$; a Newton-like method is then used for correction. The a posteriori KAM theorems of Haro et al. require the map to be analytic and exact symplectic, the frequency to satisfy a Diophantine condition, the dynamics to be nondegenerate, and $\| E_N \|$ to satisfy an explicit smallness condition. Existence of a true invariant torus can be certified only after all constants have been verified. A small residual alone supports only the statement that a numerical approximate torus has been obtained[28].

5.7. **Parameter continuation and safe operating domain.** Continuation is performed with respect to magnetic-field strength, insertion speed, flow velocity, contact clearance, and material parameters. At each parameter point, record:

- ε_{map} and the Hamiltonian learning error;
- the symplectic or K-symplectic structural defect;
- the invariance residual $\| E_N \|$;
- the minimum singular value of the twist matrix/frequency map;
- the finite-order Diophantine margin and decay of the Fourier coefficients;
- the energy balance, port power, peak contact response, and trajectory error;
- the discrepancy between the Kirchhoff model and the high-fidelity model.

Define the KAM-theoretic guarantee domain as

$$\mathcal{S}_{KAM} = \{\mu : \text{the symplecticity, nondegeneracy, nonresonance, and a posteriori smallness conditions all hold}\}. \tag{5.20}$$

The complete engineering safe domain is defined as

$$\boxed{\mathcal{S}_{safe} = \mathcal{S}_{KAM} \cap \mathcal{S}_{energy} \cap \mathcal{S}_{contact} \cap \mathcal{S}_{model} \cap \mathcal{S}_{control}.} \tag{5.21}$$

Here, KAM theory provides only a conditional long-time stability guarantee for the conservative backbone. Open ports, dissipation, contact switching, and high-fidelity model discrepancy are addressed by additional indicators. The safe domain is neither a single "chaos threshold" nor evidence of clinical efficacy.

At present, $\mathcal{S}_{KAM}$, $\mathcal{S}_{safe}$, the invariant-torus embedding K_N, and its a posteriori constants have not been instantiated numerically. The complete algorithms and criteria in this section define the subsequent numerical validation program; the current results do not constitute numerical KAM certification and do not establish a true invariant torus or an engineering safe domain.

6. Numerical Experiments and Results

6.1. **Objectives, numerical models, and evaluation methods.** The numerical experiments successively examine the low-dimensional network implementation, learning behavior on a fixed $N_{\mathrm{diag}} = 2$ diagnostic basis, and consistency of the COMSOL–Galerkin canonical data mapping. P0 uses a four-dimensional nonlinear coupled oscillator to compare an unconstrained vector-field network, a Real HNN, a standard complex-linear CHNN, and a widely linear CHNN. P1, P1.5, and P1.6 use three fixed trajectories—mode1, mode2, and mixed12—to study derivative supervision, gray-box residual learning, and hard equilibrium constraints. Each test split holds out one complete trajectory, and the five random seeds represent algorithmic repetitions rather than independent physical samples. The canonical-data experiments diagnose errors in coordinates, cotangent momenta, and generalized-force labels; the existing single sentinel is at most a candidate input to a subsequent A1 qualification experiment and does not certify multi-amplitude or cross-condition data generation.

The evaluation quantities include vector-field relative L2 error, rollout error, linearization error, energy drift, canonical closure error, and observed time-discretization order. The Bridge perturbation-energy criterion and the force-v2 joint-L2 criterion address energy drift and force closure, respectively, and use independent thresholds. The current delivery lacks auditable raw arrays, metric functions, slices, and threshold sources, so these names and historical excerpts cannot be re-run as complete G0–G6 machine gates. Training, validation, and test data are stated as partitioned by complete parameterized operating conditions rather than by random points from a shared trajectory, but fold manifests, early-stopping records, and normalization logs are not delivered. The full 96-trajectory data set, P2 model comparison, open-port analysis, high-fidelity discrepancy study, and numerical KAM validation remain subjects of subsequent numerical work; the present small-sample results are not extrapolated to those tasks.

6.2. **Numerical implementation and the COMSOL–Galerkin bridge.** The implementation was checked by source compilation, editable installation, component and interface tests, and a synthetic-data training workflow. Thirteen regression-test files were run in isolated processes for three consecutive rounds without failure. These tests assess software consistency rather than scientific model performance.

The COMSOL–Galerkin bridge comprises stationary exports, modal-reality checks on meshes 40 and 80, the first 12 production modes, and three 1 s candidate trajectories. The original perturbation-energy criterion uses a 0.5% threshold and is met only by mode2; under the exploratory 1% threshold, all three trajectories satisfy the criterion. Force-v2 tests force closure on common-support indices $2 \dots 998$, with a fixed $p[2]$ anchor and an independent 2% joint-L2 threshold; all three trajectories meet that criterion. Table 6.1 and Fig. 6.1 reports these tests separately so that the energy and force criteria are not conflated.

Table 6.1. Trajectory-level comparison of Bridge perturbation energy and force-v2 force closure. Energy drift is evaluated against the 0.5% criterion and the exploratory 1% criterion; force-v2 uses an independent joint-L2 criterion.

Trajectory	Energy drift (%)	$< 0.5\%$	$< 1\%$	force-v2 (%)	$< 2\%$
mode1	0.815090	fail	pass	1.513677	pass
mode2	0.329369	pass	pass	0.062551	pass
mixed12	0.523328	fail	pass	1.080748	pass

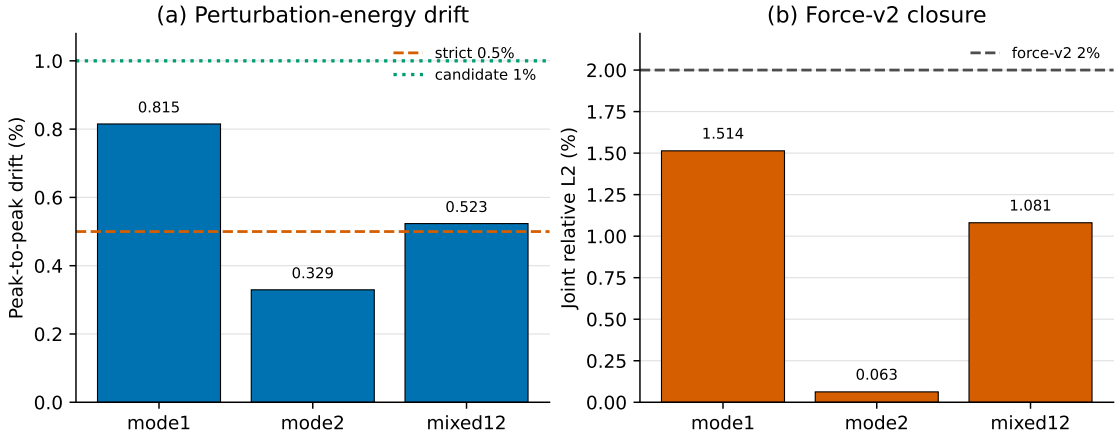


Figure 6.1. Bridge perturbation-energy criteria and the force-v2 force-closure criterion. The left panel applies the 0.5% criterion and the exploratory 1% criterion to the same perturbation-energy drift; the right panel shows the force-v2 joint-L2 error and its independent 2% threshold.

6.3. **Identification results for the conservative branch.**

6.3.1. *P0 low-dimensional architectural baseline.* P0 uses 80/16/20 training/validation/test trajectories on the common four-dimensional coupled oscillator, with 80 steps per trajectory, $\Delta t = 0.025$, 300 epochs, and five algorithmic seeds. With the quadratic physics baseline included, the standard CHNN has vector-field and rollout relative L2 errors of 0.0109144 and 0.0351959, respectively, below the Real HNN values of 0.0122422 and 0.0444270. Its energy drift, 0.00455759, is not lower than the Real HNN value 0.00402084. This experiment therefore supports low-dimensional implementation and architecture comparisons but cannot replace a P2 comparison on COMSOL data or a KAM analysis.

Table 6.2. Model comparison for the P0 quadratic-physics branch. The vector-field network does not use a physics baseline.

Model	Vector-field relative L2	Rollout relative L2	Energy drift
Vector-field network	0.0200837	0.103076	0.0245256
Real HNN	0.0122422	0.0444270	0.00402084
Standard CHNN	**0.0109144**	**0.0351959**	0.00455759
Widely linear CHNN	0.0121700	0.0458872	0.00403167

6.3.2. *P1, P1.5, and P1.6.* All 60/60 P1 runs completed. Across the three held-out trajectories, the standard CHNN has joint vector-field relative errors of 0.1417%–0.2444%, rollout errors of 0.2131%–1.5237%, and linearization errors of 0.3125%–0.3286%. In the five-seed comparisons for this small-sample experiment, its joint-field, rollout, and linearization metrics are lower than those of all three baselines in 15/15 cases. This conclusion is restricted to the three fixed small trajectories and the derivative-only objective.

All 60/60 P1.5 runs also completed. Relative to physics-only, the residual models reduce the mean joint-field error by 90.37%–97.29% and the mean rollout error by 53.40%–95.14%. Compared with the gray-box Real HNN, the standard CHNN has a lower joint-field error in 11/15 runs but a mean relative reduction of −2.195%; it has a lower rollout error in 12/15 runs with a mean reduction of 20.915%, at approximately 3.04× the training time and 2.54× the evaluation time. The benefit is therefore metric-dependent and does not establish stable one-step vector-field superiority.

All 30/30 hard-constrained P1.6 models satisfy the $f(0)$ and zero-point Jacobian conditions with zero numerical residual, but predictive accuracy deteriorates substantially. The mean joint-field/rollout errors change from $1.832 \times 10^{-3}/1.439 \times 10^{-2}$ to $2.872 \times 10^{-2}/1.011 \times 10^{-1}$ for the Real HNN and from $1.819 \times 10^{-3}/1.195 \times 10^{-2}$ to $2.623 \times 10^{-2}/7.594 \times 10^{-2}$ for the standard CHNN. Exact satisfaction of the equilibrium constraint therefore does not imply improved prediction. Table 6.3 summarizes the principal results and their domains of validity.

Table 6.3. P1-series results and interpretation boundaries. The five seeds are algorithmic repetitions; the number of physical trajectories remains three.

Experiment	Numerical result	Domain of validity
P1	Standard-CHNN joint-field, rollout, and linearization errors are 0.1417%–0.2444%, 0.2131%–1.5237%, and 0.3125%–0.3286%.	Restricted to a three-trajectory, derivative-only small-sample comparison.
P1.5	Residual-over-physics-only mean reductions are 90.37%–97.29% for the field and 53.40%–95.14% for rollout; the CHNN-over-gray-Real mean field reduction is −2.195%.	Conditional and more expensive; no stable one-step-field advantage.
P1.6	All 30/30 constrained models pass the equilibrium gates exactly, while the mean Real/CHNN joint-field errors worsen to $2.872 \times 10^{-2}/2.623 \times 10^{-2}$.	Constraint satisfaction does not imply improved accuracy.

6.3.3. *Canonical data-contract diagnosis and time-step convergence.* An initial check contained a false positive caused by a cross-trajectory comparison. With same-trajectory comparisons, mode1 met the criterion whereas mixed did not. A repeated single-mode mode1 case could also exhibit closure error, showing that modal mixing was not necessary for the anomaly. Further comparisons excluded insufficient $N = 2$ truncation as an explanation and showed better closure for projected momentum than for $M_N \dot{q}_{\text{native}}$. Because the internal generalized-alpha stage was unavailable through the public API, coordinates, momenta, and force labels could

not be guaranteed to represent the same internal solution stage. Order-2 BDF with `tsteps` was therefore used to store canonical quantities at the same solver step. BDF is used here only for state sampling and vector-field label generation; it has not been shown to be symplectic.

The mixed initial condition was subsequently found not to include the normalization/sign transform of the fixed basis. Applying the same transform to displacement and velocity yielded 12/12 consistent offline initial-condition checks and satisfied the prescribed canonical-closure criteria on the mixed short window. In an initial two-level time-step comparison, G2 increased from 0.00147337 to 0.00155680, so the predefined monotonic-convergence criterion was not met. A third time-step level was therefore introduced to resolve the trend.

The three-level study directly compares q, $p_{\text{projected}}$, $M_N^{-1}p$, B^TL, and total energy. Their observed convergence orders are 1.64186–2.00088, demonstrating stable convergence for all quantities considered. In the subsequent 1 s mode1_phi_pio2 case, G2, pointwise G3, impulse G3, and peak-to-peak perturbation-energy drift are 0.00146226, 0.000221162, 0.0000563606, and 0.00307057 $<$ 0.005, respectively, and all satisfy the predefined thresholds. Table 6.4 and Fig. 6.2 reports these values.

Table 6.4. Three-level convergence and 1 s validation results for the canonical data contract.

Case	Metric	Value	Criterion/conclusion
Three-level	Observed order of q	1.9978492	pass
Three-level	Observed order of $p_{\text{projected}}$	2.0008768	pass
Three-level	Observed order of $M_N^{-1}p$	2.0008769	pass
Three-level	Observed order of B^TL	1.9976914	pass
Three-level	Observed order of total energy	1.6418624	pass
1 s	G2 joint relative L2	0.00146226	$<$ 0.02
1 s	G3 pointwise relative L2	0.000221162	$<$ 0.02
1 s	G3 impulse relative L2	0.0000563606	$<$ 0.02
1 s	Peak-to-peak perturbation-energy drift	0.00307057	$<$ 0.005

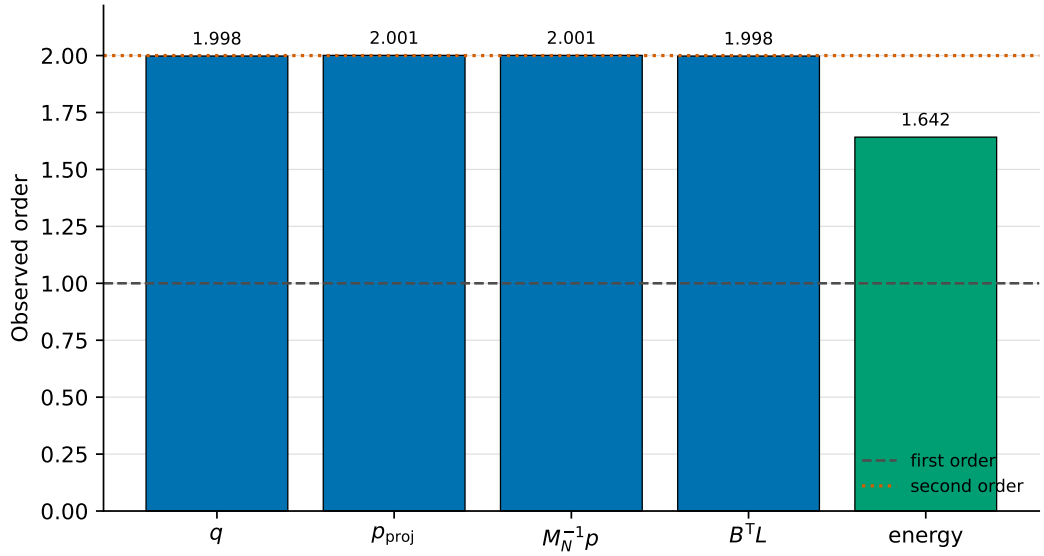


Figure 6.2. Observed convergence orders in the three-level time-discretization study of the canonical data contract. The orders range from 1.64186 to 2.00088 and all exceed one; these results do not establish BDF symplecticity or constitute numerical KAM certification.

The finite evidence supports at most this single diagnostic sentinel as a candidate input to a subsequent A1 qualification experiment; it does not certify data-generation qualification across

modes, amplitudes, or operating conditions. Multi-amplitude statistics, construction of the full 96-trajectory data set, and the full P2 model comparison will be reported in subsequent experiments. The present results do not establish that CHNN is generally superior to Real HNN on that full task.

6.4. **Open-port and high-fidelity numerical-validation plan.** The open-port branch will compare state-prediction error, componentwise generalized-force errors, power-balance residual, dissipation-sign violation rate, port leakage, and long-time trajectory error under paired operating conditions and ablations. These comparisons are designed to test the identifiability of conservative, dissipative, and externally applied forces. High-fidelity comparisons will quantify the discrepancy of the primary Kirchhoff model relative to Timoshenko-beam and solid models. The corresponding computations remain to be carried out, so this section specifies methods and metrics without asserting model superiority.

6.5. **High-order error certificates and numerical KAM validation.** Numerical KAM analysis requires a specified principal modal order, integrable reference system, and normal-form coefficients, together with the frequency-map nondegeneracy margin, a finite-order Diophantine margin, a high-order or analytic error certificate, the symplectic-map defect, a step-size nonresonance condition, and invariant-torus residuals. Numerical certification can be discussed only after the total error and the selected KAM threshold are expressed in the same norm. The present canonical-closure and time-step-convergence results do not constitute numerical KAM certification, and no invariant torus has yet been obtained.

6.6. **Parameter continuation, model discrepancy, and safe operating domain.** Subsequent studies will perform continuation in magnetic-field strength, insertion speed, flow velocity, contact clearance, and material parameters while reporting structural responses, port quantities, and discrepancies between reduced and high-fidelity models. A complete engineering safe domain must additionally incorporate open-port power exchange, peak contact response, high-fidelity discrepancy, and uncertainty propagation; it cannot be identified with the theoretical guarantee domain of the conservative KAM backbone. Because these computations have not yet been instantiated, this work does not claim an engineering safe domain.

7. Discussion and Scope of Applicability

7.1. **Theoretical foundations and methodological boundaries.** The continuum mechanics and finite-dimensional reduction are founded on geometrically exact rod theory, classical geometric mechanics, and geometric numerical integration. The HNN, dissipative/port-network, and complex-valued-network literature defines the learnable objects and function classes. Finite-dimensional continuous KAM theory, KAM theory for symplectic maps, and the parameterization method support the conservative flow, symplectic discretization, and invariant-torus validation, respectively. These bodies of literature concern different mathematical objects; software documentation or learning-model literature is not used as a substitute for the structural assumptions of continuum mechanics or KAM theorems.[1, 2, 17–29, 33]

Moser's invariant-curve theorem for annulus maps, Kuksin's infinite-dimensional KAM theory, the metriplectic framework, and theoretical results for specific complex-reaction networks delimit the applicability of two-dimensional sections, PDE limits, thermodynamic dissipation, and architecture-specific advantages of complex structures. They do not directly constitute the principal theorems for the finite-dimensional catheter system considered here.[30–34]

The canonical data-contract diagnosis shows that structural consistency must be checked componentwise in the configuration, tangent, and cotangent spaces and cannot be replaced by energy drift or a single residual. Cross-trajectory comparison, the distinction between projected momentum and the mass diagnostic, the unavailability of an internal solver stage, and omission of the fixed-basis transform from initialization represent four distinct error sources: comparison objects, cotangent projection, output semantics, and initialization. The two-level time-step study did not meet the predefined convergence criterion, whereas the three-level study resolved

the convergence trend. This sequence provides a reusable diagnostic strategy for other finite-dimensional projection problems. Force-v2 closure and Bridge perturbation-energy drift address different questions and cannot substitute for one another.

7.2. **Present scope and conditions requiring further validation.** The continuous model, joint Galerkin reduction, canonical Hamiltonian formulation, structured CHNN, error budget, and invariant-torus validation interfaces have been constructed, and the COMSOL field projection and canonical-label mapping have been instantiated on a fixed $N_{\mathrm{diag}} = 2$ basis. P0, P1, P1.5, and P1.6 provide low-dimensional and three-trajectory small-sample results; P1.6 specifically shows that exact satisfaction of hard structural constraints can coexist with degraded prediction. The three-level and 1 s studies support at most this single diagnostic sentinel as a candidate input to a subsequent A1 qualification experiment; they do not certify data-generation qualification across modes, amplitudes, or operating conditions.

The principal modal order n_{main}, full 96-trajectory data set, P2 model comparison, open-port identification, high-fidelity discrepancy, normal-form coefficients, nondegeneracy constants, statistical-learning constants, high-order error certificates, and KAM thresholds still require numerical instantiation. Seeds in the P1 series are algorithmic repetitions, and three trajectories do not support population-level physical extrapolation. The present results therefore do not establish general CHNN superiority over Real HNN on the full P2 task, invariant-torus certification for a catheter operating condition, a long-time stability guarantee for the complete open system, an engineering safe domain, or clinical efficacy. BDF remains only a state-sampling and vector-field-labeling method and has not been shown to be symplectic.

7.3. **Auxiliary models and future extensions.** A strict weak-form COMSOL implementation, Timoshenko beams, and three-dimensional solid models will support high-fidelity validation. Infinite-dimensional KAM theory, thermodynamic metriplectic structures, learning of unknown canonical coordinates, complete Maxwell-field models, and moving-wall fluid–structure interaction are retained as future extensions. Open-port studies will further examine power balance, dissipation sign, and port identifiability, while numerical KAM studies will compute nondegeneracy margins, high-order errors, and invariant-torus residuals. These models and validation tasks retain their full scientific scope but are not conflated with the present conclusions for the finite-dimensional conservative Kirchhoff branch.[4–8, 10, 11, 30, 32]

8. Conclusions

A unified framework has been developed for the multiphysics large-deformation dynamics of magnetically actuated flexible catheters, spanning a geometrically exact continuous model, joint Galerkin reduction, canonical Hamiltonian formulation, structured CHNN, and KAM stability analysis. The framework retains cross-sectional orientation, independent torsion, and rotary inertia and separates conservative action, passive dissipation, background-flow input, and control through mutually exclusive energy accounting. It thereby provides consistent state and power conventions for the continuous model, COMSOL data, and the finite-dimensional learning system.

The complex-representation-enhanced CHNN uses a real-valued physical Hamiltonian readout and independent branches for conservative correction, dissipation, and external ports. Statistical learning error is only one component of the KAM interface; complete certification additionally requires high-order or analytic error certificates, pre-existing physical perturbations, coordinate and truncation errors, symplectic-discretization error, nondegeneracy, and Diophantine conditions. The resulting analysis assigns distinct roles to continuous Hamiltonian KAM theory, KAM theory for symplectic maps, and a posteriori invariant-torus validation.

P0, P1, P1.5, and P1.6 provide low-dimensional or three-trajectory small-sample results. P1 shows a conditional CHNN advantage in the derivative-only comparison. P1.5 shows substantial residual-over-physics-only gains but no stable one-step-field advantage. P1.6 shows that all 30/30 constrained models satisfy the equilibrium conditions exactly while predictive errors

deteriorate markedly. These results cannot be extrapolated into general CHNN superiority on the full P2 task, and different random seeds are not independent physical samples.

The 0.5% COMSOL–Galerkin Bridge perturbation-energy criterion is not met by all three trajectories, whereas the independent 2% force-v2 joint-L2 criterion is met by all three; the two findings have distinct interpretations. After same-trajectory comparison, error-source isolation, identification of the cotangent projection, comparison of solver-output semantics, and correction of fixed-basis initialization, the three-level study yields observed convergence orders of 1.64186–2.00088. In the 1 s case, G2, pointwise G3, impulse G3, and peak-to-peak perturbation-energy drift are 0.00146226, 0.000221162, 0.0000563606, and $0.00307057 < 0.005$, respectively. The finite evidence supports at most this single diagnostic sentinel as a candidate input to a subsequent A1 qualification experiment; it does not certify data-generation qualification across modes, amplitudes, or operating conditions.

Multi-amplitude statistics, the full 96-trajectory data set, the P2 model comparison, open-port analysis, high-fidelity discrepancy, and numerical KAM validation remain subjects of subsequent numerical work. This study has not shown BDF to be symplectic, has not obtained numerical KAM certification, an invariant torus, or an engineering safe domain, and has not established general CHNN superiority over Real HNN on the full P2 task. The theoretical guarantees are limited to the conservative backbone under the corresponding structural, regularity, nonresonance, and small-perturbation conditions. Open ports, contact switching, model discrepancy, and engineering safety must be evaluated jointly using the energy balance, port quantities, high-fidelity validation, and experimental data.

Appendix A. Diagnostic Methods for the Canonical Data Contract

This appendix organizes the canonical data-contract diagnosis by error source and scientific interpretation.

Table 1.1. Diagnosis of anomalies and isolation of error sources in the canonical data contract.

Diagnostic step	Observation and analysis	Scientific interpretation
Same-trajectory comparison	After removing a cross-trajectory comparison, mode1 met the criterion whereas mixed did not.	The apparent positive result was caused by inconsistent comparison objects.
Repeated single-mode case	A repeated mode1 case could also exhibit closure error.	Modal mixing was not necessary for the anomaly.
Cotangent projection	Insufficient $N = 2$ truncation was not the cause, and projected momentum closed more accurately.	Momentum should be projected to the cotangent space of the fixed basis.
Solver-output semantics	The internal generalized-alpha stage was not available through the public API, whereas BDF `tsteps` returned same-solver-step states.	BDF is used only for state sampling and label generation.
Fixed-basis initialization	Applying the normalization/sign transform to displacement and velocity gave 12/12 consistent initial conditions and short-window closure for mixed.	The anomaly arose from inconsistent initialization in the fixed basis.
Two-level time-step comparison	G2 increased from 0.00147337 to 0.00155680.	Two levels did not meet the predefined monotonic-convergence criterion.
Three levels and 1 s validation	Observed orders were 1.64186–2.00088, and the extended-case closure and energy quantities met their thresholds.	The data contract meets the data-generation requirements for later multi-amplitude experiments.

The diagnosis successively examines comparison objects, modal mixing, reduced-order truncation, the cotangent-momentum definition, solver-output semantics, and fixed-basis initialization. The false positive caused by a cross-trajectory comparison shows that closure errors must be computed on the same physical trajectory and at the same sampling times. A repeated single-mode case can still exhibit the anomaly, so the error cannot be attributed directly to modal mixing. Increasing the reduced representation did not reveal the signature of an $N = 2$-truncation error; projected momentum closed more accurately than $M_N \dot{q}_{\text{native}}$.

Because the internal generalized-alpha stage is unavailable through the public API, exported displacement, velocity, and load cannot be guaranteed to correspond to the same internal solution stage. Order-2 BDF with `tsteps` is used to obtain states and vector-field labels at the same solver step. This choice does not imply that BDF is symplectic and does not interpret the discrete trajectory as a symplectic map.

Force closure and perturbation-energy drift use independent criteria. An earlier force comparison was sensitive to endpoints and the momentum anchor. On common-support indices $2 \dots 998$, with a fixed $p[2]$ anchor, the force-v2 joint L2 is below 2% for all three trajectories. This result does not alter the trajectory-level conclusions obtained with the original 0.5% Bridge perturbation-energy criterion.

Appendix B. Metric Definitions and Numerical Settings

B.1. **Canonical mapping and closure quantities.** For the fixed $N = 2$ basis, the mapping is

$$q = \Pi_{\text{conf}}(U), \qquad p = B^T P_{\text{full}}, \tag{2.1}$$

$$\dot{q} = M_N^{-1} p, \qquad \dot{p} = B^T L. \tag{2.2}$$

Here q is a configuration coordinate and p is its associated cotangent coordinate. Closure errors compare the exported $\dot{q}$ with $M_N^{-1}p$ and the exported $\dot{p}$ with $B^T L$, while the time-discretization study also examines q, p, and total energy.

The three-level study uses coarse, middle, and fine grids. If the errors on two adjacent levels are e_h and $e_{h/2}$, the observed order is

$$r_{\text{obs}} = \log_2\left(\frac{e_h}{e_{h/2}}\right).$$

The resulting observed orders span 1.64186–2.00088. In the 1 s case, the thresholds for G2, pointwise G3, and impulse G3 are all 0.02, and the peak-to-peak perturbation-energy-drift threshold is 0.005.

B.2. **Bridge and low-dimensional network experiments.** Bridge perturbation-energy drift is compared trajectory by trajectory with the 0.5% criterion and the exploratory 1% criterion. Force-v2 uses an independent 2% joint-L2 criterion. The energy and force quantities are not combined into a single conclusion.

P0 uses a four-dimensional nonlinear coupled oscillator. The training, validation, and test sets contain 80, 16, and 20 trajectories, respectively; each trajectory has 80 steps, $\Delta t = 0.025$, and the models are trained for 300 epochs. Five random seeds estimate algorithmic repeatability.

P1 and P1.5 each comprise 60/60 completed training-and-evaluation runs, and P1.6 comprises 30/30 hard-constrained models. The three physical trajectories—mode1, mode2, and mixed12—are partitioned by leave-one-whole-trajectory-out. Different random seeds share the same physical trajectory set and therefore do not increase the physical sample size. P1.5 compares physics-only and residual learning, whereas P1.6 compares predictive accuracy and structural residuals before and after imposing the hard equilibrium constraint.

B.3. **Subsequent numerical instantiation.** Multi-amplitude statistics, the full 96-trajectory data set, and the P2 model comparison will retain whole-trajectory partitioning and report cross-condition means, variances, and paired comparisons. Open-port studies additionally require power-balance, dissipation-sign, and port-leakage quantities, while high-fidelity studies must quantify the discrepancy of the Kirchhoff model relative to Timoshenko-beam and solid models. Numerical KAM analysis requires nondegeneracy and Diophantine margins, high-order errors, symplectic-map defects, and invariant-torus residuals in a common norm. Implementation code and supplementary data will be provided with the accompanying code and supplementary materials.

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